

Reasoning

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Version 2

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**“Few men think, yet all will have opinions. Hence men’s opinions are
superficial and confused”
– Bishop George Berkeley.**

0. Thinking and Reasoning

Thinking is something that we all do. This book is mainly concerned with a particular kind of thinking, namely, the activity of *reasoning*. What is reasoning? Reasoning is a kind of thinking that includes making inferences. To make an inference, or to *infer*, is to conclude something *based* on some other claim or set of claims. You *infer* a claim from, or based on, one or more claims. What is a claim? It is, roughly, a declarative sentence. (More on claims later). What do we mean by ‘based on’? We mean, roughly, that the truth of one claim depends on, or is connected to, the truth of another claim in such a way that the truth of the other would either entail, or make probable, the truth of the one. In this way, reasoning concerns special *connections* between claims, namely, those that are relevant for truth. The ultimate goal of reasoning is truth, or knowledge and understanding of the truth.

Consider the following set of sentences:

- a. The sun is shining.
- b. Sally’s favorite color is blue.
- c. There exists at least one desk in Los Angeles City Hall.

There appears to be no obvious connection between sentences a, b, and c. It appears that that set of sentences is just a set of random sentences without any clear connection between them. Although these sentences are not even *about* the same thing, the rational connection that occurs in reasoning is not merely about being about the same topic or thing. Consider this second set of sentences:

- d. Rin is a strong and independent young woman.
- e. Rin specializes in deep learning in artificial intelligence research.
- f. Rin enjoys eating Korean food.

Unlike the sentences in the first set (a-b), this set of sentences (d-f) is presumably about the same subject, namely, Rin. However, like the first set of sentence (a-b), they lack the sort of rational connection that is distinctive about lines of reasoning. In other words, there doesn’t seem to be any *inferential* relation between sentences d, e, and f. To put it another way, the truth (or falsity) of sentences d and e seem to have no direct or obvious bearing on the truth (or falsity) of sentence f. Like the first set, this set of sentences is just a list of facts. Both sets do not constitute *reasoning*. Now consider this set of sentences:

- g. Sami’s fingerprints are on the murder weapon.
- h. Sami’s DNA was found at the crime scene.
- i. Sami is the one who committed the crime.

The connection between the sentences in this set g-i stands out fairly clearly. In particular, the first two sentences, i.e. sentences g and h, if true, *support* the truth of sentence i. If sentence g and sentence h are true, then it seems that we could *reasonably* conclude or infer the truth of sentence i. Reasoning

is about finding these kind of connections in language and in thought. (We'll talk a little about the relation between those later).

There are some complications to this picture. Sometimes a person reasons – makes an inference – based on a claim or set of claims that actually amounts to something like a-c or d-f, but they *think* that it's like g-i. In other words, a person might infer a claim from a set of claims that *does not* provide any support for its truth. These are either *not arguments* or *bad arguments*. We'll talk about that later.

1. Preparation and Guarding Against Biases

Like all activities, thinking can be done well or poorly. In order to think well, we need to prepare our minds. There are many things that can be obstacles to critical thinking. One is our own attitude and mindset. Another is natural biases that dispose our minds to think in a bad way.

1.1 Attitude and Mindset

Everybody has opinions about things, and opinions are beliefs. Beliefs can be true or false. We may believe something is true when in fact it is false; and we may believe that something is false when in fact it is true. In either case, our belief would be false. Since opinions are beliefs, opinions can be true or false. We *think* that all of the opinions – beliefs – that we hold are true, for no rational, sane person holds to an opinion *knowing* and *understanding* that it is false. However, *thinking* that your opinions are true does not *make* them true. That is, *knowing* that a belief is true or false is one thing, but the *belief* being true or false is another thing entirely.

There is a popular saying that many hold: “Everyone is entitled to her/his/their own opinion.” In the literal sense, this is more or less true: you have a right to believe whatever it is you believe, and having a sincere opinion about some matter is not by itself an obvious violation of any moral principle. However, in that sense, it is *trivial*: of course you can believe whatever you believe.

However, what people who say that seem to mean is that their opinion is just as good – no better or no worse – than the opinions of others, and that they have a right not to be criticized for their belief. They hold that their belief is just as equally rational and true as any other belief. All of this is mistaken. First, everyone's opinions (beliefs) *can* be criticized, and *not* all opinions are equal, for some are false or irrational. Second, while it is true that we ought to *separate* criticizing a *person* for holding to a belief, on the one hand, and criticizing the *belief that they hold*, on the other hand, a person is nevertheless not immune from criticism for the beliefs that they hold. If a person holds to certain set of beliefs that flies in the face of logic and scientific evidence, and they are *ignoring* or *willfully choosing not to consider* the problems with their beliefs, they *can* be criticized. We should realize this: everyone's opinions (beliefs) are *not* necessarily equal – some *are* better (true or more rational) and some are worse (false, irrational, or arational), and *no* opinion is immune to critical scrutiny under the lens of reason and evidence.

Third, the consequences of having a belief is one thing, but having the belief is another thing. So, the right believe whatever you want is not necessarily violated if the consequence of having that belief is being *criticized*. In other words, having a right to believe whatever you want is not the same thing as having a right to certain consequences of having that belief.

It may be embarrassing sometimes when someone shows us that what we believe is false or lacking basis in good reason. We often say something like “it’s just my opinion” when this happens to guard ourselves. However, it is more than likely that, in the course of our life, some, if not many, of our beliefs are false. Nobody in their right mind wishes to carry false beliefs. When there are good reasons for thinking one is mistaken in one’s belief, one *ought* either to give up one’s belief, or at least suspend belief until further thought and examination. To be rational is not just to have good reasons for one’s beliefs and actions, but also to be responsive to reasons, especially when good reasons run counter to what one believes. To learn and grow requires not only jettisoning beliefs upon discovering that they are false, but understanding why they are false. To do this properly and effectively necessitates a sincere willingness to acknowledge our mistakes.

Personal pride often gets in the way of our own growth. In particular, we often wish to preserve the worth of our ego, and this could easily get in the way of not only admitting, but even recognizing, that a belief (opinion) we hold is false or our line of reasoning is faulty. Since recognition of one’s own shortcomings and mistakes is crucial for improving one’s thinking skills, pride can get in the way of growing and improving as a critical thinker.

Does this mean that one should always relent and concede to any criticism of your view or belief? Shouldn’t you stand up for what you believe? Isn’t that the best thing to do? First, firmly standing up for what you believe could be a very foolish thing to do. It may look good on a t-shirt, bumper sticker, or inspirational poster, but it is overly simplistic and in many instances ridiculous. Someone who stands up for their racist beliefs is not admirable or respectable simply because they stand up for what they believe. What is admirable is standing up *for what is true*, not simply for what *you* believe is true.

The common belief that everyone has their own ‘truth’ that is fundamentally equal to any other person’s truth is a very uncritical, lazy way of thinking. Everyone certainly *thinks* that their beliefs are true, but simply *thinking* that your beliefs are true is not necessarily imply that your beliefs are in fact true. Moreover, while not conceding and stubbornly standing for all of your beliefs in the face of any criticism is not good, so is conceding to every criticism of your beliefs. What we want is the ability to independently assess criticisms of our beliefs and reasons, and *based* on our independent assessment, reconsider our beliefs. (Methods for proper assessment are some of the things we will be learning). So, the suggestion that one ought to avoid stubbornly standing up for your beliefs even in the face of overwhelming reasons against your belief does not imply endorsing the opposite extreme of giving up all of your beliefs in the face of any criticisms of them.

We need to remember two things. First, discovering that our line of reasoning is faulty, or that a belief we hold is false, is ultimately something that is *good* for us as thinkers. We should not simply want to *appear* right in our beliefs and thinking. We should want, rather, to *be* right. Second, it is important to recognize that criticizing someone's belief as false is not necessarily insulting or criticizing the person as a person. Take it as a criticism of the belief, not your character or your intellect as a whole. So, being mistaken in your belief or reasoning does not necessarily mean that you are not intelligent.

Critical reasoning should aim at achieving *knowledge* or *understanding* of the truth. Insofar as we all value the truth, we all value reasoning well, for reasoning is how we achieve knowledge or understanding of the truth. Even those who skeptically ask "why should I care about the truth?" presuppose that they care about the truth. When anyone sincerely asks a question (e.g. "How do I get on the 210 freeway?" or "Where is the nearest restroom?"), they express that they care about a *truthful* answer. Our starting place, then, is that we all care about the truth, and reasoning is an activity that aims at the truth.

1.2 General Tips

Here are some general tips on how to pursue the truth and to approach things in life.

First, be wary of extreme views and thinking that extreme positions are the only options. Extreme views are ones that claim that *everything* is a certain way. Extreme views are not necessarily false, but they should make you pause with a healthy level of skepticism. Sometimes extreme views oversimplify matters based on very bad reasoning, and, oftentimes, they are based on lazy thinking. For instance, extreme relativism is the view that all value (truth and falsity, right or wrong, meaning) is relative to a particular culture, society or group. What we mean by 'right' and 'wrong' is different from what another culture means by 'right' and 'wrong'. Meaning is thus determined relative to different cultures or groups. Since the cultures or groups or societies differ in their way of life, it is thought that the meaning of certain words are taken to be different as well. In the most extreme form, relativism becomes subjectivism: what is true and what is false, what is right and what is wrong, etc. are solely determined relative to *individuals*. So what I mean and what you mean by 'right' and 'wrong', or 'good' and 'bad', are different. Subjectivism holds that meaning and values are determined by each person. This may seem like a deep, thoughtful, critical, and intelligent view to hold – something like an insightful and wise position to adopt. It may also seem to foster tolerance of others and their differences. However, it's actually very messy and bad thinking, and in many ways typical of lazy thinking and horrible reasoning. It's not deep, and it's actually *anti*-intelligent and very *unwise*.

Second, be skeptical of anything that lacks rational support. What I mean here by 'skeptical' is this: don't just accept something as true unless you have discovered good reasons for thinking it is true. However, one must not become a radical skeptic. Radical skepticism is the view that *everything* is

to be doubted in the most radical way, and that knowledge cannot be achieved. This is an extreme view. The obvious problem with anyone who claims “Knowledge is impossible” is that it raises the question, *are you claiming to know that knowledge is impossible?* One of the easiest ways to *feel* and *appear* like a critical thinker and good reasoner is to be a radical skeptic. Questioning *everything* in a radically skeptical way makes one feel like one is smart and critical. This is not the case. Anyone can just question everything. That’s not difficult to do, and anyone can do it. However, it takes careful thinking to know the appropriate degree, time, and place for skepticism, and how to pose a skeptical question. Skepticism is warranted under certain circumstances. When a claim is made that seems outrageous, or when a claim is made by an unreliable or non-credible (in-credible) source, then some level of skepticism is warranted. We can discuss later what makes a source credible.

Third, try to understand something before you criticize it. If you don’t understand it, you can’t rightfully criticize it. Unfortunately, most of the time people criticize before they understand. For instance, they criticize an artwork as stupid before they understand it, and perhaps because they don’t understand it. If your criticism is going to carry any weight, it must first be based on an understanding of what is being criticized. (Note also that understanding something does not equate to accepting it as true or correct).

Fourth, there are other perspectives and different points of view on most, if not all, issues. However, you shouldn’t think that entails that every perspective and point of view is true, equally accurate, for many of these views and perspective might actually be *false* or *inaccurate*. However, since there is a real possibility that we think our own perspective or view is true or the most accurate when in fact it is false or inaccurate. So, in order to make sure that you’re not believing anything false, you should try re-examine your viewpoint and beliefs, and try to examine things from a different perspective. Again, not because every perspective is equally correct – that’s certainly not true—but rather because we want to constantly make sure that our beliefs are true; that we have the right perspective on things; and that we’re not missing anything or presupposing something that is dubious or without rational support.

Fifth, when dealing with a problem, break the problem up into smaller bits, and then tackle the smaller bits. Sometimes a problem will seem overwhelming and you don’t know where or to start tackling the problem. Before you attempt a solution, understand that problem and its components. Break it into smaller problems. Once you’ve done that, you have a strategy: tackle each smaller problem, and eventually, you’ll have tackled the entire problem. When tackling the smaller problems, tackle the *easier* ones first.

1.3 Cognitive Biases

Many people typically think that things in the world are first *perceived* as they are, more or less, and then, perhaps after some thought, *evaluated and judged* by us. However, the problem is that we do not perceive things the way they are. Rather, we often perceive things according to what we already believe,

or at least in a way that is slanted and skewed towards making a particular sort of judgment.

A cognitive bias is a psychological tendency to think in a biased way. They are *cognitive* in that they have to do with perceptions, beliefs, etc. They are *biases* because they are distorted, skewed, or misleading. The cognitive bias is not consciously done, and it begins at the level of perception and initial cognitive processing. For example, suppose you learn a new word in class: “perfunctory”. After you learn it, you notice it in the following days on a regular basis: in the news, in the magazine, in a movie, being used by customers at a coffee shop, etc. The reason that it seems more common than before is most likely that you are just *noticing* it more. It was just as common before, but you just didn’t notice it. Before you knew the meaning of the word, you didn’t cognitively register it – *did not consciously perceive it*. Now that you know the meaning of the word, you are perceiving it more. This is an example of something that happens at the initial stages of perception and cognitive processing. It is not that you didn’t notice it because it wasn’t being used. Rather, knowing what it means now enables your cognitive system to register and notice uses of the word.

We are all subject to being biased. Awareness of these biases is the first step towards reducing their influence on us when we want to make the most accurate and impartial evaluation about matters. Remember, these cognitive biases happen pre-consciously: we’re not even aware that we’re being biased. We perceive the world, and then we judge what happens in it. Unfortunately, the biases affect us at the perceptual level, and that prejudices our judgment. Biases can also create problems in the social sphere.

There are too many cognitive biases to cover here, so we’ll just cover some of them.

1.3.1 Belief Bias

Belief bias is the tendency to evaluate evidence, reasons, or arguments in terms of how believable or plausible the main point that is being supported. Consider:

1. Songdo is in Incheon.
2. A city is in Incheon if it’s in South Korea.
3. Therefore, Songdo is in South Korea.

The conclusion (line 3) is very believable, and is in fact true. However, the argument/reasoning isn’t good. (We’ll get into this later). Because the main point (conclusion 3) is *believable/true* does not mean that the reasoning/argument is any good.

Belief bias often occurs when people actually agree with the main point or conclusion of an argument. If we already believe X, then we are more likely to think that evidence and arguments given *for* X are *strong evidence and arguments*. If we disagree with the conclusion of an argument, or the conclusion does *not* align with our current beliefs and values, then we tend

to do the opposite: reject the argument as flawed or at least lean heavily towards assessing the argument as weak or in an otherwise negative light. In other words, we'll already be *disposed towards* rejecting the argument *even before* understanding what the argument or evidence is.

For example, theists will be disposed towards evaluating arguments for the existence of God in a favorable way, whereas atheists will be disposed towards evaluating those arguments in a negative way, *before even fully understanding the arguments*.

A true claim could be given *bad arguments* and *weak evidence*. A false claim could be given *good arguments* and *strong evidence*. In the end, we need to assess an argument as impartially as possible, without being skewed by the plausibility and believability of the conclusion. Even if you don't have a belief either way, the plausibility or believability of the conclusion may lead one to be less careful in assessing the evidence and reasons for it.

1.3.2 Availability Heuristic

Availability heuristic is a cognitive bias where our judgment about the likelihood of something is based on what is most readily available in our mind. For example, suppose you watch a news report about a terrible tsunami and ensuing nuclear meltdown. News of this tragic event is on your mind. You are then inclined to think that tsunamis and nuclear meltdowns are highly likely to happen. You overestimate the likelihood of such events occurring based on what is on your mind

Here is another example. You read about a commercial airplane that crashed into the ocean a few months ago, killing everyone on board. Unfortunately, you read this at the airport, awaiting your flight. You board your plane, and now think that there is a good chance that your plane will crash into the ocean.

Here is another example. You watch a horror movie about a group of friends going camping in the wilderness, only to be hunted by a deranged person wielding a machete. The next day, your friends invite you to go camping with them. You decline their offer, sensing that it is likely that you will end up being hunted by murderous lunatic in the woods.

1.3.3 False Consensus Effect

False consensus effect is the tendency to assume that our personal attitudes and beliefs, or those that you share with those close to you, are shared by most everyone else.

For example, you, your family, and your closest friends believe that President X is an awful president. You meet someone new who is from outside of town, and you automatically assume that this person *also* believes that President X is bad. However, it turns out that this person believes that President X is a good president.

This happens also with music and movies. You watch a movie and find the movie to be awful. You meet someone else who also saw the movie, and you are incredibly surprised to find that this person believes the movie was excellent.

With false consensus effect, we tend to think that the *world at large* is like the *world at small*, as it were. The *local world* consisting of our self and our social group informs, unconsciously, the way we think of the *larger* world. You are inclined to think that *others* outside of your circle believe the same things. But this, again, is a bias at work: we think that our locally shared views are representative of *the wider population*.

1.3.4 Bandwagon Effect

Bandwagon effect is the unconscious tendency to act or believe in accordance with those around us. In other words, it's an inclination to follow what is popular or being done by others around you.

For example, suppose you go to the DMV. There is a long line going out the door and wrapping the parking lot. You park your car, gather your necessary documents, and proceed to the line. You wait in line for two hours, only to discover that you were in the wrong line. You just assumed, without thought, that you should be waiting in this long line.

You are not a very political person, and you also do not understand politics well. However, people around you tend overwhelmingly to believe that President X is a bad president. You yourself, without much thought, come to believe that President X is a bad president.

1.3.5 Negativity Bias

Negativity bias is the tendency people have to put more weight on negative information than positive information. Suppose you have two candidates for president: A and B. What studies have found is that the statement "Everyone likes A!" is *not* as effective in persuading people to vote for A as saying "Nobody likes B!" is. People tend to find the negative information more persuasive than the positive.

For example, suppose you are shopping online for a product. You find one that you like. Also, based on one thousand customer reviews (let's assume they're all honest), the product you have decided on purchasing has 4.5 stars out of 5. However, before purchasing the item, you wonder why it does not have 5 out of 5 stars. So, you decide to read the very few negative reviews. After reading them, you decide *not* to purchase the item. Even though the negative reviews are in the minority, you weigh them *more* than the overwhelmingly positive reviews.

1.3.6 In-Group Bias

In-group bias is a bias that we form *against* people who don't belong to a group to which we belong, and *in favor* of those people who do belong. The bias is in *favor* of perceived in-group associates, and *against* perceived out-

group others. The ‘group’ to which you perceive yourself to belong is loose. It can be your family to your nationality.

For example, you and your extended family go to a restaurant to have dinner. Your toddler nephew is jumping in his seat and making a lot of noise. You perceive your toddler as vivacious and a bundle of energetic joy. A few weeks later, you go to a restaurant to have dinner. A family is seated several tables from you, and they have a toddler son who is behaving like your nephew: jumping around in his seat and making a lot of noise. Yet, you perceive this family’s toddler son to be loud, rude, and misbehaved. Let’s look at several examples.

You hear of a building collapse in another country that tragically takes the lives of dozens of people. You immediately see this as a reflection of the poor regulations and horrible engineering of the people of that country generally. The same thing happens in your country, but you take it to be an isolated tragedy, not at all reflective of your country in any way.

You are watching ‘your’ basketball team play against a rival. A member of the rival team fouls a player from your favored team. You immediately perceive this as unfair and an act of horrible sportsmanship. A little more into the game, a member in your favored team fouls a player in the rival team, in almost exactly the same fashion. You immediately perceive the foul as an excellent strategic move.

You are supporter of political party A. News has come out that a member of opposing party B committed adultery. You take this as decisive against that politician, calling her/him an adulterous scoundrel. A few months later, news comes out that a politician in party A has been having an adulterous affair. You immediately come to that politician’s defense, pointing out that what happens in her/his personal life is none of our business, etc.

1.3.7 Fundamental Attribution Error

The *fundamental attribution error* is the tendency to explain a person’s behavior or circumstances by minimizing or dismissing factors *external* to the person while emphasizing factors *internal* to the person, such as the person’s character or personality. A very rough way of putting is this: fundamental attribution error is a tendency to resort to *psychological* explanations in explaining someone’s behavior while ignoring or not considering *sociological* and *circumstantial* factors that may be at play. Fundamental attribution error occurs especially when you are trying to explain behavior that is perceived to be negative.

Suppose someone runs a red light or cuts you off in traffic while speeding. The immediate tendency will be to explain these actions as the result of the *driver being a bad or rude driver*. Typically, people do not immediately think that the driver is in dire need of medical attention, hence running the red light or speeding and cutting people off. Rather, we simply think something about the psychology or character of the driver.

Suppose a student arrives late to class. The professor gives a disapproving look, and thinks that the student is a bad, lazy, or unintelligent student. The professor immediately believes that the student's lateness is the result of the student's characteristics as a student. The professor does not consider the possibility that external factors may have hindered the student from arriving early. (Perhaps the student was wrongfully detained by the police).

Suppose you see your professor walking down the street, flailing his hands wildly and cursing out loud. You might immediately be inclined to think that your professor has finally flipped his lid and gone crazy, when in fact he just walked into a swarm of bees.

The inclination is to resort automatically to something about the character or disposition of a person to explain his or her behavior rather than to think that factors external to the person significantly explain the person's behavior.

1.3.8 Actor-Observer Error

The *actor-observer error* is tending to explain negative behavior or circumstances with an emphasis on external factors when one is the actor (one whose behavior or circumstances is being explained) but with an emphasis on internal characteristics (e.g. personality, skills, etc.) when one is the observer (one who is observing the behavior or circumstances of another person). When the behavior or circumstance to be explained is positive, the point of emphasis switches: internal characteristics are emphasized when one is the actor, but external factors are emphasized when one is the observer.

Let's consider some examples. First, let's consider a *negative* behavior or circumstance. Suppose a person observes his friend play a video game. His friend is doing very poorly. He says to his friend "Man, you suck. You're awful. You have absolutely no skills when it comes to this game!" The person is explaining his friend's poor performance in the video game by emphasizing internal characteristics, i.e. the friend's skill and ability at playing the game. Suppose now this person takes his turn, and performs just as poorly as his friend. He says "Something is wrong with this controller. Also, I think there is serious internet lagging." Now that this person is the actor, i.e. the one playing, the person gives a different explanation: an explanation that emphasizes external factors.

Here's another example. A person learns that her friend failed a certain class. She thinks that her friend either isn't very smart or isn't very studious. In other words, she explains her friend's poor performance by emphasizing characteristics internal to the friend, viz., not being very smart, not being very studious, etc. Suppose she takes the same class with the same instructor the next semester, and she, too, fails the class. She says "The instructor was awful" and "The class is unreasonably difficult." Now that she is the 'actor' in this case, she gives an explanation of her own failing of the class by emphasizing external factors.

Next, let's consider a *positive* behavior or circumstance. Suppose a person is running in a race, but loses to a competitor. He says "My competitor was lucky, and I wonder if he was using illegal performance-enhancing drugs." Notice here that he is emphasizing external factors in explaining his competitors win. Suppose now that the person runs in another race and wins. He says "I'm naturally talented when it comes to athletics, and I'm very focused and hardworking when it comes to training." Notice in this case that the person emphasizes internal characteristics to explain his winning of the race.

Actor-observer error is similar to fundamental attribution error. The difference is that with actor-observer error, the point of emphasis in explaining behavior differs depending on whether one is the *actor* or one is the *observer*. Fundamental attribution error has to do with *observing* the action in another person only.

1.3.9 Obedience to Authority

Obedience to authority is a bias that leads us to become more acquiescent, credulous, and less critical in our thinking in the presence of someone or something we perceive to be a figure of authority. Note that this is not a tendency to believe or acquiesce to someone who is in a position of authority, but someone *we perceive* to be in a position of authority. Also note that it is a tendency to believe or acquiesce *generally*, and not just with respect to things considered to be within the area of expertise of the perceived authority figure.

For example, suppose you are at a shopping mall and someone approaches you wearing an official-looking uniform. He tells you to stop what you're doing and to sit on the ground. Because of the appearance of authority that the person exudes through the uniform, you will be inclined to acquiesce and do as he commands. Even if you hesitate, consider if the same thing happened if it was someone wearing normal street apparel. In such a case, you would be less inclined to comply with the person's commands.

Suppose you walk into a dentist's office for a teeth cleaning. As you sit, someone enters the office from the exterior door wearing a white lab coat. He approaches you and asks that you please hand over your smart phone. You will be more inclined to comply with his commands simply because of the appearance of authority he exudes by wearing a white lab coat. Had he been wearing normal street clothes, you would be less inclined to comply.

Suppose a physics professor tells the class that philosophy is a waste of time. Because the professor has a position of authority, many of the students in class will be inclined to believe what the professor is saying, or at least to put a lot of weight on it. However, notice that the professor is speaking outside of his area of expertise. (Imagine an English professor telling his students that physics is a waste of time).

Note that while obedience to authority as a cognitive bias is problematic, we should not take this to mean that we ought not to heed or respect authority whatsoever. It is prudent generally to respect and appreciate what

is said by a figure of authority within her or his area of expertise. An authority who is an expert has more knowledge and understanding in their particular field of expertise than we do. For instance, it would be foolish not to heed the authority of a medical doctor when it comes to matters pertaining to health that falls within the domain of her expertise.

1.3.10 Confirmation Bias

Confirmation bias occurs when we selectively notice only evidence or reasons that confirm what we already believe or what we already suspect to be the case. We are inclined to notice only cases that confirm or favor our prior beliefs and suspicions, or interpret data in a way that does so. Note that this is not to say that we have a tendency to *ignore* contrary evidence or data. To ignore something implies that we did notice it but choose not to consider it. When confirmation bias happens, we do not even notice the contrary cases and evidence. Like other cognitive biases, it happens unconsciously. In essence, confirmation bias is the selective filtering of data to fit and support a prejudice. The prejudicial stance or belief influences perception of cases and which cases are perceived.

Suppose your boss believes that a co-worker, Sally, is lazy. All the times that Sally is working hard, the boss will not notice or take much note. It is not consciously intentional. Rather, the prejudicial belief that Sally is lazy is affecting what he sees and does not see. However, every time she takes her official break, her boss notices and takes what he sees to be evidence that confirms his belief that Sally is lazy.

Sometimes confirmation bias affects our more active efforts in seeking evidence, or patterns in data that support our initial beliefs and suspicions. A police detective who strongly suspects that Sally is the perpetrator of a crime may seek evidence not in an impartial way, but in a way that aims to support his initial suspicions of Sally. He emphasizes evidence that suggests that Sally did it. When it comes to evidence suggesting that Sally did not commit the crime, the detective either does not even recognize or notice them, or he does but minimizes their importance.

1.3.11 Self-Serving Bias

The *self-serving bias* occurs when we perceive events in a way that protects our own egos. The tendency is to perceive negative events with which we are associated as not our fault but rather as the fault of others or accidental circumstances, but to perceive positive events with which we are associated as largely and primarily due to our own presence or actions. Generally, the self-serving bias is a tendency to think of ourselves in a favorable way, and to preserve that favorable sense of self.

For example, you take a class and you fail. You perceive your failing not as a result of your poor performance, but shift the blame to elements of the class: the material was too difficult; the textbook was very unclear; the teacher was horrible; etc.

Suppose you read a book and find it very difficult to understand. You immediately believe that the book is written in a very obscure manner, blaming the book for your lack of understanding rather than your reading comprehension skills.

Suppose you are disappointed to find that a particular dishwasher you used did not clean the dishes well. You immediately come to believe that the dishwasher is broken or not very good quality rather than thinking that you may have done something wrong in your operation of the dishwasher.

1.3.12 Bias Blindspot

Bias-blindspot is the tendency not to see or notice the influence of biases in oneself and to see or notice their influence more readily in others. After learning about the cognitive biases covered above, you may begin to notice these biases impacting other people around you: your friends, your family, etc. However, you'll likely notice their influence more in others than you do in yourself.

For example, you may notice the heavy influence of confirmation bias in your parents with respect to what they think about a friend of yours. They may think that she or he is inconsiderate and trouble. When your friend is over, she typically places any used cups in the sink after rinsing them. However, your parents do not notice those instances. One day your friend forgets to rinse out and place her used cup in the sink, and your parents take notice, pointing this out as evidence to support their belief that she is an inconsiderate person. However, you prejudicially believe that people from Los Angeles are fake and superficial, and despite meeting genuine and intelligent people from Los Angeles, you only take note of the few people from Los Angeles you have met who happened to impress you in a way that confirms your prejudicial belief. Though you readily noted the impact of confirmation bias in your parents, you do not notice the influence of that cognitive bias in your own thinking.

2. Language and Logic

One primary function of language¹ is communication: we use language to communicate with each other our thoughts, desires, beliefs, feelings, etc. We use language to talk about issues, to work through some problem, to entertain each other; and so on.

The basic rules of language are the laws of grammar, which (generally) specify how to use certain words and phrases, and how to put words and phrases together to construct sentences. The laws of grammar can differ from one language to another. For example, the rule for constructing an adjectival phrase in Spanish states that the adjective be placed after the noun that is modified by the adjective (e.g. 'la puerta verde'). In English, the rule states that the adjective come before the noun (e.g. 'the green door').

¹ I mean *natural language*. A natural language is, roughly, one that is, or was, used and spoken by a people or culture sometime in the history of humankind.

The basic rules of reasoning consist, at least in part, the laws of logic. That is, one aspect of reasoning well is following the laws of logic, i.e. being *logical*. The laws of logic are different from the laws of language (grammar). Unlike the laws of language, the laws of logic, at their core, are invariant across cultures: they do not differ from one culture to the next. An illogical argument given in one language or culture will remain illogical when translated in any language or given in any culture.²

What is the connection between language and logic? This is a difficult question to answer. For the sake of our class, let's try to keep it simple. We are interested in language insofar as it is used in reasoning and argumentation. Since reasoning and argumentation should follow the laws of logic, language will need to follow the laws of logic as well. What does that mean?

When we use language, we must try to understand the logical meaning or logical sense of the words and sentences that we use. What we also need to do is pay close attention to the linguistic meaning of each of the words, and how they figure into the meaning of a sentence.

To illustrate what I mean, let's consider some examples. Consider the following statement:

1. End racism by any means necessary.

Someone might respond to this statement by saying the following: "We must reject statement 1 because it advocates and encourages people to end racism by killing, torturing, shooting innocents, etc. That's unacceptable and clearly wrong. Hence, statement 1 is wrong."

The problem with that response is that it is failing to understand the meaning of statement 1. Statement 1 does not say, imply, or suggest that one *end racism by any means*, i.e. *end racism by any way possible*. Rather, statement 1 states that one end racism by any means *necessary*. The word 'necessary' is doing some work here, and the response above disregards this important word. "End racism by any means" differs in meaning from "End racism by any means *necessary*."

Two questions arise at this point. First, who decides what is necessary in ending racism? Though that is an important question, it's a different question. When someone says "Handle all firearms with necessary precautions", there is an important question regarding what are the necessary precautions. However, that does not cast bear on the truth of the claim. Even if we may disagree about what precautions are necessary, it is not any less true that we ought to handle all firearms with necessary precautions. Second, isn't pointing out 'necessary' a trivial point? No.

² There are different formal systems in logic, with different laws. For the sake of simplicity, we are assuming that there is one, general set of logical laws that govern correct reasoning in common discourse. While logic is a complex topic, for our class we are going to take it to be the laws of reasoning.

Reasoning carefully involves using language carefully, and when we're not careful with language – even with a single word – we can end up reasoning poorly or saying something inaccurate.

Let's consider another example. Consider the following case:

2. A Harvard study compared three leading headache medicines on their ability to quickly and effectively treat headaches, and the study found that nobody beat Brand X.

Let us assume that everything stated in this case is true, and that the study has been done with the most rigorous methods and representative samples. In reading case 2, someone might make the following inference: "Assuming that the Harvard study was conducted with as much objectivity as possible, and the results of the study are accurate and true, Brand X is better than the other headache medicines that were studied."

This inference is mistaken. Case 2 does not say that Brand X beat all the other brands in the study. Rather, it says that none beat Brand X. It could be that Brand X tied with all the other brands that were studied. Even in that case, it would still be true that none of them beat Brand X. However, in that case, it would be false to infer that Brand X beat the others.

These are just some ways that one might reason improperly largely due to not reading something carefully. What we will do next is examine language through a logical lens. In particular, we want to understand certain words that have logical import, by which I mean certain words that will be important to note when reasoning.

2.1 Sentences

There are many different kinds of sentences that can be formed in language: questions and commands are some. However, only one kind of sentence can clearly be true or false: declarative sentences. They are sentences that purport to *declare* what is the case. Declarative sentences are also called "descriptive sentences" because they purport to describe a certain state of affairs, or how the world is.³

In this section, we will focus on declarative sentences. I'll call these sentences 'statements', 'claims', 'assertions', and even 'propositions' for the sake of simplicity.⁴

The following are examples of declarative sentences:

1. France is a country.
2. Earth revolves around the Sun.

³ There is some debate on this. For instance, declarative sentences are used in fiction, but it's not at all obvious that they aim at describing or declaring the way the world is. In any case, for our purposes, this will do.

⁴ In philosophy of language, all of these are actually distinguished. However, for our purposes, we will take them as interchangeable, at least for now.

3. The moon is made of cheese.
4. Everything sucks.
5. France is a country and Earth revolves around the Sun.
6. If the moon is made of cheese, then the number one exists.
7. Although the number one exists, France is a country.
8. Either Earth revolves around the Sun, or the moon is made of cheese.
9. The moon is not made of cheese.
10. It is not the case that France is a country.

All of these are declarative sentences, i.e. claims. Let us now organize declarative sentences into logical types.

2.1.1 Logical Operators

The logical types of declarative sentences will be organized based on whether or not they contain *logical operators*, and, if so, what kind of logical operator. Logical operators are truth-functions. What is a truth-function?

First, let's discuss the concept of a function. Functions are like machines: they have a designated input (something you insert) and a designated output (something that comes out). A vending machine is something in which you insert money and some code (e.g. 'A1'), and then there is an output (e.g. a candy bar or a soda). You can define a vending machine in this way:

VendingMachine(x,y)=food/beverage item

Here, x is going to be money, and y is going to be the code you punch in. Then we define a particular vending machine like this:

VendingMachine(50¢,A-button) = Doritos
 VendingMachine(50¢,B-button) = Fun-Yuns
 VendingMachine(50¢, C-button) = Pretzels
 Etc.

Think about the proper and valid inputs and outputs of a vending machine. An elephant is not a valid input for a vending machine. Money, namely, coins and paper currency, perhaps a credit/debit card, will be the designated valid kind of input. Valid outputs depend on the vending machine. If it's a soda vending machine, it can give you whatever is in its inventory, e.g. Pepsi, Mountain Dew, etc. However, it cannot give you what's not in its inventory, e.g. a Bentley, a Gucci bag, etc.

Functions are like vending machines. There is a valid set of inputs, and a valid set of outputs. Think about a function in mathematics. A mathematical function only takes numbers as inputs, and gives only numbers as outputs. For example:

$$f(x) = 2x$$

is a mathematical function. For 'x' you input a number, and then as output you get another number. We might define this function as taking *natural*

numbers as valid inputs, and even natural numbers as valid outputs. Insert 2 (for x) and you get 4; insert 6, you get 12; etc.

The **domain and range** of a function are the set of valid inputs and the set of valid outputs, respectively. A vending machine has a domain (money) and range (food/beverages). All functions have a domain and range. The domain of the mathematical function above ($f(x) = 2x$) is the set of natural numbers, and the range is the set of even natural numbers.

A truth-function takes as valid inputs and valid outputs *truth-values*. For our purposes, there are only two truth-values: *TRUE* and *FALSE*. Truth-functions are determined by *logical operators*. There are four kinds of logical operators:

1. Negation
2. Conjunction
3. Disjunction
4. Conditional

These logical operators are the truth-functions that we will be concerned with in our class. As I said, you can think of these logical operators as functions. For example:

Negation(x)
Conjunction(x,y)
Disjunction(x,y)
Conditional(x,y)

Here, ‘ x ’ and ‘ y ’ indicate what you put ‘into’ the truth-functions, namely, a truth-value, or, a true or false sentence. What they give as outputs are truth-values, or true or false sentences. The domain of logical operators (valid inputs) consists of true or false declarative sentences, and the range of logical operators (valid outputs) consists of true or false declarative sentences.

Remember that for us, declarative sentences, and whatever has the propositional structure of declarative sentences (i.e. sentential structure), will be the primary bearers of truth-values. (Something has ‘propositional structure’ or ‘sentential structure’ if it has the structure of a proposition or sentence).

Let’s consider an example. Take negation: as mentioned above, we can think of negation as a function *Negation*(x). The input x is a true or false declarative sentence. If we insert a *true sentence*, then *Negation*($\langle$ true sentence $\rangle$) = false. For instance, “Pozole is a soup made using hominy” is a true sentence. *Negation*(“Pozole is a soup made using hominy”) is the negation of that sentence. That would give us the truth-value of *false*. To put it in common language, since “Pozole is a soup made using hominy” is *true*, the negation of that sentence, which is the sentence “It is not the case that Pozole is a soup made using hominy” is *false*. The locution “It is not the case that x ” *negates* the sentence ‘ x ’. We’ll talk in more detail later.

I stated that there are four types of logical operators: negation; conjunction; disjunction; and conditional. While logical operators are truth-functions, logical operator *words* are words that *refer to*, or *express*, logical operators. Since there are four types of logical operators, there are four types of logical operator words:

Negation Operator Words: these include words equivalent to the word “not”, e.g. “none”, “no”, or some form of negative. (Here ‘negative’ does not mean *bad*, but more in the sense of “no”).

Disjunction Operator Words: these include words equivalent to “or”.

Conjunction Operator Words: these include words equivalent to the word “and” in it, or some equivalent. Words equivalent to “and” include the following: “but”; “although”; and “while”. There are more, but this will suffice.

Conditional Operator Words: these are words that introduce a condition, and they include words equivalent to “if”. They most obviously occur in ‘if-then’ claims.

2.1.2 Equivalence

Equivalence is a relation between two things with a truth-value. That is, equivalence is a relation that holds between two things that have propositional structure. Think of equivalence as *equal value*. Two sentences are equivalent if they have the same *truth-value* under all conditions. If two sentences are equivalent, then whenever one sentence is true, the other is true; when one is false, the other is false; and *vice versa*.

We sometimes talk about *terms* being equivalent. Terms are not just sentences, but things like words and phrases. Two terms are equivalent if you can replace one word with the other in a sentence without effecting a change in the truth-value of the sentence. For example, “cat” and “feline” are equivalent words, and so are “dog” and “canine”. A *term* is just a word or a phrase.

Note that equivalence is not about linguistic *meaning*. Equivalence is about the truth-value of a sentence. Two sentences may have a different meaning, but yet still be equivalent. For instance, “the sum of the interior angles of a trilateral polygon is one hundred and eighty degrees” and “the sum of the interior angles of a triangle polygon is one hundred and eighty degrees”. These two sentences have slightly different meanings: one mentions a polygon having three *sides*, but the other mentions a polygon having three *angles*. However, they are equivalent.

2.1.3 Atomic Sentences

An **atomic**, or **simple**, declarative **sentence** is a declarative sentence that contains no logical operator word. So if it has a negation word (e.g. ‘not’) or a conjunction word (e.g. ‘and’ or ‘but’), then it is not an atomic sentence.⁵

Here are some examples of atomic claims with a subject-predicate structure (the subject part of the sentence is underlined):

- (a) The dog barks.
- (b) Whales are mammals.
- (c) Sally is studying.
- (d) John is working.

Here are some examples of atomic claims with a subject-relation-subject/object structure (the subject or object parts of the sentence are underlined):

- (e) Sally is taller than John.
- (f) John kicked the ball.
- (g) I dislike the taste of liver.
- (h) Whales eat krill.

Sometimes I will be using variables, or place-holders, for atomic sentences. Capital letters ‘P’, ‘Q’, ‘R’, ‘S’, etc. will be used to indicate a sentence. For instance, in “If you are in Oslo, Norway, then you are in Europe” might be abbreviated as “If P, then Q”, where I specify that P is the sentence “You are in Oslo, Norway”, and Q is the sentence “You are in Europe.”

2.1.4 Molecular Sentences

A *molecular* or *compound claim* is a claim that (1) contains at least one atomic claim; and (2) contains at least one logical operator word. Because there are four types of logical operator words, there are four types of molecular claims: *negations*; *disjunctions*; *conjunctions*; and *conditionals*. What we are going to do is discuss what these are and the truth-conditions for each.

Truth-conditions are conditions under which a sentence is true or false. In explaining the truth-conditions, we will be explaining the corresponding logical operator as a *truth-function*. A truth-function is a function that has truth-values as inputs and outputs.

Sometimes I will be using variables, or place-holders, for a sentence, *either atomic or molecular*. Greek letters Φ (pronounced like ‘pie’ but with an ‘f’ in place of the ‘p’), Ψ (pronounced like ‘sigh’), X (pronounced like ‘pie’ but with a ‘k’ in place of the ‘p’), etc.

2.1.4.1 Negations

A **negation** is roughly a ‘not’ claim. You take a claim (atomic or molecular), *negate* it by introducing a negation operator word, and then you have a

⁵ There is a complication to this claim when it comes to statements that express attitudes towards propositions. For instance, ‘John believes that Bill is not home’. This sentence contains a ‘not’, and appears to be *not* an atomic sentence. However, it might be argued that it is an atomic sentence since the ‘not’ appears within the belief held by John. For the sake of simplicity, I will not consider such cases for now.

negation. For our class, we are going to make the standard form of a negation to be a claim that has the term “It is not the case that” attached the front. For example, you take something like “Sally is studying” (atomic claim) and then make a negation by attaching “It is not the case that” in front of it to make a standard form negation: “It is not the case that Sally is studying.” You take the sentence “Jin is swimming and Sally is studying” (molecular claim) and then make a negation by attaching “It is not the case that” in front of it to make a standard form negation: “It is not the case that Jin is swimming and Sally is studying.”

A standard form negation will have the following structure:

It is not the case that <claim>.

Notice that ‘<claim>’ is a place-holder for *any* sentence, atomic or molecular. If the claim is “Sally is swimming”, we get “It is not the case that Sally is swimming”. If the claim is “It is not the case that Sally is swimming”, then we get “It is not the case that it is not the case that Sally is swimming.”

There are also non-standard forms of negations. “It is not the case that Sally is swimming” can be stated as “Sally is not swimming” or “Sally isn’t swimming”. These are both non-standard form. However, they are all equivalent to the standard form.

Here are some other examples of negations:

1. John isn’t going to the party.
2. It is not the case that Sally is going to the party.
3. Nobody is going to the party.
4. No humans are mammals.
5. None of the people in this room know Gödel’s Incompleteness Theorems.
6. It is not the case that John isn’t going to the party.
7. It’s false that Sally is going to the party.
8. It isn’t true that John is going to the party.
9. It is not the case that it is not the case that John has studied.
10. It is not the case that if I eat watermelon seeds a watermelon will grow in my stomach.

A negation is a truth-function. What negation does is it takes a sentence as input (you put something in the <claim> part) and then it gives you another sentence as output. In particular, here’s what negation as a truth-function does: you put in a *true* claim, and then it spits out a *false* claim; you give it a *false* claim and it spits out a *true* claim.

For example, suppose we have two claims: “Los Angeles is the capital of California” and “Tom Waits is a musician”. The first sentence is false, and the second sentence is true. If you put in the first sentence into the negation truth-function, you get “It is not the case that Los Angeles is the capital of California”. This sentence would be true. So, you put in a false claim, you got out a true claim. You put in the second claim into the negation truth-function and you get “It is not the case that Tom Waits is a musician.” This is a false claim. Hence, you put in a true sentence, and you get a false

sentence. Here is a chart that maps out the truth-conditions for the truth-function of negation:

If the input sentence is:	Then the output sentence is:
TRUE	FALSE
FALSE	TRUE

2.1.4.2 Conjunctions

A **conjunction** is roughly an ‘and’ claim. A standard form conjunction will be two claims put together using the word ‘and’. Standard form conjunctions will always have at least *two* claims (not necessarily different) put together using ‘and’ to make one claim. The claims that are conjoined are called “conjuncts”. So, when you have “John is at work and Sally is in the library”, the sentence “John is at work” is a conjunct, and so is the sentence “Sally is in the library.” Remember, a conjunction is *one sentence*, but it is constituted by two sentences being put together by the conjunction ‘and’.

A standard form conjunction has the following structure:

<Claim 1> and <claim 2>.

<Claim 1> and <claim 2> are usually different sentences, but they can be the same sentence.

You can also make a non-standard form conjunction by putting together terms using ‘and’. You can also make non-standard form conjunctions using terms equivalent to “and”, such as “but” and “although”.

Here are some examples of conjunctions:

1. John is going to the party and Sally is studying.
2. Sally and Felipe are studying.
3. Both Sally and Felipe are human beings.
4. John is not going to the party and Sally is studying in the library.
5. The Eiffel Tower is in France, and the moon is made of cheese.
6. Sally is studying but John is partying.
7. Although John is in the library, he is playing a video game on his laptop.
8. While Sally is studying, John is partying.
9. Soo is agile, quick, and very intelligent.

A conjunction is a truth-function. What conjunction does is it takes *two* sentences as input and spits out one sentence as output by conjoining those two sentences with “and”. Here is a chart that maps out the truth-conditions for the conjunction truth-function:

If the input sentences are:	Then the output sentence is:
Both TRUE	TRUE
Different truth-values	FALSE
Both FALSE	FALSE

Notice that a conjunction can be true in only one way: when the input sentences are *both TRUE*.

2.1.4.3 Disjunctions

A **disjunction** is an ‘or’ claim, and some disjunctions are sometimes called a ‘dilemma’. Like standard form conjunctions, standard form disjunctions will always have at least two claims (not necessarily different) put together using an ‘or’ to make one claim. The two claims that are put together to form a disjunction are called “disjuncts”. So, in the sentence “Sally is in the library or Sally is in class” is one sentence, namely, a disjunction, that is constituted by two claims, “Sally is in the library”, and “Sally is in class”. These two sentences are the disjuncts of the disjunction.

Sometimes it helps to use the word ‘either’ A standard form disjunction has the following structure (with or without a comma ‘,’ before the ‘or’):

<Claim 1> or <claim 2>.
Either <claim 1> or <claim 2>.

<Claim 1> and <claim 2> are usually different sentences, but they can be the same sentence.

You can make non-standard form disjunctions by putting together terms using ‘or’. Here are some examples of disjunctions:

1. John is going to the party or Sally is studying.
2. Sally is going to the party or she is studying.
3. Either Sally is going to the party or she is studying in the library.
4. Sally is either studying or partying.
5. Either Sally is not studying but partying, or Sally is in the library and studying.
6. Either the sun is out or it isn’t.
7. John or Sally is in the room.
8. Sally is either studying, partying, or sleeping.
9. John is either at the library, sleeping, or sitting at a desk.

A disjunction is a truth-function. What disjunction does is it takes *two* sentences as input and spits out one sentence as output by disjoining those two sentences with “or”. Here is a chart that maps out the truth-conditions for the disjunction truth-function:

If the input sentences are:	Then output sentence is:
Both TRUE	TRUE
Different truth-values	TRUE
Both FALSE	FALSE

Notice that a disjunction can be false in only one way: when the input sentences are *both FALSE*.

The ‘or’ in truth-functional logic is called an *inclusive disjunction* because it includes the possibility of *both* disjuncts being true. So, in our class, when

we say something like ‘Either Sally is in the library or Sally is sleeping’, there is the possibility that *both* are true.

There are cases where both cannot be true, e.g. “Either Sally is in Los Angeles, California, or Sally is in Mumbai, India.” Here, both disjuncts cannot be true. However, this is not because of the ‘or’, but because of something about medium-size objects like Sally and spatio-temporal facts. In other words, the reason is not logical.

In any case, what we know is this: even if *both* disjuncts are true, it still means that the entire disjunction is true. If only one is true, and only one can be true, it would still follow that the entire disjunction is true.

2.1.4.4 Conditionals

A **conditional** is a claim that introduces a condition, and a consequence of that condition being satisfied. A standard form conditional is an if-then claim that has this structure:

If <claim 1>, then <claim 2>.

<Claim 1> and <claim 2> are usually different sentences, but they can be the same sentence.

There are two parts to a conditional: the *antecedent* and the *consequent*. The antecedent is the claim that comes after ‘if’ and before ‘then’; the consequent is the claim that comes after ‘then’.

The antecedent specifies a *sufficient condition*, and the consequent specifies a *necessary condition*. A sufficient condition is a condition that, *if satisfied/met*, would be *sufficient* or *enough* to make the consequent true. A necessary condition is a *necessary* or *required* condition for the antecedent to be true.

A necessary condition may not be a sufficient condition, and a sufficient condition may not be a necessary condition. For example, consider the following:

If a person is in Los Angeles, California, then that person is in the U.S.A.

The antecedent of this sentence is “A person is in Los Angeles, California.” The consequent is “That person is in the U.S.A.” If the antecedent is *true*, then that would be *enough*, or *sufficient*, to imply the truth of the consequent “That person is in the U.S.A.” Is it *necessary*? Is it *required* or *necessary* that a person be in Los Angeles, California in order to be in the U.S.A.? No: you can be in Helena, Montana, and many other cities in the U.S.A. So, while the antecedent is sufficient for being in the U.S.A., it is not necessary. Is it necessary that you be in the U.S.A. in order to be in Los Angeles, California? Yes. So, the consequent specifies a *necessary* condition.

When dealing with conditionals, first try to identify the antecedent and the consequent, and that will tell you which one is the necessary condition and which one is the sufficient condition.

We stated that the standard form conditional is this: *if <claim 1>, then <claim 2>*. There are two equivalent ways of putting this that we should note. Let ‘ $\equiv$ ’ mean *is equivalent to*.

$$\begin{aligned} & \text{“If <claim 1>, then <claim 2>”} \\ & \equiv \\ & \text{“<Claim 1> only if <claim 2>”} \\ & \equiv \\ & \text{“<Claim 2> if <claim 1>”} \end{aligned}$$

Note the order. For example, “<Claim 1> if <Claim 2>” is NOT EQUIVALENT to “If <claim 1>, then <claim 2>”.

Non-standard form conditionals can be made using the terms “if”, “only if”, “whenever”, and “when”. Here are some other examples of conditionals:

1. If a person is in Beijing, China, then that person is in the continent of Asia.
2. A person is near the East Sea if a person is in Busan, South Korea.
3. When you are in class, you are able to learn interesting things.
4. Sean will get burned only if there’s fire or something hot.
5. If you are not in Los Angeles County, then you are not in Pasadena.
6. If you study hard, attend every class, and do all the homework, then you will do well on the exam.
7. You will graduate only if you take all the required classes and pass those classes.
8. If you are in Pasadena or in Alta Dena, then you are in California.

A conditional is a truth-function. What conditional does is it takes *two* sentences as input and spits out one sentence as output connecting those two sentences with “if” and “then”. Here is a chart that maps out the truth-conditions for the conditional truth-function:

If the input antecedent sentence (i.e. <i>if</i> part) is:	And the input consequent sentence (i.e. <i>then</i> part) is:	Then output sentence is:
TRUE	TRUE	TRUE
TRUE	FALSE	FALSE
FALSE	TRUE	TRUE
FALSE	FALSE	TRUE

Notice that a conditional can be false in only one way: when the input sentence for the antecedent is *TRUE* and the input sentence for the consequent is *FALSE*.

The conditional truth-function is not difficult to define, but it is difficult to understand and, hence, not easy to remember. However, you can easily

remember it by considering a very simple case as a guide. Suppose I make the following promise to you:

PROMISE: If you pass the class, then I will buy you dinner.

What would make this claim false? In other words, what would it take for me to break my promise?

Now, if you pass the class and Q is true, then obviously I've fulfilled my promise. What about if you pass the class and Q is false? Well, if I promised to buy you dinner if you pass the class, and you did pass the class, but I didn't buy you dinner, then I'd be breaking my promise. Hence, my promise would be broken.

If you did *not* pass the class, and I didn't buy you dinner, does that mean that I broke my promise? Of course not – I only promised to buy you dinner *if* you passed the class.

But suppose you did not pass the class but I still bought you dinner. Of course not. I'm just being a nice person, not a promise-breaker.

Hence, the only way for me to break my promise, i.e. for the promise above to be false, is if the antecedent is true and the consequent is false.

So, think of a conditional being false like a conditional promise being broken. The only way to break a conditional promise is when you fulfill the condition, and I don't follow through with I promised to do.

In the same way, the only way for a conditional to be false, in the way that a conditional promise can be broken, is if the antecedent is true and the consequent is false.

2.1.5 Truth-Conditions

Here is a chart that summarizes the truth-conditions we just discussed:

Negations

If sentence 'P' is	Then ' <i>It is not the case that P</i> ' is
TRUE	FALSE
FALSE	TRUE

Conjunctions

If sentence 'P' is	If sentence 'Q' is	Then sentence 'P <i>and</i> Q' is
TRUE	TRUE	TRUE
TRUE	FALSE	FALSE

FALSE	TRUE	FALSE
FALSE	FALSE	FALSE

Disjunctions

If sentence 'P' is	If sentence 'Q' is	Then sentence 'P <i>or</i> Q' is
TRUE	TRUE	TRUE
TRUE	FALSE	TRUE
FALSE	TRUE	TRUE
FALSE	FALSE	FALSE

Conditionals

If sentence 'P' is	If sentence 'Q' is	Then sentence " <i>If</i> P, <i>then</i> Q" is
TRUE	TRUE	TRUE
TRUE	FALSE	FALSE
FALSE	TRUE	TRUE
FALSE	FALSE	TRUE

2.1.6 Composing Molecular Claims

In many cases you'll have sentences that have more than one logical operator word in it. In such cases, it is difficult to discern what kind of molecular claim it is. Whether or not something is a negation, conjunction, disjunction depends on the *scope* of the logical operator word. We will call molecular claims with just one logical operator "basic molecular claims", and molecular claims with more than one logical operator "composed molecular claims".

Since logical operators operate on *sentences*, when you are presented with a molecular sentence, circle all of the logical operator words and determine the sentences on which those logical operators are operating. If there are more than one logical operator words, then there will be a *main* logical operator, and the others will be *embedded* logical operators. In natural language sentences, sometimes it is difficult to determine what the main operator is. Because of this, it is sometimes difficult to determine what kind of molecular sentence you're dealing with. For example, consider the following sentence: You need a U.S. Passport or a valid U.S. Driver's License and U.S. Social Security Card. Is this a disjunction with one of its disjuncts being a conjunction, or is it a conjunction with one of its conjuncts being a disjunction?

What makes a logical operator the main operator is that its operation extends through the entire sentence as a whole. What makes a logical operator the embedded operator is that its operation does not extend to at least one part of the entire sentence as a whole. To figure this out, try to determine what the atomic sentences are. Then, slowly build the molecular sentence in question. The last logical operator you use to build the molecular sentence in question is the main logical operator, and that will tell you the kind of molecular claim it is.

For example, consider the following molecular sentence:

Bill is sick and it is not the case that he is coming out with us.

This sentence has two instances of logical operator words, ‘and’ and ‘it is not the case that’, and it has two atomic sentences:

- Bill is sick.
- He is coming out with us.

The negation is clearly operating on the second atomic sentence. So: let’s make that:

- *It is not the case that* he is coming out with us.

Next, this negation is connected with the first atomic sentence using the conjunction word ‘and’:

- Bill is sick *and* it is not the case that he is coming out with us.

We are done building the molecular sentence under examination. What was the last logical operator that was used to build this molecular sentence? It was the conjunction word ‘and’. Hence, the sentence is a conjunction, not a negation.

To determine what sort of molecular claim a composed molecular claim is, one must determine the main logical operator for the claim. To determine this, here is one method. First, circle the logical operator words. Second, underline the sentences that a logical operator word operates on. Second, determine which logical operator underlines the entire sentence. This will be the main logical operator. For instance, take the sentence above: *Bill is sick and it is not the case that he is coming out with us*. Using this underlining method, we get something like this (instead of circling, I will put the logical operator words in **boldface**):

*Bill is sick **and it is not the case that** he is coming out with us.*

Notice here that the conjunction *and* underlines all the sentences. Hence, this sentence is a conjunction.

Sometimes it’s not clear what kind of molecular sentence it is. This can happen in natural language. This what we call an ambiguous sentence (namely, a syntactically ambiguous sentence). We will talk about ambiguity (and syntactic ambiguity) later.

We will adopt some conventions in order to make sentences less ambiguous. Here are the conventions:

2.1.6.1 Negations of Composed Sentences

When negating a conjunction, we will use ‘both...and...’ after the negation ‘It is not the case that...’ in order to indicate that the negation is operating on the *entire conjunction*. For example:

It is not the case that both P and Q.

When negating a disjunction, we will use ‘either...or...’ after the negation ‘It is not the case that...’ in order to indicate that the negation is operating on the *entire disjunction*. For example:

It is not the case that either P or Q.

When negating a conditional, we will place the negation ‘It is not the case that...’ immediately before the ‘if’ that begins the conditional. For example:

It is not the case that if P, then Q.

Note that one can negate negations. For example:

It is not the case that it is not the case that the sun is shining.

2.1.6.2 Conjoining Composed Sentences

When conjoining composed sentences, we will demarcate the conjuncts using a comma immediately after the (left) conjunct followed by ‘and’. If needed, we will also use the expression “both...and...”. We will also use commas to make it clear where the (right) conjunct ends. For instance, the following are conjunctions:

*It is not the case that P, and Q.
P or Q, and R.
Both P or Q and R or S.*

Note that one can conjoin conjunctions. For example:

P, and both Q and R.

2.1.6.3 Disjoining Composed Sentences

When disjoining composed sentences, we will demarcate disjuncts using a comma immediately after the (left) disjunct followed by ‘or’. We also use commas to make it clear where the (right) disjunct ends. If needed, we will also use the expression “either...or...”. For instance, the following are disjunctions:

*It is not the case that P, or Q.
P, or Q and R.
Either P or Q and R.
Either P, or Q and R.
Either P and Q or R and S.*

Note that one can disjoin disjunctions. For example:

P, or either Q or R.

2.1.6.4 Making a Conditional Using Composed Sentences

When making a conditional using composed sentences, we will demarcate the antecedent and consequent using ‘if...then’ or ‘if..., then...’. What comes between the ‘if’ and ‘then’ is the antecedent, and what comes after ‘then’ is the consequent. We will also use commas to make it clear where the consequent ends. For instance, the following are all conditionals:

If it is not the case that P, then Q.
If it is not the case that P, then Q and R.
If P and Q, then R or S.

Note that one can make conditionals out of other conditionals. For example:

If P, then If Q, then if R, then S.
If if P then Q, then if R, then S.

2.2 Clear and Unclear Sentences

Clarity in thinking, speaking, and writing is a very difficult thing to achieve and requires practice. It’s difficult to define clarity, but we can give a rough idea of what it is. To speak and write clearly is to convey intended information in a way that makes understanding accessible. Note that I did not say *easy*. Something can be clearly stated *without* being easy to understand. So, a complex physics problem can be clearly stated, but yet difficult to understand. A complex argument for something might be clearly written but it might nevertheless be very difficult to understand. So, don’t think that if you cannot understand something, then that means that you’re dealing with something unclear. However, if you are dealing with something unclear, then you are going to have a difficult time understanding it.

To get a better sense of clarity, it may help to get a sense of what *unclear* is. Let’s look at some common sources of unclarity.

Vagueness

Vagueness is a common cause of unclarity. Vagueness occurs when a word or phrase refers to things that have borderline cases. If a word, concept, or phrase is vague, then it is difficult to determine whether or not the word, concept, or phrase applies to certain things. In other words, if X is a vague concept, word, or phrase, then there are some things that aren’t so clearly X and not so clearly not-X. There may be things that are *clearly* X, and some things that are *clearly* not-X, but, if X is a vague concept, there are also things that aren’t so clearly either one.

For example, the concept of baldness is a vague concept. Clearly, Patrick Stewart is bald. You can give examples of people who are no doubt clearly bald. You can also give very clear examples of people who are definitely *not* bald, e.g. Paris Hilton and Carrot Top. But then there are others who aren't so clearly bald or not bald. Perhaps they are *going* bald, but whether they *are* bald is hard to determine.

Consider another vague concept: red. Now, you probably can point to things and say that they are clearly red. For example, a stop sign is clearly red. Other things you can point to and rightly judge as definitely *not* red, such as the color of the Statue of Liberty. But consider a color continuum between red and yellow:



Can you say exactly what spot in this continuum the concept of red applies and then ceases to apply? Probably not – it's a continuum. It seems to be a *vague concept* because the concept isn't precise or fine-tuned enough to allow you to determine where the concept applies and where it ceases to apply in *all points* of the color continuum.

Consider the concept of a heap. You have a heap of sand in front of you. If you take one pebble away from the heap, will you still have a heap? Well, if you have a big enough heap, certainly taking away a single pebble won't make it cease to be a heap. But if you keep doing that – if you keep taking a pebble away from the heap – you'll no longer have a heap but just a few granules of sand. When did the concept of heap stop applying? After taking away 100 granules? After taking away half of all the granules? It's not clear. The concept of a heap isn't precise or fine-tuned enough to help us determine that. In other words, the concept of a heap is a *vague concept*.

Now, vagueness may not be something you can avoid completely, but it is something that you should watch out for, because excessive vagueness is something that can seriously decrease the clarity of an argument or a piece of writing. (Note here that the concept of *excessive* vagueness is itself vague). As a general rule, a level of vagueness is acceptable and okay if a claim is not *so* vague as to prevent it from conveying appropriately useful information in a way. This will depend on the context in many cases. If you were in a car lot helping a friend look for a car, and she told you that she wants a red car, that would probably be clear enough for you to understand. If, however, you were told to go into a storeroom that contained hard candy that ranged in colors like the color continuum above, and was asked to pick out only red candy, this probably wouldn't be conveying "appropriately useful information". You'd probably want to ask "Do you mean really dark red like the color of red-velvet cake? Or something lighter?" The context will help you determine how much vagueness is acceptable.

One way to avoid vagueness is to avoid excessive generality. So, to say "I met a thing" is very vague: what *thing*? The concept of a thing is just way too general, and in that sense it is vague. It would be better to say something like "I met a friend" or "I met a colleague." Of course, these are

somewhat vague too in terms of generality: which friend did you meet? Which colleague? However, depending on the context, you should be able to determine most of the time what level of vagueness is acceptable. In certain contexts (e.g. writing a scientific paper, for instance), you should avoid general vagueness and try to be as specific as possible. However, in informal settings, you can probably be a little more general and not be specific (e.g. “The salt is in the cabinet next to the dishwasher” vs. “The salt is located at the following geo-spatial coordinates: 34°08'49.8"N 118°06'29.0"W”).

Ambiguity

Ambiguity is another common cause of unclarity. A word, phrase, concept, or sentence is ambiguous when it has different meanings within a context of use. Ambiguity comes in three forms.

Semantic Ambiguity

Semantic ambiguity occurs when the word or sentence can *mean* different things. The word “semantic” means *having to do with linguistic meaning*. So, when something is semantically ambiguous, it is unclear because there are two distinct meanings associated with a word or phrase. Take for example “The room is cool.” The word ‘cool’ can be associated with two different meanings. The common one has to do with temperature. The other meaning is a slang meaning, which means something like *nice* or *hip* or *good*. When someone says “The room is cool”, it may be difficult to determine whether or not what they’re saying is true or false because you don’t know which meaning of “cool” they have in mind. Consider another one: “The people are revolting”. Here, “revolting” can mean either of two things. It can mean the act of revolution, in which case the person making that statement is claiming that the people are currently rebelling and in revolution. It can also mean that the people are *disgusting* or *despicable*, as in when we say something like “The taste of bird turd is revolting”. The word “revolting” is ambiguous between two meanings, and in order to determine whether or not the statement is true, we need to determine what the statement means. And to determine that, we need to know which of the two meanings the speaker intends.

Grouping Ambiguity

Grouping ambiguity occurs when it is not exactly clear whether or not a word or phrase is being used to refer to a group collectively or the members of a group individually. For example, if I said “UC Berkeley students get the highest scores on the SAT in the nation”, you might wonder whether I am using “UC Berkeley students” to refer to the students there *collectively*, or whether I mean to refer to the students there *individually*. If I mean it collectively, I mean that overall the students get higher SAT scores than anyone else. But if I mean that phrase to refer to the Berkeley students individually, I mean to say that any student at UC Berkeley will have a score on the SAT that’s higher than any other student in the nation. Here’s another one: “The marbles are heavy.” If someone were to say this, they

might mean that the marbles as a group, collectively, is heavy. However, they might also mean that the marbles *individually* are heavy.

Syntactic Ambiguity

Syntactic ambiguity occurs as a result of an ambiguity in the *syntactic structure* of the sentence. Consider the following sentence: John did not go to the party because Sally was there. This sentence is ambiguous in its structure. It can mean that the reason that John didn't go to the party is that Sally was there (and he wanted to avoid seeing her). But the sentence can also be parsed to mean that Sally's presence at the party was not a reason for John attending – he might have gone to the party because he knew there would be free food and refreshments. The syntactic structure of the sentence “John did not go to the party because Sally was there” is ambiguous between two different meanings, and hence it's not clear what's being said. Another example: “Players with beginners' skills only may use Court 1”. The syntactic ambiguity occurs with the placement of the word “only”. The sentence could mean that players that have beginners' skills *and no skills beyond that* may use Court 1. It could also mean that players that have beginner's skills may use Court 1 *and players with higher level skills may not use Court 1*. It could also mean that players with beginners' skill may use Court 1, *but may not use any other Court*. Each of the italicized phrases helps clarify what the ‘only’ might be doing syntactically. Is ‘only’ modifying “beginners' skills”? If so, then the first interpretation seems to be the right one. Is ‘only’ modifying “players with beginners' skills”? If so, then the second interpretation seems to be the right one. Is ‘only’ modifying ‘Court 1’? If so, then the third interpretation seems the right one. (There's at least one other interpretation of the syntactic function of ‘only’ – can you figure it out?).

2.3 Language, Mind, and World

The relationship between language and the world for our class is simply this: a sentence (language) is *true* when it agrees with the facts (world); otherwise, the sentence is false. Facts are just facts: what is the case in the world.

This is roughly the correspondence theory of truth. This is somewhat controversial, since there are other theories of truth (coherence and pragmatist theories, for example). However, for our class, we will assume the correspondence theory in order to keep things simple, and since the correspondence theory is fairly intuitive.

Claims can be used to express our thoughts. That is, we can use language to express what we are thinking. However, language is not necessarily *about* thoughts. They can be, but they need not be. For example, if I am thinking that it is a nice day, then the sentence “It is a nice day” expresses my thought, but it is not about my thought – rather, it is about the weather today. However, I can use language to talk about my thoughts and express my thoughts. For instance, “My thoughts are very grim today.” This sentence what I am thinking, and the sentence is also about my thoughts.

Many of our beliefs are *about* the world. A belief is a kind of thought, and it has a sentential structure, i.e. the structure of a sentence. Just like claims, our beliefs can be true or they can be false. They are true when they agree with the facts (world), and they are false when they do not.

Opinions are beliefs. Hence, opinions can be true or false. When someone says “I’m entitled to my own opinion”, one thing they are saying is that they have the right to believe what they want. However, sometimes people mean that their opinion cannot be judged in any way. That’s wrong. If you have the opinion that U.S.A. is in the southern hemisphere, then your opinion is wrong. Sure, you are entitled to believe something false, but that still doesn’t change the fact that your belief is false. Anyone would be certainly correct to judge your belief as false. Beliefs are *not* necessarily immune to falsity, and so neither are opinions.

2.4 Subjective and Objective Truths and Claims

Subjectivity typically refers to the *subjective states* of an individual, or the subjective point of view. When we talk about perspective, or a point of view, we’re talking about how things are or seem from a particular standpoint. Subjective states include such things as emotions, instances of thoughts, perceptual perspective, or appearance to something. The subjective standpoint is *biased* or *partial* because it is rooted in *a particular standpoint*. Objectivity, on the other hand, is *unbiased* or *impartial* in that it is *regardless* of any particular standpoint, where this standpoint refers specifically the subjective standpoint, i.e. the standpoint of some particular person. For something to be *subjective* is for it to depend on a particular subject’s *mind* in some way. For something to be objective is to depend on the *object*. This is vague, so let’s try to make this clearer.

Objects in the world have properties. Properties are just states or ways about an object. So, a mango has the property of *being a fruit*. When we say “Mango is a fruit”, we are saying that mangoes have the property of being a fruit. Sometimes, when we say that something has a property, whether or not what we are saying is true depends on the object, regardless of what we think. So, for example, the claim “Mango is a fruit” ascribes the property of being a fruit to the mango object. Whether or not this claim is true depends solely on the object mango, namely, whether a mango in fact has that property. This is what makes the claim an *objective claim*, or an *objective truth*.

A subjective claim is, roughly, a claim that is either true or false depending on which person is making the claim and on what that person making claim thinks, feels, experiences things in the world, etc. So, sometimes when we ascribe a property to something, whether or not that object in fact has that property does not depend on the object alone, but on the mind of the person making the claim. For instance, “Mangos are delicious” is a subjective claim. It ascribes the property of being delicious to mangos, but whether or not mangos in fact have the property of being delicious does not depend solely on the mango. In fact, it seems that whether or not mangos have that property depends solely on whether or not the person making the claim *has a mind that has pleasant taste experiences* when they eat mangos. In this sense, the truth or falsity of the claim depends on the

subject's experiences. In this sense, it's truth-value depends on who you ask and who says it. In particular, whether or not it is true depends on whether or not s/he *likes* the taste of mangos.

This can get confusing if you're not careful. So, for example, "Mangos are delicious" is subjective. For you it may be false, but for me it is definitely true. Why? Because *for me* mangos produce a very pleasant taste experience, whereas for you, it may produce an unpleasant one. "Mangos are delicious to Joe" – is this a subjective or objective claim? This is actually an *objective claim*. The reason is that it doesn't matter if you *think* that mangos are delicious to Joe. Whether or not mangos are delicious to Joe does not depend on the mind of each subject who makes that claim. Rather, it depends on one object, namely, Joe, and *his experiences*.

3. Arguments

All reasoning takes the form of an *argument*. An argument tries to show that a claim is true by offering other claims. These other claims are given to *back up* or *show* that a certain claim is true. Arguments show an *inference* relation or an *entailment* relation between one set of sentences to another set of sentences. This inference or entailment relation is the mark of *reasoning*.

3.1 Structure of Arguments: Premises and Conclusion

A claim in an argument is either a *premise* or a *conclusion*. Both a premise and a conclusion is a sentence. A premise is a sentence *from which* an inference is made, whereas a conclusion is a sentence *that is inferred*.

For example, "John lives in the city of Amsterdam located in the Netherlands; therefore, John resides in Europe" is an example of an argument that consists of a premise and a conclusion. The conclusion is the sentence "John resides in Europe." This conclusion is inferred from the premise "John lives in the city of Amsterdam located in the Netherlands." In short, an argument tries to *support* its conclusion through its premises, or it tries to *prove*, *demonstrate*, or *entail* its conclusion through its premises. We'll talk about that difference later.

A sentence can be a premise and a conclusion. In other words, being a premise and being a conclusion aren't mutually exclusive. When one sentence is inferred from another sentence(s) *and* is itself used to infer a different sentence, that one sentence is both a premise and a conclusion. In this case, we call that sentence a *sub-conclusion* when discussing it in the context of the entire argument. When we're just talking about in relation to the premise(s) from which it was inferred, we call it just a conclusion. The final conclusion at the end an entire argument we'll call 'main conclusion'. An argument must have at least one main conclusion and at least one premise.

3.1.1 Premise and Conclusion Indicator Words

How do you know which sentences are the premises and which ones are the conclusions? Well, the argument holds that its conclusion is true *because* of the premises. So, when you have two sentences, ask yourself: is the first sentence presented as true *because* of the second sentence, or is the second sentence presented as true *because* of the first sentence? Other ways to determine whether or not a sentence is a premise or a conclusion is through indicators. Here are some common ones:

Premise Indicators	Conclusion Indicators
Since...	Therefore...
For the reason that...	Thus...
<i>What comes after "because"</i>	Hence...
For...	It follows that...
	We can conclude that...
	<i>What comes before "because"</i>
	So...

3.2 Identifying Arguments

As we stated, arguments consist of sentences, and in ideal cases, some of these sentences are premises, and at least one is a conclusion. Sometimes when we state a sentence, we are not actually asserting any actual belief. However, sometimes we are. When we *assume something for the sake of argument*, we are not necessarily believing that it is true, but simply supposing, or pretending, that it is true. When someone states something as an actual belief that they have, they are making a claim. A claim, then, is a sentence, and making a claim is expressing one's belief. However, sometimes a bunch of claims are made *without* there being any argument given for them. If this is the case, the person is just spouting out claims (their beliefs) without providing any rational support for them. Sometimes it's difficult to discern whether or not an argument is actually being given. Here are some practical tips.

Identify the main issue. Ask yourself: what is the main question that is trying to be answered or addressed? An issue is a question. If the question is a 'how' question or a 'where' question, then there is a chance that you're not dealing with an argument. But that's not always the case.

Identify the conclusion. Once you identify the main issue, you'll be able to identify what the conclusion is, or at least be in a better position to do so. The conclusion is typically the proposed answer to the question. Look for the conclusion indicators listed above to help you locate the conclusion.

Identify the premises. Premises are hard to identify sometimes. In many cases, an argument will include many sentences that are not relevant to the argument itself. So, what you need to discern is whether or not a sentence is actually relevant to the issue at hand. One thing to do is to ask yourself whether or not taking the sentence out would disrupt the rational force of the argument. You want to identify what work the premise is doing: is it just giving the author's feeling about something? Is it just an example? Also, look for premise indicators to help you locate any premises.

3.2.1 What an Argument is Not

Sometimes the best way to understand an argument and clarify its structure is to take it apart and see how these parts hang/fit together. Generally, there are three steps in doing so. First, find the conclusion, main point, or main thesis of the passage. Second, look for the reason or reasons offered for the conclusion. In other words, look for the *premises* given in support of the conclusion. Third, look for any further reasons for those reasons, i.e. look for sub-arguments for sub-conclusions, and identify what the reasons are supposed to support.

First, an argument, in our sense, does *not* mean a verbal fight. For example, when someone says that they got into an argument with their husband, wife, mother, father, brother, sister, or friend, what they usually mean is that they got into a verbal fight or altercation. This is *not* what we mean by ‘argument’ in this class. While a verbal fight *can* involve arguments in our sense, they need not.

Second, to argue is not exactly the same as to disagree. Disagreement in itself is *not* an argument. If Sally said “Los Angeles has more people than Seoul” and Wanda replies “No, you’re wrong! Seoul has more people than Los Angeles”, Wanda is disagreeing with Sally, but Wanda is not thereby *arguing* with Sally. To argue is to support or prove some conclusion on the basis of some premises. Hence, to argue *against* someone is not simply to disagree with them.

Third, to reason and argue successfully is not the same as to persuade or convince successfully. To persuade or convince is to get someone to believe something. To argue or to reason is to show that something is true based on some premises. The former (persuasion/convincing) is not necessarily concerned with truth, but the latter (argument/reasoning) is concerned with truth, namely, the truth of the conclusion *from* the truth of the premises. An argument might fail to show that its conclusion is true, and yet it might nevertheless succeed in persuading someone that the conclusion is true. An argument might succeed in showing that its conclusion is true, and yet it might not persuade anyone at all. The art of persuasion is part of rhetoric, not the art of reasoning. To reason and to persuade are different. For us, rhetoric is the enemy. However, we will study some rhetorical techniques in the art of persuasion in order to know what it means to argue *badly*.

Fourth, to reason and argue is not the same as to explain. An explanation of something is supposed to provide us with knowledge of what something is, how to do something, where it comes from, and how something came to be the case. However, it need not imply or support the truth of something. Explanations further our knowledge about something, whereas reasoning and arguments are supposed to help us come to know some truth.

Fifth, hypothetical examples, analogies, and rhetorical questions are not arguments. To give a hypothetical example that illustrates a claim does not show that the claim is true. Examples are given in order to show or to *illustrate* something in order to help someone understand that thing better. Sometimes examples are actual cases that serve as evidence for the truth of

something. Such examples might then serve as premises in an argument. However, in that case, we should describe this as *giving evidence*. Remember, providing example cases for evidence *does* count as giving an example, but it doesn't work the other way around: giving examples does *not* necessarily count as giving evidence for something. Analogies are also good ways to explain something and help someone understand something. Analogies are *likenings* to something else, often in a metaphorical way. Asking rhetorical questions is one of the most common ways that people try to argue for or against something, but in itself it's not an argument. So, don't do this: "If God is all good, knows everything, and is all powerful and really exists, then why doesn't God stop all of the evils in the world from happening?" This is raising a question – it doesn't give an argument at all.

Sixth, pictures and movies are *not* arguments. They may be compelling, persuasive, and moving, and they can certainly incite emotions within us. But these are not necessary characteristics of arguments. An argument has premises and a conclusion, and the premises support or prove the conclusion. Picture and movies, however, don't have that. This is not to say that some movies cannot *contain* arguments, like some documentaries. But the images themselves are *not* arguments. The picture is just a picture, and it doesn't have a conclusion built into it. Arguments, remember, have premises and a conclusion, and these constituents of an argument are things that can have a truth-value.

Seventh, conditionals (if-then statements) by themselves are *not* arguments. "If you wash your car now, then it will get spots" is not an argument. There are no premises and there is no conclusion. It can be a premise of an argument, and thus be *part* of an argument. It can also be the conclusion of an argument. However, in itself it is not an argument.

Eighth, lists of facts are not by themselves an argument. Consider the following two cases:

- (A) Identity theft is up at least tenfold over last year. More people have learned how easy it is to get hold of another's Social Security number, bank account numbers, and such. The local police department reminds everyone to keep close watch on who has access to such information.
- (B) The number of people who have learned how to steal identities has doubled in the last year. So you are now more likely to become a victim of identity theft than you were a year ago.

In (A), you have no argument. The reason is that none of the sentences in (A) support any of the other sentences. Rather, they are just different facts loosely centering on the same topic. In order to be an argument, at least one of the sentences must be supporting the truth of another statement. But if you consider (B), we have a different story. In (B), there *is* an argument. The reason is that the first sentence of (B) is given to support the second sentence in (B). Hence, one sentence is given to provide support for the truth of another sentence, thereby making (B) an argument.

Ninth, statements of the form "A because B" *could* be giving an argument, but not necessarily. It could be that such a statement is giving an explanation, which is not an argument. 'Because' ought to be understood

carefully. Sometimes what follows the ‘because’ is an *explanation* of what is before it, as in “His head hurts because he’s been pounding his own head with a hammer”. In this case, the explanation specifies the *cause* of the head hurting. Sometimes what follows the ‘because’ is a *reason* given as a premise to support what is before it, as in “Murdering people is wrong *because* people have an intrinsic value that is to be recognized and respected.” Here, a reason is given to show why it’s true that murdering people is wrong. An explanation for *why* something is true is *not necessarily* an argument that shows *that it is true*, and an argument that shows something to be true does *not necessarily* explain how it came to be true. The difference between an argument and an explanation is that an argument attempts to show that the conclusion is true by giving premises, whereas an explanation attempts to specify, identify, or clarify the *cause* of something, *how* something works or came about, *what* something is made of, what something *means*, or the *intention* behind something, and so on. In short, an argument tries to show that some claim is *true*, whereas an explanation tries to *clarify* a claim in some way.

3.3 Arguments with Unstated Premises and Unstated Conclusions

Sometimes arguments have a missing premise. A missing premise is a premise that is not explicitly stated, but that is assumed or presupposed in the argument. These are also called “hidden premises”, “unstated premises”, and “implicit premises”. (They’re called other things too). Typically, you’ll be able to tell that there is a missing premise when the explicitly given premise or set of premises provide only *partial* support for the conclusion. For example, consider the following argument:

John lives in the city of Amsterdam located in the Netherlands;
Therefore, John resides in Europe.

This argument has a missing premise. The missing premise *might* be “Anything that is in the city of Amsterdam located in the Netherlands is also in Europe”, or something like that. This premise is not explicitly stated in the argument but is assumed by the argument.

Sometimes arguments have missing conclusions. A missing conclusion is a sentence that is entailed by or suggested by certain premises, but is not explicitly stated. For example, consider the following argument: “Anyone who has a rash and has difficulty breathing needs to see a doctor as soon as possible. *You* have a bad rash and are having a hard time breathing.” Here, there is a hidden or missing conclusion, which is (obviously) “You need to see a doctor ASAP.” These missing conclusions are sometimes called “unstated”, “hidden”, “implicit”, “

When you’re faced with an argument that is missing a premise, you need to think about several things in order to figure out what the missing premise is. One way to do this is to try and figure out from the context what kind of argument the person is trying to do: are they trying to provide evidence merely to show that the conclusion is highly likely to be true, or are they trying to prove or demonstrate that the conclusion necessarily follows from

the premises? Once you've determined this, you can then try to figure out what premise they are assuming when giving the argument.

For example, suppose your friend says to you "Stacy and Justin are on the brink of divorce. They're always fighting." Now, your friend is arguing for the conclusion that Stacy and Justin are on the brink of divorce, and the *reason* she offers *explicitly* is that Stacy and Justin are always fighting. But besides this explicit premise, there is also a missing premise. What is that missing premise? You need to think about what your friend is trying to say. Is she trying to say that always fighting *necessarily* leads to divorce? If that's the case, then the missing premise would be this:

Stacy and Justin are always fighting.

[Any couple that always fights is on the brink of divorce.]

Therefore, Stacy and Justin are on the brink of divorce.

The missing premise (stated in the brackets above) *would* make the argument successfully prove its conclusion, if the premises were in fact true. However, do you think that it is true that *any* couple that fights all the time is on the brink of divorce? That is a very strong claim, leaving no room for exceptions. Hence, it's probably safe to say that it isn't true. It's too strong – there are probably lots of exceptions. So, your friend is probably *not* assuming that *any* couple that always fights is on the brink of divorce. Rather, your friend is likely assuming implicitly that *most* couples who always fight are on the brink of divorce. If that's the missing premise, then the argument would *not* prove the conclusion, but it would support it, if indeed the premises are true. We'll talk more about assessing argument later.

Here's another example. Suppose someone says "You should definitely break up with him. He got physically abusive with you", the person is most likely giving an argument with the unstated premise "You should always break up with someone who gets physically abusive with you." The person is not saying that it's *likely* that you should break up with him, or that it is *highly probable* that you should break up with him, but that you should *definitely* break up with him. So, the person is giving this argument is probably assuming the stronger version of the unstated premise.

Sometimes, an unstated premise is a *presupposition*. A presupposition is a claim where the truth of that claim is either tacitly assumed to be true. A presupposition can also be a *necessary* condition for a statement, where this necessary statement is implied by the explicitly made claim. What exactly a presupposition is can be a much more complex business. We're not going into that. For us, we will just rest with the idea that a presupposition is something that is tacitly assumed to be true when making a claim. Here are some examples of presuppositions:

1. The man responsible for the recent string of burglaries is 175 centimeters tall and wears a baseball hat.
 - a. *Presupposition: the perpetrator is a male.*
2. Language is used to communicate with each other in such a way that can foster mutual understanding.
 - a. *Presupposition: linguistic meaning remains somewhat constant.*
3. Jill is in a lot of pain and is reasoning about what she ought to do.
 - a. *Presupposition: Jill has a rational mind.*

3.4 Principle of Charity

When engaged in discussion, one of the basic and most important things to be established and maintained is understanding. It's easy to misunderstand what someone else means, and the reason for the misunderstanding varies: it can be the speaker's lack of clarity, or oversight in providing key information, or it can be the listener's failure to understand or grasp certain parts of what the speaker has said. There are many reasons that misunderstanding can happen. It is very important when arguing that we always try to understand what the other is saying, and also to try to be clear so that the other can understand what we are saying. A discussion is a two-way street, and that doesn't mean that each person should be equally able to speak freely. It means that we should try to *listen* to the one speaking and try to understand what they are saying, or trying to say. This allows us to formulate our thoughts and words *in response* to what they have said, not simply as a *reaction* to what has been said. The latter is not a discussion, the former is.

People are often quick to criticize before taking the time to understand what someone is saying. But criticism based on misunderstanding is obviously not well-founded, nor is it fair. Good criticism always presupposes good understanding. In order to achieve good understanding, you need to be careful when you try to figure out what the person is saying. If you're not sure, ask questions. Give your fellow interlocutor the benefit of the doubt. You cannot criticize (well) what you do not understand.

One of the most important principles in both philosophy and practical everyday life is the *principle of charity*. This principle basically states that you should try to give the *best* and *most charitable* interpretation of someone's statements or someone's argument. When dealing with an argument, assume that the person who gave the argument is rational, thoughtful, and has given a good argument. Then, interpret the argument in a way that presents the author's position and argument in the most plausible and reasonable light. Following the principle of charity is a way of being intellectually generous.

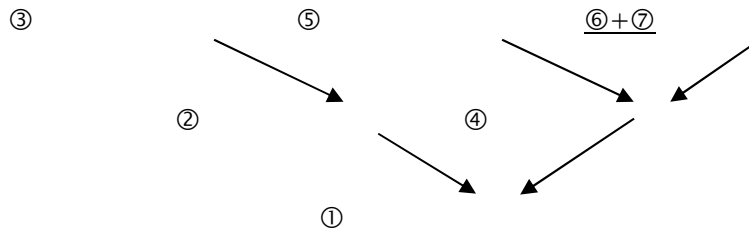
3.5 Diagramming Arguments

Diagramming arguments is a way to make the *structure* of an argument clear and explicit. When you diagram a given argument, you are trying to diagram how an argument goes, not necessarily how an argument *should* go. Sometimes, people give an argument that isn't very good. When you diagram it, you might be tempted to diagram how you think it *should* go. That's not what you're trying to do here. You're trying to diagram the given argument *as is*, and if it is a bad argument, then the badness of the argument will show in the diagram. Diagramming will help you understand the structure of an argument.

The tree-diagram method is one method of diagramming an argument. Here is how to do it:

1. Circle the conclusion indicators and premise indicators, if there are any.
2. Place each statement in [brackets] if it is a premise or a conclusion.
3. Go through and number each statement consecutively.
4. Draw a tree diagram using the following symbols to show the relationship between each of the statements:
 - a. “ $\rightarrow$ ” between numbers indicates that one statement is a premise supporting the other statement, e.g. $\textcircled{1} \rightarrow \textcircled{2}$ indicates that statement 1 is a premise that supports statement 2.
 - b. “+” between numbers and a horizontal line that underscores them indicates that the statements jointly support another statement, e.g. $\textcircled{1} + \textcircled{2} \rightarrow \textcircled{3}$ indicates that statement 1 together with statement 2 entail or support or demonstrate statement 3. When two claims jointly support another claim, each of the two supporting claims cannot independently, on their own, support the claim, but they need each other to provide that support.

A sample tree diagram will look something like this:



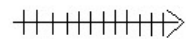
So this tree diagram states the following. Statement ① is the main conclusion, and it's directly supported by two premises, statements ② and ④. Each of these premises is a sub-conclusion of a sub-argument (an argument within an argument). Statement ② is supported by statement ③, and statement ④ is supported by statement ⑤ and statements ⑥ and ⑦ jointly.

Why isn't ② and ④ connected by a "+" and underscored? Aren't they jointly supporting statement ①? Same with statement ⑤ and statements ⑥ and ⑦: aren't those three jointly supporting statement ④? According to the diagram, statement ② is represented as giving support to statement ① independently of statement ④, and *vice versa*. Statements ⑥ and ⑦ are represented as relying on each other in order to support statement ④ – statement ⑥ alone does not give support for statement ④, and neither does statement ⑦ alone. Statement ⑤, however, is represented as giving independent support to statement ④.

There are some complications that you'll need to keep in mind when evaluating arguments. Sometimes an argument will mention some reason that may go against the conclusion of the argument. So, for example, suppose your position on education is that college education should be mandatory for all citizens. You might reason as follows:

- a. A democracy will work only if its citizens are intellectually developed and mature.
- b. College education is where people develop and mature intellectually.
- c. Making college education mandatory may lead to increased taxes, but because it is necessary for a healthy democracy, college education is vital.
- d. Therefore, college education should be mandatory for all citizens.

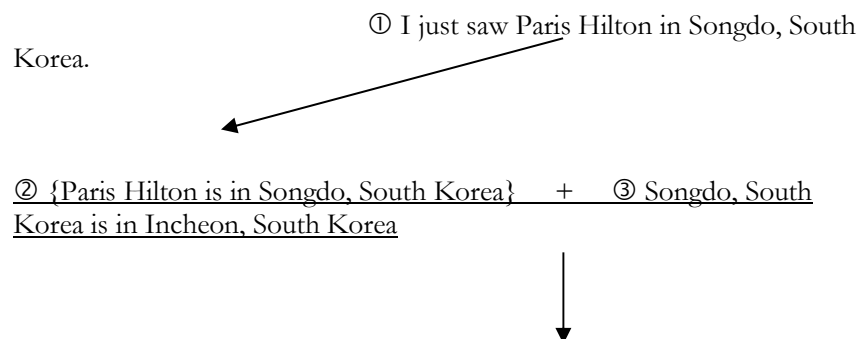
Notice that statement c. does not offer direct support for the conclusion d. Rather, it considers a counter-argument, or counter-statement, that goes against the position of the person giving the argument. It considers it, and then dismisses it as not a good reason against the conclusion. When a reason is given as a consideration *against* a claim, then you indicate that with an arrow that is crossed out, like this



Another complication to look out for are sentences without any direct argumentative function. Arguments will often have statements that don't serve any purpose in the argument, i.e. is neither a premise nor a conclusion. These are *window-dressings*, added to make the argument appear nice or eloquent. Sometimes, however, sentences that do not serve any direct argumentative function may help the argument's clarity by explaining or clarifying a premise, a relevant concept or principle, or even the conclusion. You'll have to keep an eye out for these because they can be confused as a premise or a conclusion if you're not careful. For example, consider the following:

"Paris Hilton is definitely in Incheon, South Korea. I just saw her here in Songdo, South Korea, which is part of Incheon, South Korea. She's probably here for a movie or something. She's one of the many celebrities in Los Angeles and has gained a lot of notoriety in the media."

In this passage, there is an argument given, and there are unstated premises. Here is a tree of the argument, with unstated premises put in {curly brackets/braces}:



④ Paris Hilton is in Incheon, South Korea.

Statement ① supports statement ②. Statement ② is an implicit, unstated sub-conclusion that is also a premise that, jointly with statement ③, supports the conclusion ④ I didn't mention anything about her being a celebuntante in the argument tree because that statement just tries to explain who Paris Hilton is in case you didn't know. The statement that speculates that she may be here to shoot a movie is also left out of the argument tree because it just gives a guess as to why she's here in the first place. Both of these statements are not part of the argument proper, but just clarifications and speculations – window-dressing.

Remember, you need to determine first whether or not you're even dealing with an argument. That means that you need to know that what you're dealing with is not a report, description, explanation, or summary of something.

3.6 Two Kinds of Argument: Deductive and Inductive Arguments

The relationship between the premises and the conclusion in an argument can be of two sorts. One sort is that of *offering support* or *evidence*. This is the sort of relationship we mean when we talk about a conclusion being *supported* by the premises in an argument, or the premises offering *evidence* for the conclusion. Another sort of relationship between premises and conclusion is one where the conclusion *follows* or *is entailed by* the conclusion, or where the premises *prove* or *demonstrate* the conclusion.

The second sort of relationship is stronger than the first. When we're talking about the premises *supporting* or *giving evidence* for the conclusion, there is still the possibility that the premises may be true while the conclusion is false. When we say that the premises *prove* or *demonstrate* the conclusion, then there is no room for that gap: if the premises really do prove the conclusion, then the conclusion *necessarily* follows from the premises.

The first relation between premises and conclusion might be called 'inductive support', whereas the second sort of relation might be called 'deductive' or 'logical support. When an argument gives inductive support for its conclusion, it still grants that there may be exceptions. For example, consider the following argument:

A study of five thousand randomly selected individuals, half of which were smokers and the other half non-smokers, showed that those who smoked tobacco cigarettes, all things being equal (diet, exercise, etc.), had a significantly higher incidence of heart disease and cancer. Therefore, smoking tobacco cigarettes causes cancer.

The first sentence is the premise, and the second sentence is clearly the conclusion. The premise seems to offer pretty strong support for the conclusion. Does that mean that you will *never* find anyone who has been smoking for a long time that won't get cancer or heart disease? No, of course not. So, the premises offer *strong support* the conclusion, but they don't guarantee it.

When we're talking about *proof* or *demonstration* of a conclusion, we are talking about *logical entailment*. What this means is that the conclusion is *guaranteed* to be true if the premises are true. What makes logical entailment possible is the *logical structure* of the argument. We'll talk about that in more detail later.

The distinction between inductive support and logical/deductive support roughly and generally captures the distinction between two kinds of argument, inductive arguments and deductive arguments, which we'll talk about next.

There are two kinds of arguments: deductive arguments and inductive arguments. Let's start with deductive arguments – what they are and how to assess them.

3.6.1 Deductive Arguments

Deductive arguments are ones that attempt to *demonstrate* their conclusions. By demonstrate, we mean something like *logically entail*, *logically imply*, or *logically follows*. Another way of putting this is this: deductive arguments try to *prove* their conclusions based on their premises. Good deductive arguments will show that the conclusion *necessarily* follows from its premises. (By 'necessarily', we mean roughly 'without exception or possibility of being otherwise').

Assessing/evaluating deductive arguments comprises two parts. The first part is logical assessment, or formal assessment. This part of the evaluation process tries to determine whether or not the argument is any good in supporting or proving its conclusion *if* the premises are true. Hence, at this state, we're not at all concerned with whether or not the premises of the argument are *actually* true. In fact, we're not concerned so much with what the statements say or mean. Rather, we're interested more in the bare, logical structure of the argument: what kind of claims are the premises: atomic or molecular? If molecular, disjunction? Conjunction? Etc. We're concerned with whether or not the premises of the argument prove or support the conclusion *if they are true*, whatever they may say. So, we're looking to see if the argument is *valid*. What is validity?

Validity

An argument is *valid* if the premises of the argument *guarantee* the conclusion. By "guarantee" I mean *logically entails* or *logically implies*. What do these mean? Again, this is a difficult question, but we can get at the general idea in the following way. If $P_0, P_1, \dots, P_n$ are the premises of an argument A , and C is the conclusion of A , then argument A is valid *if and only if* given $P_0, P_1, \dots$ and P_n , *necessarily* C . Another way of putting this is in terms of truth: A is valid if and only if the conclusion must be true *if* its premises are true. Notice that validity does not require that the premises of an argument *actually* be true. Rather, all it requires is that the premises, *if true*, necessitate the truth of the conclusion. Validity is simply about the form or structure of the argument. Let's state the definition of validity here:

An argument is valid if and only if the conclusion must be true if all the premises are true.

Notice here two things. Statements are not valid – *arguments* are valid. Validity is a property of arguments, not statements. Moreover, validity doesn't care if the premises are actually true. Validity only applies when the conclusion's truth follows necessarily *if the premises are true*. The premises may in fact be true, they may not be, it doesn't matter to validity.

Soundness

The second part to assessing/evaluating deductive arguments is truth assessment. This part of the evaluation process tries to determine whether or not the premises of the argument *are actually true*. For validity (the first step discussed previously), we simply assumed the premises are true and then looked to see if the truth of the conclusion necessarily followed. In this step, we are inquiring about the actual truth-value of the premises: are they really true? Hence, this will require some background knowledge, doing some research, checking sources for credibility, etc. The question of what sentential/propositional truth is is a difficult question to answer. There are different theories of truth, but for now, we are just going to go with the most common understanding. A sentence is *true* when what it says is actually the case. It is false, however, when what it says is not actually the case. That is, a sentence is true if it corresponds to a fact. So, sound arguments are truth preserving in the sense that truth is preserved from premises to conclusion, and that preservation is guaranteed.

After you do these two steps, you may find that a deductive argument is *valid* and that all of its premises are *in fact/actually true*. When this occurs, the argument is *sound*. *Soundness* is a property of deductive arguments that have two things: validity, and all true premises. Let's define soundness explicitly here:

An argument is sound if and only if the argument is valid and all of its premises are true.

Remember: an argument can be valid *without* being sound, but if an argument is sound, it must be valid.

Valid Deductive Inferences

Here are some *valid* forms of deductive inferences. If these inferences are the only inferences made in a deductive argument, then it is a valid argument. (There are plenty more).

Suppose ' Φ ', ' Ψ ', and ' X ' are variables for some claim (atomic or molecular). Suppose also that ' $\sim$ ' is the symbol for negation, meaning *It is not the case that*. The symbol ' $\therefore$ ' means *Therefore*.

1. Modus Ponens (MP)	2. Modus Tollens (MT)	3. Transitivity (T)
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If Φ , then Ψ . <u>Φ.</u> $\therefore \Psi$.	If Φ , then Ψ . <u>$\sim\Psi$.</u> $\therefore \sim \Phi$.	If Φ , then Ψ . <u>If Ψ, then X.</u> $\therefore$ If Φ , then X .
4. Disjunctive Argument (DA) Φ or Ψ . Φ or Ψ . <u>$\sim\Phi$.</u> <u>$\sim\Psi$.</u> $\therefore \Psi$. $\therefore \Phi$.	5. Simplification (SIM) <u>Φ and Ψ.</u> <u>Φ and Ψ.</u> $\therefore \Phi$. $\therefore \Psi$.	6. Conjunction (CON) Φ . <u>Ψ.</u> $\therefore \Phi$ and Ψ .
7. Addition (ADD) <u>Φ.</u> <u>Ψ.</u> $\therefore \Phi$ or Ψ . $\therefore \Phi$ or Ψ .		

In each of these valid forms of arguments, it doesn't matter what Φ , Ψ , and X say. Φ could be the sentence 'Joe is an oily prime number'. Remember, validity doesn't care about what the sentences say. *If* the premises are true in each of these forms, then the conclusion *necessarily follows*. This is what makes them valid forms of deductive arguments. (Question: 'How can the premise 'Joe is an oily prime number' be true at all?' For assessing validity, it doesn't matter: we're just *assuming* that it's true for the sake of assessing the argument, while disregarding *how* it could or would be true).

Note on how to use the chart: Whenever you come across an argument, try to symbolize or abbreviate all the atomic claims within it. For example, if you have:

Sally is a student.
 If Sally is a student, then the sun is an ice cream
 cone.
 Therefore, the sun is an ice cream cone.

You can abbreviate:

Student.
 If Student, then Ice Cream Cone.
 Therefore, Ice Cream Cone.

Or:

S
 If S, then I.
 Therefore, I

This way, you can see easily whether it matches one of the forms above in the chart. Another thing to note is that the lines can be switched around. As you can see from the previous example, the argument seems to match Modus Ponens, but the lines of the premises are switched. In the chart, the conditional comes first, but in this example, the conditional comes second. That's fine. In this example, it still matches the chart in the right way: you have a conditional statement, and then you have another premise that states the antecedent of the conditional; and you have a conclusion which is the consequent of the conditional. So, while the line-ordering does not necessarily have to be the same, the inferential relation has to be the same.

Keep in mind that these valid forms of arguments are valid *inferences*. So, inferring the conclusion from each of the premises in each of these forms is a *valid inference*. That is, no matter what, if the premises are true, then the

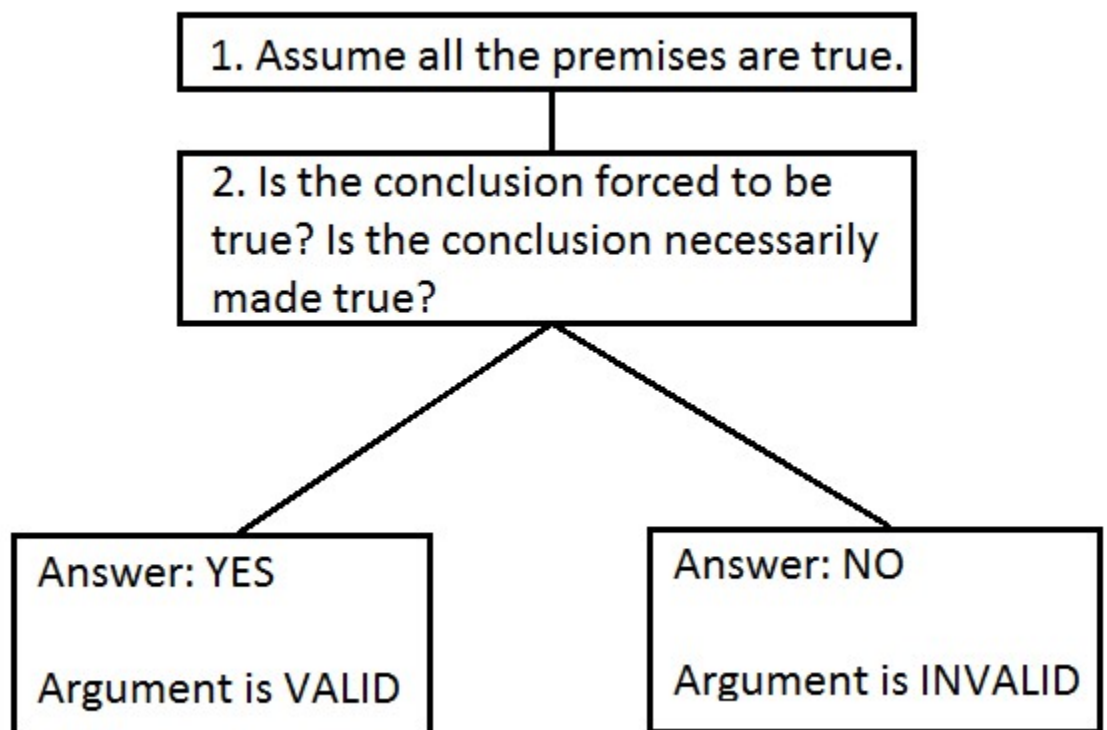
conclusion you infer from them following the forms in the chart will necessarily be true.

Testing for Validity

How do you assess whether or not a deductive argument is valid? One way to do is to try to show that it is *invalid*: try to show that it is possible for all premises to be true *and* conclusion false. If, however, it is impossible for all the premises to be true and the conclusion false, then it must be valid.

There are two methods you can take. The first method is this: (1) Assume all the premises are true; (2) Is the conclusion *forced* to be true? In other words, is the conclusion *necessarily* made true no matter what, given the premises are all true? If the answer is 'yes', then you've shown that *if the premises are all true, then the conclusion must be true*. In other words, you've shown that the argument is valid. If the answer is 'no', i.e. the conclusion isn't necessarily made true, then you have shown that if the premises are all true, then conclusion is *not necessarily true*. In other words, you've shown that the argument is invalid. Here is a flow chart for this method:

VALIDITY TESTING METHOD 1:



Let's do an example. Consider the following:

- a. If P, then Q and R.
- b. P.
- c. Therefore, Q and R.

Using this first method, assume first that the premises are both true. So, we assume that the premise 'If P, then Q and R' is true, and that the premise 'P' is true:

Assumed TRUE: a. If P, then Q and R.

Assumed TRUE: b. P

We've assumed that 'P' is true, and that 'If P, then Q and R' is true. Since 'P' is assumed true, we know that 'If P, then Q and R' has a true antecedent. Remember, we've assumed 'If P, then Q and R' is true. So, that means that the consequent of that conditional, 'Q and R', *cannot* be false. Why? If 'Q and R' was false, then that means 'If P, then Q and R' would have a true antecedent and a false consequent, making the whole thing false. (Recall that conditionals are false in only one way: the antecedent is true and consequent is false). But since we assumed that the whole conditional is true, 'Q and R' cannot be false: it has to be true. By assuming a and b to be true, we discovered that 'P' is true, and 'Q and R' is true. So, that means that the conclusion, 'Q and R' is true. By using our method, did assuming the premises are true *force* the conclusion to be true? Did assuming the premises are true *necessarily make* the conclusion true? Yes. Therefore, this argument is *valid*.

Let's try it with an invalid argument:

- a. If P, then Q.
- b. Q.
- c. Therefore, P

Following validity testing method 1, we assume that the premises are true:

Assumed TRUE: a. If P, then Q.

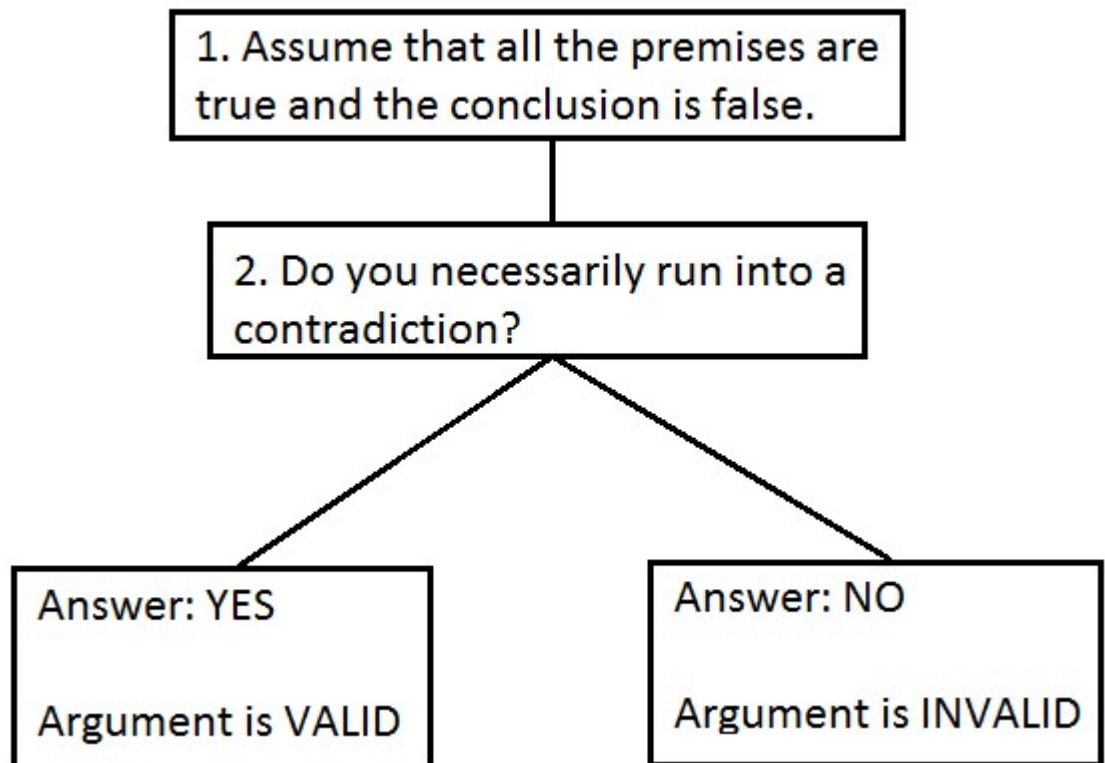
Assumed TRUE: b. Q

We've assumed 'Q' is true, and that 'If P, then Q' is true. Since 'Q' is assumed true, that means the consequent of 'If P, then Q' is true. Remember, we've assumed 'If P, then Q' is true. What, then, must P be? We know that the consequent 'Q' is true and that the whole conditional 'If P, then Q' is true. Can P be true? Yes: if P is true, then the antecedent and consequent would both be true, and that would fit our assumption that the whole conditional 'If P, then Q' is true. Can P be false? Yes: if P is false, then the antecedent would be false and the consequent would be true, and that would fit our assumption that the whole conditional 'If P, then Q' is true. (Remember, a conditional is false in only one way: true antecedent and false consequent). So, if we assume both premises to be true, P can turn out as TRUE or it can turn out as FALSE: not one way is forced or necessitated. By using our method, did assuming the premises are true *force* conclusion to be true? Did assuming the premises are true *necessarily make* the conclusion true? No. Therefore, this argument is *invalid*.

VALIDITY TESTING METHOD 2

The second method is similar to the first method. Here is the second method for testing whether or not an argument is valid: (1) Assume all the premises are true *and* assume that the conclusion is false; (2) Does a contradiction *necessarily result*? If the answer is 'yes', then you've shown that it is *impossible* for all premises to be true and the conclusion false (since contradictions are impossible). In other words, you have shown the

argument to be valid. If the answer is 'no', then you've shown that it is *possible* for all premises to be true and the conclusion false. In other words, you've shown the argument to be invalid. Here is a flowchart for this method:



Let's do an example. Consider the following:

- a. P or Q.
- b. It is not the case that P.
- c. Therefore, Q

Following validity testing method 2, we assume that the premises are true *and* the conclusion is false:

Assumed TRUE: a. P or Q

Assumed TRUE: b. It is not the case that P

Assumed FALSE: c. Therefore, Q.

Now, given our assumptions, we get that 'Q' is false, and we also get 'P' is false. How did we get to know that 'P' is false? We assumed 'It is not the case that P' is TRUE. So, that means that 'P' is FALSE.

We assumed that 'P or Q' is TRUE. However, we also found that 'P' is false and 'Q' is false. Well, if both of them are false, then 'P or Q' must be false, since both disjuncts would then be false. However, we assumed 'P or Q' is TRUE. So, we run into a contradiction. Hence, since contradictions are impossible, it is impossible for this argument to have all true premises and false conclusion, making this argument *valid*.

Let's try method 2 with an invalid argument. Consider the following:

- a. P or Q.
- b. P.
- c. Therefore, it is not the case that Q.

Following validity testing method 2, we assume that the premises are true *and* the conclusion is false:

Assumed TRUE: a. P or Q

Assumed TRUE: b. P

Assumed FALSE: c. Therefore, it is not the case that Q.

Given our assumptions, we know that 'P' is true and 'Q' is true. How did we get that 'Q' is true? We assumed that 'It is not the case that Q' is false. That means that 'Q' is true.

Since 'P' is true and 'Q' is true, that means that 'P or Q' is true. We also assumed that 'P or Q' is true. Is there any contradiction? Nope. Hence, this argument *can* have all true premises and false conclusion, making this argument *invalid*.

NOTES ON VALIDITY TESTING METHODS

It's up to you which method to use. You'll need to have practice in order to get better at using these methods. Getting to know both methods would be best, but if you had to choose one, I would go for method 2. Sometimes, method 1 will take longer, so going with method 2 will be best. Also, note that you will NEVER get different results when you apply both methods to an argument.

3.6.2 Inductive Arguments

Inductive arguments are ones that attempt to *support* their conclusions, *without demonstration*, so that the conclusion is *probable* given the premises. A *strong* inductive argument is one that has premises that would make it *probable* or *likely* that the conclusion is true, if the premises are true. A *weak* inductive argument is one that has premises that, if true, would *not* make the conclusion probable or likely. An inductive argument cannot provide an absolute guarantee that its conclusion is true if the premises are true, no matter how strong the argument is, i.e. no matter how probable the premises make the conclusion. Since the notions of strong and weak are relative, admitting of degrees, there are stronger and weaker inductive arguments. This is unlike deductive arguments, which are either sound or not sound, valid or not valid, with nothing in-between. Let's state it here:

**An inductive argument is strong if and only if
the premises of the argument, if true, would make the conclusion
probably true.**

**An inductive argument is weak if and only if
the premises of the argument, if true, would not make the conclusion
probably true.**

Notice that a weak inductive argument is not one where the premises, if true, would make the conclusion false. Lacking support for the truth of a claim does not thereby make the claim likely to be false. It might still be true, but with weak inductive arguments, the conclusion is just not supported by the premises.

If an inductive argument is strong *and* has premises that are all actually true, then the inductive argument is *cogent*. So, the cogency of an inductive argument requires the satisfaction of two conditions: strength and all true premises. Otherwise, the inductive argument is *not* cogent. Let's define it explicitly here:

An inductive argument is cogent if and only if two conditions are satisfied:

(a) the argument is strong; and (b) all of the premises of the argument are actually true.

In many cases, inductive arguments infer conclusions based on past cases. For example, "In the past, this car stalled whenever it went faster than 50 km/h. Hence, going over 50 km/h now will make the car stall." Now, the premise doesn't *prove* the conclusion. It provides some support for inferring the conclusion, but the conclusion can only be inferred as *probable*. Compare this with the following: "If Sally is human, then Sally is a mammal; Sally is a human; therefore, Sally is a mammal". The conclusion "Sally is a mammal" isn't just merely *probable* given the premises, it is *proved* by the premises, i.e. it necessarily follows from them.

There are different kinds of inductive arguments. I will discuss just three types in a very, very rough and general way: generalizations; inferences to particulars; and inference to the best explanations.

Inductive Generalization

An inductive generalization, often called "enumerative induction" or "induction by enumeration", is when you start by observing particular instances of members of a group *G* having some property *P*, and then make an inference to all the members of *G* having *P*. This is rough, and there are lots of nuances in this kind of inference, but that's the general idea. For example, you observe a thousand swans, and all of those swans are white. So, the group here is *swans*, the particular instances observed are the thousand that you observed, and the property is *whiteness*. From these observations, you might generalize to all swans: therefore, *all* swans are white. This is a generalization about *all swans*, observed and not observed, based on observing *some swans*. To be a little more precise, we can formulate the general structure of inductive generalizations like this:

1. Some members of group *G* have been observed, and *N*% of those observed have some property *P*.
2. Therefore, *N*% of all members of *G*, observed or not, have *P*.

In the swan example, *G*=swans, *P*=the property of whiteness, and *N*=100. Note that the premise begins with members of *G* that have been *observed* or

studied. The conclusion is supposed to be a generalization to *all members of G, observed and unobserved*.

There are many factors that need to be considered when making an inductive generalization. One of the most important is the sample of observed instances. One important factor that determine whether or not your sample provides a sufficient basis for making an inductive inference is its *size*. If we observe just four swans, and all four happen to be white, this might serve as a very weak basis for making the inductive generalization that *all swans are white*. Having just a small sample size will not provide enough information about the entire population. Another important factor is the sample's representativeness of the whole population. The sample size can affect this: if the sample size is too small, then it will likely not be representative of the population. Also, you want to make sure that the sample was not chosen in a biased way, but instead determined in a random manner. This will help to achieve representativeness.

For instance, if I decide to observe 1000 swans, but only in a particular private zoo, then perhaps my sample is not representative of swans as a whole. The private zoo keeper may have something for white swans, or maybe the private zoo keeper started with ten white swans, and they bred until there were twenty, all coming from a line of white swans. Inductive generalization is much more complicated than this, and there are many, many other factors that need to be considered, and considered more precisely. This, however, is enough to give you a rough idea.

Inductive Inference to Particulars

An inductive inference to particulars is an inference that goes from a general statement about members of a group *G* having some property *P*, where this statement is an inductive generalization, to the conclusion that some particular members of *G* have *P*. For instance, suppose a large percentage of people in the world have been found to be right-handed. From this, you might then infer that the person sitting next to you is right-handed. This inference goes from a general claim, itself established inductively, to a claim about a particular individual.

Inference to the Best Explanation

Inference to the best explanation, sometimes called 'abduction', is a form of inductive reasoning that infers that an explanation, and all the concepts it employs, is true provided that the explanation is the *best* explanation for some phenomenon. The form of this is as follows:

1. There is some phenomenon *P*.
2. An explanation *E* is the best explanation of *P*.
3. *E* posits the existence of (or employs the concepts of) $a_1, a_2, \dots a_n$.
4. Therefore, *E* is probably true, and $a_1, a_2, \dots a_n$ probably exist (are probably concepts of real things).

This is roughly how scientists come to conclude that certain things exist. They observe something happening in the world. Then, given what they

know, they try to come up with an explanation of that phenomenon. It sometimes happens that the concepts that they have aren't enough to explain the phenomenon, and positing the existence of a new thing *does* provide a complete and powerful explanation. If this is the best explanation, then scientists accept it as true, or probably true.

Again, this is rough, but you get the idea. This form of reasoning is quite common. You wake up in the morning and you cannot find your keys. One explanation is that you misplaced them when you came in. The other explanation is that someone has taken them while you were asleep. Supposing that all the doors to the house are locked and there is no sign of a break-in, and supposing that you remember using your keys to unlock the front door last night to enter the home, the best explanation may perhaps be that you simply misplaced them.

What counts as a best explanation? That's a difficult question, and the answer is quite complicated and a debated matter. Several things that have been noted are the following: *ontological parsimony*; *explanatory simplicity*; *consistency*; and *predictability*. Ontological parsimony is achieved by an explanation if it posits the least amount of (new) entities. For instance, suppose a clock falls into the ocean, and after being retrieved, it fails to work properly. One explanation of its failure to operate is that its gears are being jammed by salt and fine sediment, or its internal parts have rusted out. Another explanation is that tiny gremlins have entered the watch and gnawed away at its gears and springs to the point of inoperability. The first explanation is better than the second, since it's ontologically parsimonious in comparison: the second requires you to accept the existence of a new kind of entity, namely, gremlins. Explanatory simplicity is just simplicity of the explanation: how many steps are in the explanation – how many concepts are employed and how many assumptions made, etc. This would be like the misplaced key case. Other ones would be mundane explanations vs. conspiracy-type theories. The latter tend to assume a lot about the intentions of many different individuals and institutions, thus growing in complexity. Another factor to consider when determining which explanation is the best is consistency. There are at least two sorts of consistency to consider. One is local consistency: is the explanation itself logically consistent, or does it make claims that are inconsistent with each other? Does it entail or presuppose claims that are inconsistent with what is said in the explanation, or with other claims entailed or presupposed by the explanation? For example, you might explain why the cup tipped over and spilt its content by the following explanation: a non-physical ghost pushed the top of the cup with great force and it tipped over. This explanation runs the risk of inconsistency: does *pushing* and *great force* invoke physical concepts? Does the use of 'pushed *at the top*' suggest location in space? If these are the case, then would they be inconsistent with the ghost being *non-physical*? The other sort of consistency is holistic: is the explanation consistent with the concepts, principles, and other statements of our well-established, accepted theories? Does it conflict, say, with physics? Biology? Chemistry? The last factor to consider in determining which explanation is the best is predictability: which explanation provides us with a way to predict future occurrences of similar phenomenon under similar

circumstances? Again, all of these are rough, and an entire semester can be devoted to inference to the best explanation.

Causal Reasoning

The last one I'll talk about is causal reasoning, or causal inference, or causal induction. Certain instances of this would actually fall under inductive generalization, but not all instances. Causal reasoning starts with observing *correlation* between two things. These things can be properties of objects, objects, events, people, etc. Two things are correlated when they co-occur, either at the same time, or at different times but with some regularity. For example, drinking gasoline (one thing) and then getting sick (another thing). The first act is correlated with the second state. If the correlation is observed in repeated instances, we might infer that the one *causes* the other. Another, perhaps more specific, way of putting causal reasoning is this:

1. A thing X having property P_X is observed in many instances to be correlated with a thing Y having property P_Y .
2. Therefore, P_X caused P_Y .

(Here, X and Y can be the same, or they can be different). Now, while this is a very, very rough description of a causal inference based on correlation, there needs to be a lot more done to make this inference strong. For one, we need to make sure that there are no other factors that are influencing Y to have P_Y . Two, we want to make sure that X having P_X and Y having P_Y isn't caused by a common cause. We'll talk about these more in the section on inductive fallacies.

Causal reasoning occurs in other forms, so there is no one way that a causal inference is drawn. However, what most causal inferences have in common, if not all, is that they conclude a causal relation based on an observed correlation, and, like all inductive reasoning, the conclusion that is inferred is always *probable* but never guaranteed. In other words, causal inferences are not truth-preserving.

There are other forms of inductive inferences: by analogy, from individual to individual, from general to general, etc. We will not go into these.

4. Argumentative Writing

Argumentative writing has one primary function: to argue for the truth of a position on some issue. You don't need to go into a bunch of history or other fancy stuff. Avoid trying to make it 'sound academic' or trying to make it sound poetic and eloquent. The fancy decorations can be put in later if you want, but you need to be a master at that in order to do it right. Many people try to make the paper sound intellectual and eloquent in a poetic, literary way, but many often end up just obfuscating the paper. You have to approach it like building a house: you want it to have a good foundation and solid structure before you put on all the window-dressings and other stuff.

4.1 Explanation vs. Argument

When asked to argue for a position, many people describe and explain their position but provide a very thin argument for the truth of their position, if they provide an argument at all. Explanations and arguments are different things. Sometimes they coincide, but in many cases they do not. An explanation has the goal of teaching a concept, principle, or a passage so that one can understand the concept, principle, or passage at a deeper level. “Deeper” here is vague, but roughly what is meant is that it provides sufficient understanding that enables you to apply the concept or principle, or to understand the point, role, and meaning of a passage. An argument, however, is different. What an argument tries to do – the aim of an argument – is to establish, prove, or support the truth of a claim or set of claims.

4.2 Basic Structure of Argument Essay

Given this, there are three things that you need to do in any argumentative essay:

1. **State clearly what the issue is.** What is the question that you are addressing? In other words, what is the main question that your essay will answer? In your explanation, you want to orient your reader towards this question, and help them understand what it is. The issue can be any of the following: Is abortion immoral? Should the death penalty be legal? Is so-and-so’s argument for the existence of God a good argument? Etc.
 - a. In the paper, you should also explain any technical terms. If the paper is on abortion, explain what abortion is; etc.
2. **State clearly your position on the issue.** What is your answer to the issue you’re addressing? What is your position? You want to make this very clear. This is a statement of your thesis: what you are trying to show or prove is *true* in the paper. The entire paper is aimed at *this one thing: your position, i.e. your thesis*. Following the examples given above, your thesis might be: Abortion is immoral; the death penalty should not be made legal; so-and-so’s argument for the existence of God fails to adequately establish its conclusion.
3. **State and explain your reasons and reasoning.** These are the main premises of your argument. The reasons may be independent of each other, or they could work together. Either way, they must contribute towards supporting or proving your position.
 - a. When you give your reasons, you should *state* the reason *and explain it*. To state something is to just say it: what is it that you want to say? State it simply and clearly. To explain something is to clarify its meaning and its relevance.
 - i. If any of your supporting reasons (premises) are somewhat controversial, you’ll also need to back up your reasons/premises with further reasons. In other words, you should try to provide some reasons to show that *that reason is true*. So, there might be sub-argument within your paper: an argument within an argument. So, if T is my thesis, I might give a reason R to show that T is true. But then, after stating and explaining R, I might

provide a reason S to show that R is true. So, the paper would go like this: T is true. Why? Because R. Why is R true? S. Of course, you could keep going: giving reasons for your reasons for your reasons, *ad infinitum*. Where should you stop? Roughly, a general rule is to stop when the reason you give is somewhat intuitive and commonplace, and *not* controversial.

- b. Explain the reasoning. That, is how do the reasons fit together? Are they independent reasons? Are the reasons working together to form one argument? The reasons are like components of a machine. You want to explain not just the components themselves, but how they come together to make the machine work. Similarly, you want to explain how the reasons come together to make the argument work.

4.3 Form and Content

In writing an argument paper, you need to consider two things common to writing *any* paper: content and form. Content is *what* you say. Form is the *structure* or *organization* and *the order* of what you say. It is how the content is organized. We split things up in terms of content and form in almost everything that we do.

4.3.1 Content

In the least, the content of your argument essay should consist of the three things stated above: a statement (and explanation) of the issue; a statement of your position (thesis); and a statement and explanation of your reasons and reasoning. So, in addition to establishing the truth of your position by providing reasons for its truth, what you want is to make it clear enough for someone to read your essay and understand the reasoning. In other words, you need to take into consideration your audience. One general rule of thumb is this: imagine that your reader does not know anything about the topic you're writing on, and imagine that your reader is a little hostile, wanting to criticize your paper for being unclear and wanting to tear apart your argument or wanting to point out how the reasoning doesn't make sense or why a particular reason isn't a reason at all, etc. If you're writing with such a person in mind, you'll make sure to write very clearly, and to be thorough in your explanations and reasoning. *State* your reason; *explain* it further; and then *provide an example*, if possible, to help further clarify what you have in mind.

Every sentence in an argument paper is doing a specific job. The paper, if it's done right, should be written in such a way that every single sentence in the essay has a purpose, and taking it out would make the entire paper incomplete. This is what you should aim for. So, in an argument paper, each sentence should be either a *claim to be proved or shown true*, a *claim that helps prove or show another statement to be true*, or a *claim that is explaining another statement*. Let's talk about these three in brief detail.

First, a claim in an argument paper could be a claim that is being proved or shown true. This could be the thesis/position statement, or it could be a reason for the thesis/position statement (since sometimes you will not just give a reason for your thesis, but also a reason for the reason). Second, a claim in an argument paper could be a claim that helps prove or show another claim to be true. This is going to be a reason or premise in the argument. As you can see, this statement could also be a sentence that is being proved or shown true, since a reason might also be given a reason. Third, a claim in an argument paper could be a claim that explains or clarifies another statement; or it will try to elaborate on another statement and bring out its meaning and relevance; or it might be an example, or a different way of saying something, that helps the reader understand what you mean. Sometimes, it is an explanation of an important concept or principle, or a definition of a technical term that will be important for the reader to know. *Note well: in every argument essay, you will need to explain something. You will always need to explain concepts, premises, an issue, principles, definitions, the argument, etc. ALWAYS EXPLAIN WHAT YOU MEAN!*

So how should you present the content? In an argument paper, the main thing here is to focus on *clarity* and *relevance*. When it comes to clarity, you want to make sure that the meaning of what you say is clear. There are many things that can get in the way of clarity. One is flowery language. Flowery language can seem poetic, and make it seem like you're a good writer, but it really takes *lots of practice and training* to be able to write eloquently without obscuring the meaning. Most times, your attempt at being eloquent will obscure the meaning, so avoid it at least when writing the first draft. For instance, consider the following sentence:

"The elusive nature of mind has vexed and perplexed brilliant thinkers from time immemorial."

This sentence may sound nice and eloquent, and may appear intelligent, etc. However, you have to remember that you have to be good at language to do things like that well. Most people don't read that much and don't write that much, and they don't get critiqued on their writing that much. If you read and write a lot, but don't get critiqued, that's not enough. You have to learn how to write in a basic and simple way first: master the basics. You can't go directly to creating a Picasso or Da Vinci without first learning the basics of art. So, instead of the above sentence, just say something plainly and simply like this:

"The nature of mind is difficult to explain."

This sentence, at heart, is saying the same thing as the other sentence. However, this second sentence is a lot clearer and more to the point. It's also shorter, which is good. When you write, start with the *basic*. The window dressings are not necessary, and if you want them, put them in later, and make sure that they don't obscure the intended meaning!

Another thing that can get in the way of clarity is excessive vagueness. This one is a bit harder to avoid, but you need to be alert to it. For instance, the following is vague:

“Descartes believes that the mind and body are related to each other.”

This is not a good sentence. When you say ‘related’, you are just saying that there is a relation between them. But what is the relation? How are they related? Is the relation an identity relation, i.e. are they *identical*? Is the relation a causal relation, i.e. is one the cause of the other? You need to be more specific and less vague.

Relevance is basically what we've already discussed. What I mean here by 'relevance' is just what we've already discussed earlier: that every sentence in the essay serves a function, and the function of every sentence is determined by one thing and one thing only: the end or goal of establishing the truth of the main conclusion of the paper, and facilitating understanding of the truth of the conclusion.

What often gets in the way of having only relevant sentences is the following: (1) poor planning; and (2) failure to focus and keep in mind the conclusion of the paper and the primary issue at hand. Always remember what you're trying to show to be true in the end, and *always* keep in mind the main question that you're trying to answer in the paper.

In writing your paper, it will be very helpful to engage in some preliminary brainstorming/note-writing in order to prepare for the paper. It might help to write an outline or a rough map, or even a diagram. For instance, suppose the prompt for an essay is this:

“Critically examine and discuss the thesis that morality is culturally relative.”

Before you respond to this prompt, it would help to take notes or write out a map. One way to do this is just start with headings that lays out what the paper will talk about. For instance, in your paper, you might just type the following headings:

Cultural relativism: what is it?

What is the argument for cultural relativism?

What are some problems with this argument?

What's my position on cultural relativism and this argument?

How did I come up with these points? I look at the prompt and I try to determine what is needed to respond to the prompt in a complete way. For instance, I see that I have to examine critically the *thesis* that morality is relative. So, it would help to answer the question: what is moral cultural relativism? Next, it might help to look at the reasons offered to support or prove moral cultural relativism. So the second question is what the argument for cultural relativism is. The prompt asks that we *critically* examine the position. So, the third part will be a *criticism* or *attack* on the argument. Since I am to discuss cultural relativism, I will then present my own thoughts on cultural relativism and the argument for it, presenting what *I* think about it, and *the reasons* that support my position. That gives us the fourth question.

These four questions break down the prompt. These four questions also mark at least four parts to your paper. So, by breaking down *what* you want to discuss, you have also provided yourself with a rough structure to your paper. We'll talk about form later. Once you have this, you can probably add more and further break down the content of the paper. For instance, you add to the above and get this:

Cultural relativism: what is it?

What is the argument for cultural relativism?

- Different cultures disagree about what is morally right or wrong. Hence, there is no absolute moral right or wrong.

What are some problems with this argument?

- The argument assumes that if A and B disagree about the answer to a question, then there is no absolute answer to the question.
- The argument exaggerates the extent of disagreement between cultures about what is morally right and morally wrong.

What's my position on cultural relativism and this argument?

- Morality is not culturally relative.
 - o Reason: Morality depends on our nature, and since we are all humans, there is a common basis for morality.
 - o Reason: Disagreement is only possible if there is a shared concept of morality.

Notice now that I've filled it out a little more. I can fill it out even more after this, and the paper is going to have a clear structure. Again, this is where trying to think about what the paper is going to talk about helps to generate the *form* that the paper will take – how the discussion will be ordered. We'll talk about that next. But first, I want to talk about two more things: exposition and argument.

To exposit an argument, or to provide an exposition of an argument, is to *state* and *explain* an argument given by someone. Here, you are not providing an argument necessarily, but rather *explaining* what the argument given by someone else is, and how that argument goes. An exposition is *not* a summary. A summary just goes over what was said in a passage or chapter or book. An exposition explains the reasoning, concepts, or principles in a passage or chapter or book. So, the aim and function of an exposition is different from the aim and function of a summary. A summary aims at acquainting and familiarizing the reader with the topics covered in something and what occurs in it. An exposition, however, aims at providing the reader with an explanation of the reasoning or discussion that occurs within a passage. The latter (exposition) aims at helping the reader understand the passage, chapter, or book, whereas the former (summary) aims at giving the reader knowledge of what happened in a passage, chapter, or book. For instance, consider the following passage from Aristotle's *De Anima*:

“The modifications of the soul present a problem: are they all shared by what has soul, or are some proper to the soul alone?

“It is necessary indeed, but not easy, to deal with this problem. For in most cases there is, apparently, no action or being acted on without the body; as in danger, desire, confidence, and sensation in general. Understanding, however, would seem especially proper to the soul. Yet if this too is a sort of imagination, or never occurs without it, not even this exists, in fact, apart from the body.

“But if the soul has some operation or affection exclusive to itself, then it could exist as a separate entity. If, however, there is nothing thus proper to it, then it is not separable, but is like a straight line, which has, as such, many properties—such as being able to touch a bronze sphere at a given point; but straightness separated does not touch it; not being in fact separable, since it is always with a bodily subject.” Book I, Chapter I of *De Anima*, by Aristotle.

Now consider the following two brief essays:

1. Aristotle first presents a question: are modification of the soul shared by what has soul, or are there modifications of soul that are proper to the soul? Aristotle then claims that this is not an easy question, and then states that most of the time, there is no acting or being acted upon without the body. Examples he gives are things like danger, desire, confidence, and sensation in general. He then considers understanding, and then presents this to be something that seems to be proper to the soul. However, Aristotle then backs off a little, stating that if understanding is like imagination, then understanding cannot happen without the body.. Next, Aristotle claims that if there is an operation of the soul that can take place without the body, then the soul might be able to exist as a separate thing. He states that if the soul has nothing proper to it, then it is not separable, and then compares it to a straight line that touches a bronze sphere, and the impossibility of this if the straightness was separated from the line...
2. In this passage Aristotle is considering whether or not the soul could exist without the body. By *soul*, Aristotle means *life*, and by ‘body’, Aristotle means just the organic, physical body. Hence, the question Aristotle is asking is whether or not life can continue in an individual apart from the body. Aristotle holds that one way that it *would* be able to do so is if there is an operation of the soul that can take place without the body. This is what he means when he asks about an act or operation that is *proper to the soul*. Such an act is something that *the soul alone* does, *without* the body. The reason that finding an act that is proper to the soul would allow for the possibility of the soul continuing to exist separated from the body is that an operation of the soul that actually takes place is an act, and for the soul to be acting is for it to exist, since nothing that acts does not exist. Aristotle grants that many operations or acts of the soul require the body. To see requires the eyes, and to imagine requires the phantasms in the body, where ‘phantasms’ are representations stored in the body gotten from our sensory experience of things in the world. Since these operations are

bodily operations, vision and imagination cannot occur without the body. The likely candidate for an operation or act of soul that might be proper to soul is *understanding*. However, he hesitates. If understanding happens like imagination, then it would be an act that requires a body. The analogy of the straight line and bronze sphere illustrates this: the living human being is a soul unified with a body, and this is what the straight line represents. Actions of the living human are like the straight line touching a bronze sphere. The straightness separated from the line is like the soul separated from the body: the former cannot touch the bronze sphere, and the latter cannot act in the world...

Which of these is an exposition of Aristotle's argument, and which is a summary? The first is a summary, and the second is an exposition. Notice that the first talks about what Aristotle says, and what he concludes, but it doesn't explain how the argument actually goes. The second, however, gives a detailed account of the reasoning offered by Aristotle. It actually *gives* the reasoning, and *explains* the reasoning (argument). The summary in the first passage does not do that – it just gives an outline of what Aristotle talks about. So a summary only tells you that a certain argument was given in the position, but it does not explain to you how that argument goes or is supposed to go. It doesn't care if you understand the argument. It only cares that you know that some argument was given. An exposition, on the other hand, cares that you understand how the argument is supposed to go. So, an explanation will lay out the reasoning, explain the reasoning, and fill it out to help you understand *why* or *how* the position or thesis claim is true. An explanation teaches you the argument and thus helps you understand the reasoning, whereas a summary just informs or acquaints you with the argument so that you know that there is an argument. You can give a summary of an argument without understanding it, but you cannot give an exposition of an argument without understanding it.

So where does argument fit in? Sometimes you will *exposit* another person's argument, and then *argue against that person's argument*. This is done quite often in a philosophy paper. You *exposit* another person's position and their argument for their position. Here, you are not criticizing or making any evaluations. You are just trying to give the most accurate account of the person's argument. This is the expository section of the paper. The reader should be able to read what you write and understand what the person's argument is: what the issue is; what that person's position on the issue is; and then what that person's reasoning is. Then, you might attack that person's reasoning by attacking at least one of the premises, or by showing that the argument's form is bad (i.e. showing that it's invalid; showing that it's fallacious; etc.). In this kind of paper, you will have two parts: first the exposition of someone's argument (e.g. "First, I am going to exposit the common argument for cultural relativism."); second, your critical argument against that argument (e.g. "Second, I am going to critically examine that argument and show that two of its premises are false, and also that the argument itself is invalid"). Where you make criticisms and evaluations is in the second part of the paper, and where you try to give an unbiased, objective account of the person's view and argument is in doing the first part of the paper.

4.3.2 Form

By now you should be getting a sense of what the form or structure of the paper is. The form or structure is just the organization of the paper. So, there's *what* you say (content) and then there's *how* it is organized (form). You have to think of a paper kind of like an organic body: *what* the body has are organs, e.g. heart, kidneys, brain, etc.; *how* they are organized is the form, e.g. heart is next to the lungs, and the brain is up in the head, etc. Similarly, a paper needs organization. You don't want to spray the page with sentences: you want to *organize it*. In just the way that the body cannot live with just *any* organization of its organs, but requires a specific organization in order to sustain life, an argument paper cannot sustain an argument well with just *any* organization of its contents, but requires a specific organization in order to sustain and convey its argument well.

The basic organization of an argument paper will be this: an introduction; the argument; and conclusion. An introduction will include 1 and 2 above (viz., a statement of the issue and a statement of the thesis). You can also state what you're going to do in the paper. For argument papers, at least in our class, it's totally fine to say something like this in the introduction: "In this paper, I will address the issue of X. First, I will explain an argument that shows that X is wrong. Second, I will present my own argument, showing why that argument is bad. Third, I will conclude with some further thoughts about the issue." Doing this in the introduction can clarify things and guide the reader. If you read an article that started out that way, it would be very helpful, for it will let you know what the paper is about and what the author is going to do. You should try to do the same thing. Think of your reader as someone who is generous enough to give your paper time to read, and so you want to return the generosity by trying to make sure that you make good of the readers time by making the paper as clear as possible.

The argument section of the paper should be organized clearly as well. So, you might organize this section according to the premises: one or two paragraphs talking about the first premise; one or two paragraphs talking about the second premise; etc. At the end of discussing the premises, you want to tie together the premises you have offered and explain to the reader how they fit together. You've given them the pieces of your argument, but you want to show how they come together to form a coherent, strong or valid argument. Use words like "The first reason..." or even "First, ...", "Second...", etc. This will help you organize your paper, and also help the reader navigate the argument.

In the conclusion of the paper, you can address possible objections that might be made against you. This will help strengthen your argument. In the conclusion, you can also present further issues that you did not address in the paper, but that should be addressed. This will provide the reader with where research should go, or what is left to do beyond the paper. The conclusion can also be a summary of the paper: what you did and what your argument was. So, you don't have to do all of these things in a conclusion, but at least one will do. (There are many other ways to conclude a paper – these are just some common ones).

Here are some basic structures for argument papers. These aren't exhaustive, but it can be helpful to know.

1. Argument Paper Structure 1
 - a. Introduction
 - b. Exposition of another person's argument for some position P.
 - c. Author's counter-argument against that person's argument.
 - d. Conclusion
2. Argument Paper Structure 2
 - a. Introduction
 - b. Author's argument for her or his position P on some issue.
 - c. Conclusion: one or two possible objections against author's argument or position, and then author's counter-response to those objections.
3. Argument Paper Structure 3
 - a. Introduction
 - b. Exposition of an argument for position P1 on some issue.
 - c. Exposition of an argument for position P2 on the same issue (or against position P1).
 - d. Author argues that one of the arguments is better than the other.
 - e. Conclusion.
4. Argument Paper Structure 4
 - a. Introduction
 - b. Exposition of someone's argument for position P on some issue. (Example: you might give an account of Spinoza's argument that God is the only substance that exists).
 - c. Author gives reasons to argue that her/his exposition is an *accurate* account of that person's argument. (Example: next, you say that your account, i.e. interpretation, of Spinoza's argument that God is the only existing substance is the right interpretation, providing textual evidence and other reasons).
 - d. Conclusion.
5. Argument Paper Structure 5
 - a. Introduction
 - b. Exposition of someone else's exposition of an argument for position. (For example, "According to Jamie Lee, Aristotle's argument runs as follows...").
 - c. Argument against that person's exposition. (For example, "I believe that Jamie Lee provides an inaccurate account of Aristotle's argument. The reason is that ...").
 - d. Conclusion

Notice that you can mix these up. For instance, Argument Paper Structures 1 and 2 can be combined. What is key in all of these is that the author presents an argument – provides reasons – for her position on some issue.

4.4 Arguing Against

There are two common ways to argue against someone's position on some issue. One is to present an argument that shows their position is wrong. This is a straightforward argument: someone says P; you want to argue that P is false; so you give reasons and organize them to show that P is false. This is directly attacking a position as false. Another thing you might do is leave P, and not try to argue that P is false. Rather, what you might do is try to show that the person's *argument* for P is a bad argument. This doesn't show that P is false, nor that P is true, but just that the argument for P is not good. This is criticizing a person's argument for P. Again, it isn't attacking P, but just showing that the argument that the person gave for P is a bad argument.

When you criticize an argument, what you should aim for is either of two things (or both – inclusive disjunctions!). One, you might argue that at least one of the premises in the argument is *false* or *not established*. If you argue that it's false, you want to present reasons or evidence that shows it to be false. If you want to show that it has not been established, what you want to do is argue that there is not enough evidence or reasons to accept it as true (which doesn't mean that it is false). Two, you might argue that the *form* of the argument is bad. So, if it is a deductive argument, what you might argue is that the argument is *invalid*; if it is an inductive argument, what you might argue is that the argument is *weak*. So, what you are doing, in essence, when you argue *against* someone's argument is to attack its soundness (if it's a deductive argument) or its cogency (if it's an inductive argument).

Another point to keep in mind: Suppose someone gives an argument for some position P and I criticize that argument and try to show that the argument is bad. Does this mean that I reject P, i.e. think that P is false? No. I may believe that P is true. What I am doing is arguing against the person's argument, not against P. So, for instance, suppose someone says "God exists, and my argument is this: The moon is made of cheese, and frozen yogurt is better than ice cream; therefore God exists." I may criticize this argument and try to show that it's bad, but that doesn't mean that I am arguing that God does not exist, nor does it mean that I personally believe that God doesn't exist. Many people have given arguments to show that God exists throughout history, and many have criticized those arguments, arguing that those arguments are bad arguments. Who gave these criticisms and arguments against arguments for the existence of God? Other people who believe that God exists! So, just because you disagree with an argument, and just because you give an argument *against* someone's argument, that doesn't mean that you are also rejecting the conclusion. So, remember, you can argue *against* someone's position by presenting an argument that attempts to show that the position is *wrong*, or you can argue against someone's argument. When you do the latter, you are not necessarily rejecting the person's position. Instead, you are rejecting their argument. In other words, you are not saying that their thesis is false, but just that they failed to provide a good argument for it.

Another point to keep in mind: Suppose that you believe that God does not exist, and your friend believes that God exists. Suppose you give an argument to prove that God does not exist, and then your friend criticizes your argument and successfully shows that it's an invalid argument, or that

one of the premises is false. Does this mean that your friend's position on the issue of God's existence is right? In other words, if your friend has successfully shown that your argument is bad, then does that mean that God *does* exist? No. To make that conclusion would be to engage in a fallacy. Just because an argument for P is a bad argument does not show that P is false.

5. Rhetorical Devices and Techniques

Rhetoric is the art or study of persuasion. To persuade is the same as to convince. One persuades or convinces someone of *something*, which is to make that person believe that thing. Clearly, the act of persuading or convincing requires three things: what does the persuading/convincing, what is persuaded or convinced, and what the person who is persuaded or convinced is persuaded or convinced of. Strictly speaking, arguments are different. They are lines of reasoning that involve two things: premises and a conclusion. Although we're interested in constructing good arguments for our claims, we also want to guard ourselves against attempts at persuasion that do not offer good arguments.

The famous ancient philosopher Aristotle discusses the art of rhetoric, and distinguishes three ways that the art of rhetoric aims at persuasion. *Ethos* is persuasion by a speaker's personal attributes, including such things as his or her background, reputation, accomplishments, expertise, and similar things. For example, you might be persuaded to believe something a speaker is saying because the speaker has a really impressive pedigree, or because they are known to be highly respected in their field. *Pathos* is persuasion by personal connection or emotional arousal. For example, a politician may say something like "Don't support the proposed healthcare plan – your grandmother will be euthanized by the state if you do!" This is a kind of scare tactic that tries to evoke a certain emotion in you and make you convinced that you should believe the speaker. *Logos* is persuasion through information and logical argumentation. That is, it is to persuade someone through reasoning, by appealing to their rationality.

Rhetoric often uses certain devices and techniques in order to persuade people to believe something to be true. The first thing to note is that using rhetoric may be *psychologically effective*, but that in no way translates into being *logically effective*. You may do well in persuading others into your view by employing rhetorical techniques that psychologically sway someone into believing you, but this alone would amount to manipulation. Furthermore, you may have still left your own position on some matter completely unsupported. You need to be able to separate the *psychological* force of a statement from its *logical force*. We are solely concerned with improving the latter, while helping you develop your skills in spotting and avoiding the former. Moreover, you want to be *logically sound* thinkers, not simply rhetorically effective speakers. The key is to be *rational*, not necessarily persuasive. The second thing to note is that using rhetoric is not in itself a bad thing. Sometimes employing a rhetorical technique may spice things up, give a statement or speech or writing a certain stylistic flare or panache, etc. So, if you have rhetorical elements interlaced with an argument, that doesn't mean that the argument is bad. In other words, no point-deduction occurs simply by the presence of rhetorical devices. Point-deduction (in terms of how critical and well-put together an argument is) only occurs when the *argument* is bad or faulty. But while employing rhetorical devices does not in itself detract from the argument (unless the devices make the arguments

unclear or convoluted), their use hardly warrants *gaining* points. Your argument does not become a *better argument* when you use rhetorical devices. Whether an argument stands or falls depends on whether or not the reasoning is good, *not* on how persuasive the argument is based on rhetorical techniques. While rhetoric can be used in addition to good reasoning in a way that makes an argument amusing, fun, inspiring, etc., it should *never* be used *in place of* rational argumentation.

So what are rhetorical techniques? The ones we will look at here are ways of persuading that fall short of giving an argument. They are not arguments at all. No attempt at an argument is given. Instead, they are attempts to *color* a claim in a way that makes it look good or bad. These rhetorical techniques are like clothing: claims are dressed in rhetorical garb, as it were, to make them *look good* (i.e. persuasive) or *look bad* (i.e. warranting rejection). Rhetorical techniques often try to persuade you by appealing to your emotions or skewing and distorting your perception on some issue.

5.1 Slanters

Some rhetorical techniques aim at persuasion by trying to give something that *looks* like a good argument or *looks* like good reasoning. One common sort of rhetorical devices is a *slanter*. Slanters are most commonly used to give a statement or position on an issue either a positive or negative slant. So, if a statement P has a positive slant, it attempts to get the reader to accept or have a more accepting attitude towards P. If P has a negative slant, it attempts to get the reader to reject or have a negative attitude towards P. Slanters, then, try to manipulate you into thinking a certain way just by slanting an issue in a positive or negative light.

5.1.1 Type-I Rhetorical Devices

There are rhetorical devices often in the form of a single word or short phrase. They are slanters in that they cast a negative or positive slant to the sentence in which they occur, leading one to accept or reject the claim. Remember, no argument is given for accepting or rejecting the claim. That's how these rhetorical devices work.

Euphemism/Dysphemism

A euphemism is a way of naming something or calling something in terms that gives a more positive slant to it than it would normally have. They may be used to refer to something in such a way as to associate the thing with something positive, cast it in a positive light, or make it more neutral. A dysphemism is an expression that is used to produce a negative effect on a listener's or reader's attitude toward something, or to minimize the positive associations it may have by casting it in a negative light or associating it with something negative. When someone uses a euphemism or dysphemism, they are trying to sway you into thinking positively or negatively about something. Some examples of euphemisms include "reurbished", "pre-owned", and "freedom fighter". Some examples of examples of dysphemisms would be something like "second-hand", "used", "terrorist", etc. Here's another example: suppose your romantic partner is going to

break up with you. One way of making this break-up sound *more* awful is by saying that they are *dumping you*. To refer to someone breaking up with another as *dumping* the other is dysphemistic. A euphemistic way of referring to that act would be something like *freeing the other person of any of the customary romantic obligations and giving the person the full freedom to see or not to see anyone else*. So, if you're breaking up with someone, a euphemistic way of putting it is "Hey, I'm going to free you of any romantic obligations or expectations with respect to me."

It's important to note that not all euphemisms are bad. Sometimes their use is appropriate. When dealing with a sensitive matter, it might be best to use a euphemism that is neutral. For example, it might be more polite to say that someone "passed on" or "passed away" instead of saying that they're "dead" or that they "died". You don't say "he's crazy" or "she's psychotic", but rather that he or she is "mentally ill". These are legitimate uses of euphemisms. Notice that the use of euphemism in these cases is to be polite, not necessarily to convince or manipulate anyone into believing something.

Weasellers

These are linguistic methods for "hedging a bet", or, to protect a claim from criticism by weakening it. Weasellers are used to get someone to think a certain way while leaving a way for the speaker to *weasel* his/her way out if challenged. To an uncritical listener, statements using weasellers may sound very convincing when in fact the statements are weak or unsubstantiated. Furthermore, the use of weasellers make it difficult to charge the person of being deceptive because technically they *are not* lying, but rather banking on the listener to not be a critical thinker. For example, if you look at shampoos or lotions, you'll often read things like "Guaranteed to give you healthy-looking hair after just one week of regular use!" or something like that. Now, suppose you used the product and your hair came to have a healthy shine, but it actually damaged your hair beyond repair. Did they lie to you? No – they said that their product is guaranteed to give you healthy *looking* hair. They never promised or guaranteed to give you *healthy* hair. Things like "healthy-looking skin" is like this: healthy-looking skin isn't the same as healthy skin. Other words that *might* indicate weaseling include "maybe", "perhaps", "appears", "seems", "up to", and "virtually".

Sometimes we do have a legitimate need to use words like "seem" or "appears" or "maybe", etc. When you're not sure whether or not the person in the distance is your friend Maddy, you might say "that seems like Maddy" or "that may be Maddy" or "that person over there appears to be Maddy". You might also describe some sort of energy drink as making you genuinely *feel* more energetic, or doing something to make you *appear* healthier. These are all fine, and aren't necessarily weasellers. It really depends on the context and motives of the one using these words. In legitimate cases, these wouldn't be called 'weasellers', since the intention wouldn't be to weasel. Weasellers are found in places where there is an attempt at persuading someone into doing or believing something, and the weasellers themselves are playing a key role in the effort at persuasion.

Downplayers

To downplay something is to make someone or something look less important or less significant, or to make someone look less deserving of praise or recognition. This can be done with words like “mere”, “merely”, “only”, “just”, and even by putting certain words in “quotes”. Conjunctions can also be used in subtle ways to downplay things, such as the conjunctions “however”, “although”, “but” and others. When someone downplays something, the person is trying to make you believe that you shouldn’t think too highly or that you should think negatively about something. But if that’s the case, the burden is up to the person to give you reasons. Consider the following examples:

1. “Although the company polluted the town’s water supply, they provided jobs to thousands of its residents.”
2. “Although the company provided jobs to thousands of its residents, the company polluted the town’s water supply.”

Notice here that the first downplay the significance or importance of the company polluting the water supply, whereas the second downplays the significance or importance of the company providing jobs to thousands of residents. Here are some other examples:

3. Don’t listen to Ms. Lee, she’s *just* a teacher.
4. Don’t listen to Ms. Lee, she (only) thinks that she is a teacher.
5. Yeah, okay, so she graduated from a “university”.
6. Whatever, she *only* has a bachelor’s degree.

In statement 3, ‘just’ downplays that she is a teacher, as if being a teacher is insignificant or an unimportant role, or at least a role that is taken to be so lowly that Ms. Lee ought not to be taken seriously. In statement 4, to say that she *thinks* that she is a teacher, or to say that she *only thinks* that she is a teacher downplays that she is in fact a teacher. In fact, it downplays it to the point of diminishing any credibility that Ms. Lee may have as a teacher: she isn’t one genuinely, just in her dreams! Statement 5 uses quotation marks to downplay the university that she graduate from, making it seem that it’s just a university in name, but really not a bona fide university. Statement 6 downplays a bachelor’s degree, making it seem that it’s insignificant, or at least not strong enough to take what she says seriously.

The words “mere”, “only”, etc., and the use of conjunctions and quotes aren’t always used to downplay. There certainly are many cases in which their use is appropriate. I may say something like “Although John had good intentions, he nevertheless still made a terrible mistake.” This use of a conjunction does downplay John’s having good intentions. But this usage may be fine, since it may be aimed at trying to focus our attention on the main issue. Of course, all of this depends on whether or not *further reasons* are given for thinking that the focus ought to be on John’s making a terrible mistake and not on his having good intentions. Also, someone might say something like “I am “working” over the weekend” in order to make a funny remark. Perhaps the person is doing something like video-game testing, which the person finds extremely fun, making it difficult to see it as work. Lastly, we might use “only” to restrict the scope of something, such as “He satisfied only the first condition.” This use of ‘only’ isn’t downplaying, but only saying that no other condition but the first was satisfied by the person. “Mere” is used in a similar way as “only” sometimes.

5.1.2 Type-II Rhetorical Devices

These rhetorical devices are ones that make or rest upon some unwarranted assumption. The assumption tends to be subtle, and it makes you think that the assumption is true, when in fact no argument was ever given for it.

Stereotypes

A stereotype is a thought or image about a group of people based on little or no evidence. In other words, stereotypes are unwarranted and oversimplified generalizations about the members of some group or class. First, since stereotypes are based on little or no evidence, they may be entirely false or they may be bad reasons. Second, stereotypes group people into simple categories, which may lead one to make an unjustified and unfounded judgment. In other words, one is lead by stereotypes to *prejudge* before carefully thinking about the issue at hand, i.e. accept something on *prejudice*.

Is there some kernel of truth to all stereotypes? People often defend stereotypes by claiming that there is. Let's think critically about this. First, the adoption of a stereotype might be based on personal experience. However, just because you've experienced five, or ten, or even fifty members of some particular group, it doesn't warrant a stereotype. People tend to form a stereotype after just the first encounter, and then *expect* and hence *look for* similar traits or characteristics in other members of the group. Hence, it biases one's outlook and really distorts one's judgment. Second, many times people fail to see other, more significant factors because they are blinded by some stereotype. For example, I've heard people say "Asians are good at math." The problem here is that it doesn't allow for variety: people are individuals and different from each other. Each person is his or her own individual, and to lump them all together under one category is to simplify them and fail to recognize the complexity and richness of each individual as an individual. Moreover, it prevents deeper, careful, and critical thinking. The stereotype could lead someone to think that being good at math is something that is *inherent* or *necessary* to being Asian. But that's clearly preposterous. So, even if it *were* true that Asians did better at math on average, the answer may be more complex. For instance, it may be that *any* immigrant group that, one, culturally tends to emphasize the value of education, and two, does not speak English well, will tend to do especially well in math.

Innuendo

An innuendo is something that is *subtly suggested or insinuated*. In other words, an innuendo uses words with neutral or positive or negative associations to insinuate something deprecatory or even positive. A *significant mention*, also called a *paralipsis*, is one kind of innuendo, where a speaker makes a subtle suggestion that something is the case, but without explicitly committing to the suggestion. When someone makes an innuendo, what happens is that the innuendo subtly works to persuade you into thinking a certain way

unless you're careful and astute. Many times these innuendos are completely unsubstantiated, and it's a tactic used by a speaker to persuade someone while avoiding giving reasons to support his or her view.

Not all things that is suggested is actually an innuendo. Sometimes certain things are implied or presupposed by people when they say something. This is almost always the case. For example, suppose you went to a party really late and find that nobody but the host is there. You ask "Where did everyone go?" The host may say "Well, some people have probably gone back home to England after a vacation in the Bahamas, whereas other people in China have gone to work after a good night's rest. And still other people...". Obviously, the host is being obnoxious and sarcastic. You didn't want to know where *everybody* in the *world* has gone, but just the people who were at the party. Take another example. Suppose you went into a chatroom and someone asks "Is anyone from Los Angeles?" Another chatter replies "Yes, there are many people from Los Angeles – in fact, probably over ten million people are from Los Angeles." Obviously, this is *not* what the person was asking. They were presupposing or assuming that "anyone" referred to the people present in the chatroom. Making assumptions or presuppositions are fine for the most part. But when someone *insinuates* something by using an innuendo for the purpose of swaying you into thinking a certain way, without providing an argument, then you need to be guarded and demand for reasons.

Loaded Questions

A loaded question is a kind of innuendo. It is a question that rests on one or more unwarranted or unjustified assumptions. It *assumes* something in the question, and answering the question will lead to confirming this unwarranted assumption. However way you answer the loaded question, you end up accepting the assumption that's *loaded* into the question, as it were. If the assumption is false, then you should not answer the question, but *reject* the question instead. One of the most popular examples of a loaded question is: "Have you stopped beating your wife?" Notice that this question is a yes-no question, and that whatever answer someone gives, he ends up accepting that he *at one point in the past* beat his wife. If that assumption is false, the best thing to do is to say "I reject the question, for I never have in the past or in the present beat my wife."

5.1.3 Type-III Rhetorical Devices

These rhetorical devices use humor and extreme exaggeration to cast a negative light on a claim so as to sway and influence people towards rejecting the claim

Ridicule / Sarcasm

These are basically ways of mocking a view in order to make others reject it. It's basically *making fun of* a view or claim in order to sway others into thinking that the view or claim being mocked is false. When you mock or ridicule or laugh at a statement, you're basically saying that you think that the statement is false. This method is a kind of bullying. When someone

laughs, mocks, ridicules or responds sarcastically to something (“yeah, like THAT’S really true, pfff”), the person is suggesting not only that the claim is false, but that it and those who hold it deserve ridicule. Nobody wants to be ridiculed, so this method may work in persuading people: they move *away* from the view being ridiculed because they don’t want to be associated with it and also become the object of ridicule. Just because someone *mocks* or *makes fun of* a claim or view does not thereby make it false.

a. Examples – There are plenty of examples of this. Here are just a few:

- “Welfare is a necessity for a healthy state? HA! Whatever...”.
- “Pff, as if.”

Hyperbole

This is an extravagant overstatement or over-the-top exaggeration. Typically, hyperboles are used with excessively emotive words or phrases. The emotive words appeal to your emotion, and try to slant you in a certain direction regarding some issue. By presenting something in an extreme way is in itself something that is intended to evoke an emotional response, since typically people find extremes *highly* implausible and are likely to react with extreme surprise and total rejection. Hyperboles may either try to get you to buy into the hyperbole and think about something in an extreme way, or the hyperbole may be aimed at getting you to think highly or negatively about something in a milder way.

b. Examples – Consider the following four examples:

- “You can’t trust John because he’s the evil devil incarnated as a young man.” This is a hyperbole used with an emotive phrase. It’s colorful, somewhat amusing, and interesting, but an extreme exaggeration. It makes use of emotive words: “devil” and “evil” evoke a negative feeling in the audience.
- “Sarah is feeling sick because she ate thousands of hot dogs.” Unless Sarah is a competitive eater, she probably did not eat a thousand hot dogs in one sitting. It’s an over-exaggeration – a hyperbole. Note that ‘thousands’ is the hyperbolic word and that it’s non-emotive, i.e. it’s fairly neutral.
- “Rudy is a nazi terrorist.” This is a hyperbole that tries to get you to think that Rudy is an extremely horrible person. This hyperbole tries to get you to have an extreme attitude towards or view of Rudy.
- “Everyone says that our pizzas are *by far the best pizzas in the whole world.*” This statement is like the third one. But the pizza store that advertises this may not actually expect anyone to believe what they say. They may, however, try to get you to think that while it may be an over-exaggeration that their pizzas are best in the world, their pizzas

are nevertheless very good. Notice that the milder claim (very good pizzas) is what they want you to believe, and they do this by making the hyperbole. You might think “Well, there must be something to their pizzas if they’re saying everyone is giving them these rave reviews”.

5.1.4 Type-IV Rhetorical Devices

These rhetorical devices put a positive or negative slant to definitions, explanations, analogies, or comparisons in order to persuade someone.

Rhetorical Definitions and Rhetorical Explanations

Rhetorical definitions try to define a word or phrase using emotively charged language in order to express or elicit an attitude about something. In other words, rhetorical definitions try to sway you into thinking a certain way by defining something with a positive or negative slant. *Rhetorical explanations* provide an explanation of something but with a positive or negative slant in order to express or elicit an attitude about something, or to get you to think a certain way. Both try to manipulate through psychological association. In both cases, the issue is presented in a way that *slants* it in one position, typically through evaluative or emotive language.

Rhetorical Analogies and Misleading Comparisons

A *rhetorical analogy* is a comparison between two things or a likening of one thing to another in order to make one of them appear better or worse than it might be. A *misleading comparison* are comparisons that leave out important information, usually making them unclear and vague, in order to mislead you into believing something. When someone makes a rhetorical analogy with something positive or negative, basically what they’re trying to do is get your mind to associate something positive or negative with some claim. But associating a claim with something negative or positive does not in itself provide any reason for thinking that one should in fact think positively or negatively of the claim. A misleading comparison does a similar thing: it misleads you into thinking that something is bad or good simply because it is being compared with something bad or good, respectively.

Sometimes it’s okay to make analogies and to make comparisons. Many times doing so can help the reader understand a complicated point or to clarify an idea. Using analogies and comparisons is bad when they are used to influence someone to think positively or negatively about something, or to get someone to reject or accept something, without giving any reasons whatsoever. When you are given a comparison, you should keep in mind several questions.

- Is important information missing?
- Is the same standard of comparison used?
- Are the compared items really comparable?
- Is the comparison expressed as an average?

5.2 Proof Surrogates and Repetition

A *proof surrogate* is an expression used to suggest that there is evidence or authority for a claim without actually citing such evidence or authority. It *appears* that statements using proof surrogates are supporting their view with evidence, but in fact they aren't. To mention "some study" is hardly giving support for the view. "Informed sources say" or "studies show" or even "it's obvious that" can all be used as proof surrogates, and they try to dupe someone into thinking that what is being said has a lot of evidence backing it up, without actually giving you any of the evidence or reasons. This tactic tries to make someone think, unquestioningly, that the truth of something is well-established by important studies or by experts in the field, or that something is so obvious that if you disagree then you must be ignorant or highly misinformed. Of course, this is a *rhetorical* tactic that does not give you any genuine reasons.

It's okay to say things like "Studies have shown" or "scholars have argued that" *as long as you cite or reference the supporting sources*. In addition, sometimes it's warranted to say "obviously" or "it's obvious that" depending on the circumstances. For example, I may say that "If you're reading and understanding this very sentence, then obviously you are somewhat fluent in the English language." Here, the use of "obviously" is okay, since *reading and understanding* that sentence *presupposes that you know how to read English*. However, in any case that you use "obvious" or "obviously" or "clearly", it's best to have reasons ready to back up your claim. For example, I may say "It's obvious that John is here at this party" and, when asked how it's obvious, I might just point to John and say "There he is wearing a bright fluorescent-pink body suit."

The technique of *repetition* seems ridiculous but surprisingly it can be quite effective in a strange, brain-washing kind of way. Someone might repeat the same point over and over again, but in different ways, and in this way persuade a lot of people into believing the point. In many cases, and to make it more effect, repetition works by repeating the same point in different ways. Of course, repeating something over and over again doesn't make it more likely to be true.

5.3 Visual Persuasion

Persuasion through the use of visual media is the use, alteration, or editing of images for the purpose of manipulating you into thinking a certain way, typically by evoking certain emotions. Persuasive use of images can range anywhere between being very manipulative to being outright deceitful. Pictures may be good evidence for thinking that something is the case, and sometimes may be helpful tools for clarifying something. Sometimes the picture may have no relation whatsoever to the claim being made, but somehow makes the viewer come to accept or reject the claim. In our digital age, persuasion through visual media is pervasive. Many of you have encountered it, and perhaps without your knowing. Beauty magazines often present models in a way that highlights their 'natural' beauty, when in fact the images you see have been modified and edited in a way to enhance

certain features, cover others, etc., usually in an attempt to *persuade* you into buying a certain product.

For example, suppose there was a picture of a man dressed in a BP uniform shaking hands with some fishers off the coast of Louisiana, and everyone is smiling warmly. The caption may read “BP: We care for everyone like no other.” The picture is loosely connected to the statement, and it may even evoke a positive, warm feeling in you about BP. But the picture might be staged, or it might be a photo of an initial meeting, after which a big fight broke out. Who knows. Just because the photo shows what appears to be a nice gathering of people with a BP official does not mean that BP does care about the well-being of other people.

Here’s another example of something that actually happened. Consider the following two images:



Steven Reddicliffe and Jacques Steinberg criticized the legitimacy of Fox News as a serious news program. So, Fox News picked up the story and, in return, criticized Reddicliffe and Steinberg. The right-side photo of each image is what Fox News *actually* showed on their news program. But notice

that the *right* side photo of each image is *manipulated* to make Reddicliffe and Steinberg look bad (big forehead, exaggerated receding hair line, egg-shaped head, big nose, yellow teeth). The left-side photo in each image is the real, untouched photo of them, and was *not* shown in the program. No disclosure was made that the images had been altered. Fox News was exposed for manipulating the images and received a lot of criticism for childishly altering images to make people think negatively of Reddicliffe and Steinberg. This was done in order to put a negative (ugly?) slant to them.

Here's another case:



In this picture, it certainly appears that the female contestant on this popular game show thinks that an elephant is the largest of the four things, including the moon. Is it true? Can a grown woman really believe that an elephant is bigger than the moon? Well, this may seem funny but it's probably not funny to her, nor to anyone who finds themselves the victim of some manipulated image. Here's the real image:



Turns out that the question and the answers were all manipulated and changed. Very deceiving – you initially would have thought that this woman needed to go back to school, but it turns out, you were being misled and deceived by a manipulated photo.

5.4 Extreme Rhetoric of Demagoguery

A demagogue is someone who does the following in order to gain popularity and garner support from people:

- Exploits the ignorance, prejudices, and anger of a people
- Attempts to stir the emotions, especially anger, of the people by using incendiary language that feed into other people's prejudices and anger, thereby contributing to the growth of fanatic zeal
- Disregarding and even dismissing rational discourse, opting instead for extreme rhetoric in order to propagate false ideas and establish fringe theories.

Four broad rhetorical techniques of demagogues include the following: otherizing; demonizing; fostering xenophobia; and fear and hate mongering. Otherizing is the dividing of people into two groups, *us* and *them*, and then portraying the latter as suspicious, dangerous, repulsive, etc. *They* are then blamed for *our* problems. Demonizing is the act of trying to induce loathing of someone or something by portraying the person or thing as evil. Fostering xenophobia is the act of instilling xenophobia in people and doing and saying things to encourage its growth. Xenophobia is the fear or dislike of what is foreign or strange. Doing this often helps to further otherize and demonize. Fear and hate mongering is the act of stimulating fear, resentment and hatred, and encouraging their growth.

Typically what happens is that the rhetoric of extreme demagogues create an identity for themselves and define their worldview largely in *negative* and reactionary terms. What is primary in their view is not what they love, but

what they hate. This is why their world is so full of hate: everything they believe is defined *in opposition* to something, in *hatred* of something, and so even what they love is defined and constructed based on first what they *hate*. What I love is understood as *what is not what I hate*.

6. Fallacies

Some rhetorical techniques are attempts to give an argument, or at least something that looks like an argument. However, their aim is to persuade rather than argue for the truth of a conclusion. However, sometimes we just try to make an argument, with the intention of doing so, and then mess up in the reasoning. *Fallacies* are errors in reasoning, or arguments where the premises fail to establish or to provide support for the truth of the conclusion. These may be committed with good intentions, viz., with the intention to argue for the truth of the conclusion, but these may also be committed with the intention to persuade and not necessarily to reason well.

6.1 Informal Fallacies

Before going on, we need to keep in mind three important points. First, you need to be very careful when assessing an argument to see whether or not it's a fallacy. More times than not, students who learn about fallacies for the first time end up mistaking and dismissing *legitimate* arguments for fallacies. Sometimes good arguments may look like a fallacy. Don't make the mistake of rejecting a good argument. Second, most fallacies have this in common: what pretends to be a premise of an argument turns out to be actually irrelevant to the conclusion. Almost all fallacies are such that even if the premise(s) is (are) true, no good reason has been given for believing that the conclusion is also true. In other words, almost *all* fallacies are invalid. Third, many fallacies, though not all, make an appeal to emotions rather than reasons to support a conclusion. Expressing emotions, however, doesn't amount to giving reasons.

6.1.1 Ad Hominem Fallacies

This fallacy stems from the Latin phrase *argumentum ad hominem*, which means *argument to or towards the man*. This fallacy attempts to refute the *claim* a person is making claims about the *person* making the claim. It's a fallacy because the claims a person makes are one thing, the person making the claims is another. While certain qualities of a person may affect her or his credibility, they do not necessarily translate make the argument bad. You need to assess the argument itself for its own merits.⁶ Consider the following case:

⁶ Credibility is whether the source of a claim is reliable. Credibility has mainly to do with whether there are grounds for being skeptical about what someone says

Matt: Did you hear what Jane said? She said that the University plans to renovate the freshman dorms.

Sarah: Jane is a mean and cold person. Forget what she says.

Notice that in this exchange, Sarah is trying to dismiss what Jane says as false or unfounded by attacking Jane. But suppose Jane is in fact mean and a cold person. Does it follow that what she says is going to be false? Of course not – she may say “ $2+2=4$ ” and she would be saying something true, despite her personality. So the qualities of a person and the qualities of a statement or claim ought to be kept distinct. The Ad Hominem Fallacy fails to keep this distinction. Let’s discuss four different kinds of Ad Hominem fallacies.

Personal Attack Ad Hominem—The personal attack ad hominem fallacy runs as follows:

1. Person P said that S;
2. P has a negative quality or feature;
3. Therefore, it is false that S.

In short, this fallacy infers that a person’s beliefs are refuted by the person’s defects as a person. This is bad reasoning. Attacking a person or showing that a person has some sort of negative quality does not by itself *refute* what that person is saying or show that what they’re saying is false. Even if someone is a criminal and is known to have lied on many occasions does not *by itself* prove that whatever the person says is false. It may make you *suspicious* of what they say and lessen their *credibility*, but it does not thereby conclusively show that what they say is *false*.

Inconsistency Ad Hominem—There are two versions of this. First, the inconsistency ad hominem might focus on some inconsistency between two claims made by a person. Its form is as follows:

1. Person P says S_1 .
2. Person P says S_2 , which is inconsistent with S_1 .
3. Therefore, it is false that S_2 (or, that S_1).

When someone says something that was inconsistent with what they said some time before, that does not mean that what they’re saying now is false. The person may have changed her mind. The inconsistency ad hominem is committed when someone thinks that a person’s inconsistencies refute his or her beliefs. Now, when someone contradicts herself in one sitting, that is grounds for criticism. If I was expressing my views on something, and I contradicted myself by saying “A” and then later “not A”, then I am being inconsistent. However, that doesn’t by itself *show* that “not A” is false, or that “A” is false. I may have changed my mind, or I may be messy in my thinking. (If, however, you hold a view that *logically entails* a contradiction, then the original view must be false).

Inconsistency with behavior—This kind of inconsistency is called “*tu quoque*”, coming from the Latin expression meaning “and you too”. This is an inconsistency between what a speaker says and what a speaker does (behavior). The form of the argument is roughly this:

1. Person P says S.
2. Person P behaves in a way inconsistent with S.
3. Therefore, it is false that S.

If I was smoking a cigarette and I told someone next to me “Smoking is bad for your health. You should quit”, the other person may look at me and say “But you’re smoking too!” However, just because I am also smoking does not mean that smoking is not bad for your health or that you shouldn’t quit smoking. At best, inconsistency with behavior may make someone a hypocrite, but it doesn’t refute what they are claiming.

Circumstantial Ad Hominem—When someone tries to show that a claim is false on the basis of the circumstances of the person making the claim, then a circumstantial ad hominem fallacy is committed. The form of the argument is roughly this:

1. Person P says that S.
2. P is associated with circumstance C.
3. Therefore, it is false that S.

For example, suppose John says that lying is wrong. Bill tells Sarah “John grew up in a religious family and is religious himself. So of course he’s going to say that lying is wrong.” Notice that Bill is describing John’s circumstances in order to argue that lying is not wrong, or to argue that what John said about lying is false. Bill is reasoning badly. The circumstances are irrelevant to the truth-value of the statement in this case. If I grew up in a family of mathematicians, and I said “The square-root of an imaginary number is negative one”, will you dismiss what I say as false simply because of my circumstances? Of course not. Someone’s upbringing may bias or prejudice him (i.e. someone who is a Christian will say that God exists), but that does mean that what they say is false. It may be an *expected* belief and *not surprising* that someone under certain circumstances will believe something, but that hardly shows that what they say is false.

Poisoning the Well—This fallacy is more of a rhetorical tactic. In form, it’s an ad hominem fallacy. However, the difference is that in poisoning the well, the ad hominem occurs *before* the speaker says something. In this way, it is a sort of *preemptive* ad hominem.

1. Person P is attacked at time t_1 .
2. At some time *after* t_1 , P says S.
3. Therefore, it is false that S.

Poisoning the well is encouraging others to dismiss what someone will say preemptively by citing the speaker’s defects, inconsistencies, circumstances, or other personal attributes beforehand. If Bill makes Sarah have an unfavorable impression of John, then Bill is “poisoning Sarah’s mind” about John, and if Sarah is not careful, then Sarah is going to think that whatever John says is false. This poisoning can be done by attacking John’s character, or it can be done by simply associating John with negative things without actually attacking his character.

Pro Homine Fallacy (Positive Ad Hominem)—Strictly speaking, this is not an ad hominem fallacy, since it is not *against the person*. Rather, it is *for the person*, so I call it “pro homine fallacy”. People you admire might say awful or false things. Don’t confuse the favorable qualities of a person or group with the qualities of the person’s or group’s assertions/statements/claims. While it is a fallacy to infer that some claim is false or bad simply because the person making the claim or the source of the claim is bad, it is equally fallacious to infer that some claim is true or good simply because the person

making the claim or the source of the claim is good. The form of this argument is this:

1. Person P says S.
2. Person P has a positive quality or feature.
3. Therefore, it is true that S.

6.1.2 Guilt by Association

This fallacy is similar to an *ad hominem* fallacy. Basically, it's a kind of 'falsity-by-association' move, where one holds a claim P to be false because it is held or endorsed by someone that one does not like or someone that is ill reputed in other ways. The general form of this fallacy is as follows:

1. Person P₁ says that S.
2. Person P₂ also said/believed S.
3. P₂ is disliked or unseemly for having quality Q.
4. Therefore, it is false that S.

An example of this is as follows: You say that vital services should be placed in government control, such as healthcare. Joseph Stalin also believed that vital services should be placed in government control, such as healthcare. Joseph Stalin was a murderous tyrant. Therefore, it is false that vital services such as healthcare should be placed in government control.

6.1.3 Genetic Fallacy

The genetic fallacy occurs when a claim is argued to be false or a practice to be bad based on the origin of the claim or practice. (It could happen in the positive way too, when someone tries to get someone to accept something). The genetic fallacy is typically restricted to those cases where person A tries to refute person B's claim by disparaging the source of the claim, where the source is restricted to some kind of entity other than person B. In short, the genetic fallacy is thinking that the origin or the history of a belief refutes it. So, here are several cases of genetic fallacies:

1. *P is a policy that originated from the conservative party. The conservative party is stupid. Hence, P is false.*
2. *People during the medieval times were superstitious and unscientific. People during the medieval times also believed in God. The notion of God originated as a means of social control and superstitious explanation. Hence, it's false that God exists.*
3. *Equal rights and benefits is a social idea that originated from leftist groups. Leftist groups are a bunch of liberal freaks. Hence, equal rights and benefits is no good.*
4. *The modern notion of justice originated from the same group of thinkers whose ideas led to and perpetuated imperialistic occupation of other countries. Therefore, there is no such thing as justice in the modern sense.*
5. *The practice of using wedding rings originated from a long ago practice of placing shackles on women as property. Therefore, the practice of wedding rings is sexist or bad.*

All of these try to show that a claim is false or a practice is bad by disparaging the source of the claim or showing the sordid history of a

practice. It doesn't matter if the disparaging comment or sordid history is true.

6.1.4 Straw Man Fallacy

This happens when a speaker or writer distorts, exaggerates, or otherwise misrepresents an opponent's position, attacks that distorted, exaggerated, or misrepresented version of the position, and then concludes that the opponent's position is false. The general form of this is as follows

1. Person P says S.
2. Reasons R given against S_D , where $S_D \neq S$, and S_D is a distorted, exaggerated, or misrepresented version of S.
3. Therefore, it is false that S.

Here is an example. Jenny says that women should not be paid less than men simply because they are women. Bill claims that not paying someone fairly for the experience that they have is wrong, and that unqualified people shouldn't get paid equally as qualified people. Therefore, Jenny is wrong: it is false that women should not be paid less than men simply because they are women. Notice here that the reasons that Bill gives attacks the position that women should be paid equally as men regardless of qualifications. This, however, is not Jenny's position. Here is another example:

1. Sally: Gun laws need to make it more difficult for criminals and mentally ill people to buy guns.
2. Rufus: It is wrong make it more difficult for good, law-abiding citizens to buy guns. Doing so goes against the Second Amendment, and this also wrongly takes away the livelihood of long time hunters who are responsible and pose no threat to the American communities. Therefore, you're wrong Sally.

Notice that Rufus is totally missing Sally's position. He presents her view as 'we should make it more difficult to buy guns'. That's not Sally's position. Rufus fails in his reasoning.

Here's another example:

1. Shelly: The Arizona Immigration Law 1070B is a bad law.
2. Eliot: Letting illegal immigrants into this country and allowing them to use up our resources during a tough economic time is completely unacceptable. The reason is that we are already struggling as it is, and we don't have enough to support all of the illegal immigrants.

Notice here that Eli doesn't even address Shelly's position. Eli attributes to Shelly the belief that illegal immigrants should be allowed in the country to use up all of our resources. Shelly did not say that. She simply said that the immigration law is a bad one.

Here's another example:

1. Jerry: We should not cut taxes for the people of California.
2. Meg: In these tough economic times, we need to stimulate the consumer market. But raising taxes is just not going to do it – we should raise taxes when our citizens are already have a difficult time financially. We need to cut taxes, not raise them! Jerry is wrong to raise them.

Jerry never said that we should raise taxes. He said that we shouldn't *cut* taxes.

Another common one:

1. Jerry: We should raise taxes for individuals who make more than \$350,000 annually."
2. Meg: Raising taxes is going to put an undue burden on hard working Americans. If your taxes go up, you won't be able to afford the things you are already struggling to afford.

Meg is distorting Jerry's position, making it appear to be the position: raise taxes on all Americans. Clearly, that's not what Jerry said.

6.1.5 False Dilemma

The false dilemma fallacy occurs when you limit considerations to only two alternatives although other alternatives may be available. In particular, and strictly speaking, a false dilemma is one where the either-or statement is false because some other alternative is available. It is the erroneous narrowing down of the range of alternatives; saying we have to accept X or Y (and omitting that we might do Z). False dilemmas always have this sense of an exaggerated ultimatum: Either X or Y, PERIOD! This is a false dilemma: it's not at all clear why only these are the options. Other options may still be available. One form of a false dilemma is this:

1. Person P_1 says that it is false that S.
2. Person P_2 assumes that either S or T.
3. P_2 tries to show that it is absurd, false, bad, etc. that T.
4. Therefore, it is true that S (i.e. P_1 is wrong).

Here is an example of a false dilemma:

1. Grace: We are not cleaning the car today.
2. Ben: Either we clean the car today or we leave it filthy for another six months. We are *not*, and *cannot*, leave it filthy for another six months. Hence, we *are* cleaning the car today.

Sometimes false dilemmas come up more subtly. Consider the following example:

1. John: Affirmative action can help with inequality
2. Sarah: I'm against Affirmative Action
3. John: Oh, so you're a racist.

John concludes that Sarah is a racist, and it's based on the following assumption: either you're for affirmative action or you are a racist. That's a false dilemma: being for Affirmative Action and being racist are not the only options. The form of this argument is slightly different from the one before. The form of this false dilemma goes like this:

1. Person P_1 claims that S_1 .
2. Person P_2 assumes the false dilemma: either not- S_1 , or S_2 .
3. P_2 concludes that P_1 also holds S_2 .

Notice that this is the same as:

1. Person P_1 claims that not- S_1 .
2. Person P_2 assumes the false dilemma: either S_1 , or S_2 .
3. P_2 concludes that P_1 also holds S_2 .

In either case, the inference made in 3 is based on assuming the disjunction in 2. The disjunction, however, is a false dilemma, since it may be that both disjuncts are false.

Here are some examples of true dilemmas (i.e. true disjunctions):

1. Either the car is red or the car is not red.
2. Either the killer is a man or it is not the case that the killer is a man.
3. Either John makes fifty-five thousand dollars a year or more, or that is not the case.

Notice that these three dilemmas are such that *one* and *only one* must be true. If one is false, then the other must be true. Other dilemmas might be genuine given certain circumstances. For example, suppose Sarah flew out of the local airport to go to San Francisco. The local airport has flights only to San Francisco and Los Angeles. However, her flight lands in San Francisco and she is not on board. So, we might say that *Sarah is either in Los Angeles or she is in San Francisco*. Under these circumstances, this dilemma would be quite correct – or at least very plausible. However, under other circumstances, the dilemma would be a false one. Suppose she was supposed to come to your birthday party and she never showed up. You might ask “Where’s Sarah?” If someone said “she’s either in Los Angeles or in San Francisco”, then you’d want to know why only one of those cities.

In general, a false dilemma is one where the either/or statement is such that both alternatives are false given the circumstances. “Either the car is red or the car is yellow” This is a false dilemma, since the car may in fact turn out to be blue or some color other than red or yellow.

False dilemma fallacies and strawman fallacies are often difficult to distinguish. The difference is that in a strawman, the opponents view is misrepresented or distorted, and that misrepresentation or distorted version is taken to be false, and then the opponent is taken to be refuted. In a false dilemma, an unwarranted or false disjunction is assumed. Ask yourself: is there a misrepresentation or distortion? Is there an assumed either-or statement?

One thing to note is that false dilemmas tend to be *valid arguments*.

6.1.5.1 Perfectionist Fallacy

One type of false dilemma is the *Perfectionist Fallacy*: The perfectionist fallacy offers the false dilemma that we ought to do something perfectly/completely or not at all. It has the following form:

1. Person P_1 says it is true/right that S.
2. Person P_2 says that there is no way to be perfect with respect to S.
3. Person P_2 concludes that not S.

Here is an example:

1. Sally: Drunk driving laws should be enforced.
2. Wayne: You cannot enforce drunk driving laws perfectly to the point of catching all drunk drivers. Therefore, drunk driving laws should not be enforced.

Here is another example:

1. Sally: We need laws to prevent criminals and mentally ill people from buying guns.
2. Wayne: We can’t prevent all criminals and mentally ill people from buying guns no matter what law we place. Hence, it is pointless to

have laws to prevent criminals and mentally ill people from buying guns.

6.1.5.2 Line-Drawing Fallacy

Another type of false dilemma is the *Line-Drawing Fallacy*: This fallacy is committed when it is insisted that either a clear, distinguishing line can be drawn, or no distinction is to be made. The line drawing fallacy mistakenly requires that a precise line be drawn someplace on a scale or continuum, when no such precise line can be drawn, or drawing such a line is not necessary given the context. This usually occurs when a vague concept is treated like a precise one. Basically, the fallacy has this form:

1. Either there is a precise line that is drawn to distinguish between A and B, or there is no difference between A and B.
2. No precise line can be drawn to distinguish A and B.
3. Therefore, there is no distinction between A and B (or, we should not treat A and B differentially).

For example:

1. Jin: School administrators are paid too much.
2. Jack: What is too much? You can't determine exactly what is too much and what isn't. So, stop saying that they're paid too much. That just isn't true.

Here's another example: Abortion is wrong; the reason that abortion is wrong is that you cannot distinguish in the development of a fetus where it is not a person and where it is. When it comes to fetuses, there is no such distinction: they are just persons the moment of conception. Therefore, abortion is wrong.

These kinds of arguments rest on the idea that you cannot make a sharp, distinguishing line that is non-arbitrary, so the distinction does not exist or the distinction is not to be taken seriously. This is a bad form of argument. To see why it's bad, consider the color continuum:



It's quite plausible to think that we cannot draw a distinct line where redness ends and yellowness begins. This certainly does not mean that there is *no difference* between red and yellow. Many times, our ability to make distinctions might be limited, and sometimes two things may be distinct but nevertheless may *appear* indistinguishable to us. But just because we can't distinguish them doesn't mean that there is no difference between them.

6.1.6 Argument from Emotion

Emotions tend to prevent us from reasoning in a sound and logical manner. Heightened emotions tend to blind us from getting the most objective, sensible assessment of a situation. Generally, there are two ways that our emotions can make us illogical and irrational. First, when something happens to make us have strong emotions, we may think that we have been given a reason when in fact we have not. You should remember that being

caused to feel an emotion is one thing, but being given a *reason* is another thing. For instance, to *anger* someone is one thing, to *wrong* someone is a completely different thing. If you feel angry, that doesn't mean that you have been wronged. Second, when we get very emotional because of one thing, our emotion may influence our evaluations of something else that's totally unrelated. For instance, when we get angry at someone, we not only think that the person is wrong, but we are likely to be led by our anger into thinking that whatever that person says is wrong. This is a mistake. Anger tends to prejudice us towards interpreting things unfairly in a negative light. It tends to be pervasive and cloud our judgment on other issues.

6.1.6.1 Argument from Outrage

Argument from outrage is the fallacy of giving inflammatory words that just express anger and outrage, followed by a conclusion. The general form of argument from outrage is as follows:

1. P (where P is intended to induce anger and outrage).
2. Therefore Q.

One particular type of argument from outrage that is especially dangerous is *scapegoating*: blaming a certain group of people or a single person for all of life's troubles. This is what Adolf Hitler did to the Jewish community: he blamed all of the hardships and problems that the German people faced, especially after World War I, on the Jewish people. Scapegoating doesn't have to blame someone or some group of *all* of life's ills; it may just involve one bad thing. For example, after the 9-11 attacks in 2001, Jerry Falwell blamed the tragic event on the gay people of America, saying that *they* were to blame for this horrific tragedy. That's a case of scapegoating, and it's *very* dangerous because it can result in innocent people being hurt or even killed.

6.1.6.2 Scare Tactics

A *scare tactic* is the use of fear to get people to do something or accept a position on some issue. For example, someone might say that the government is going to take everyone's house away and that citizens will have to give up all their personal belongings if X becomes president. This tries to scare people into thinking that X should not be president. It's not that what is being said is false or wrong. Rather, it's that the scare tactic is a bad form of reasoning that does *not* support the claims being made.

Generally, scare tactics have this general form:

1. P (where P is a claim that is aimed at scaring)
2. Therefore Q (where Q is an alternative to P, but somehow related).

A special kind of scare tactic is *argument by force*, which is simply threatening the well-being of a person to get him/her into believing something. For example, a person says that he is going to kill me and torture my family if I don't believe what he says about gun control laws. This is argument by force: it's forcing me to accept something as true by direct threat.⁷

⁷ One question about this is whether or not someone can *truly* come to believe something as true when they are threatened. If someone was pointing a gun to my head and told me to accept that all educated people are evil, I might *say* it, but I don't think I'd truly believe it.

Peer Pressure Argument

This is a particular form of scare tactic. Many times we are swayed into believing something or thinking that something is true because we want to be accepted by our peers or by some group. Many people have done things or come to believe things because of peer pressure, whether it's thinking that someone is wrong, or something is okay, or whether it's doing something bad against someone else. The general form of this fallacy is as follows:

1. Peers will reject me from their group if I do not accept P as true/false.
2. Therefore, (I should accept) P is true/false.

6.1.6.3 Argument from Pity

The *argument from pity* is a fallacy that happens when pitying or feeling sorry for someone pushes us towards accepting a position on an unrelated matter or issue. Suppose Jill wants to be a lawyer and knows that she needs to get good grades in order to get into a good law school. For her critical reasoning class, she ends up getting a grade D. She may try to make the instructor feel bad for her: she struggled with cancer when she was young and barely came through alive; her parents passed away a few years ago; and her only sibling stopped talking to her. Does that mean that Jill *deserves* a higher grade? Does that mean that she *should* get a grade higher than a D? Of course not. The emotion of pity can be very powerful and strong, and it can easily cloud your judgment on things, just like anger and fear can. Arguments from pity (or appeal to pity) will have the following general form:

1. P (where P is a claim that is intended to induce pity).
2. Therefore C.

6.1.6.4 Apple Polishing

The fallacy of *apple polishing* is the fallacy of accepting a certain position on an issue or making a certain judgment or claim based on a pumped-up ego (i.e. on pride). This is also called “appeal to flattery”. Allowing praise of oneself to cloud one's judgment about the truth of a claim, or trying to get others to do this, is the apple polishing fallacy. For example, suppose I am trying to convince you that Jane's position on some issue is mistaken. In order to do this, I say “It takes a very reasonable person with a sharp intellect to examine the issue and see that Jane's position is wrong. The people in this audience are certainly way beyond average in intelligence, and I wouldn't be surprised if many here are at the genius level. In any case, because of the stellar audience we have here today, there probably is no need for me to tell you that Jane's position on this issue is wrong. Clearly, her position states that P, Q, and R, but only the naïve among us will fail to see the numerous problems with P, Q, and R.” The speaker here is trying to massage the pride of the audience in order to get you to accept a position. The pseudo-reasoning is this: *X is true, and you should believe that X is true. Why? Because you're smart, and you're astute, and you're not a naïve, misinformed person, and*

so of course you'll believe that X is true. The form of this fallacy is roughly as follows:

1. Person A claims P
2. Person A flatters person B.
3. Therefore, P is true.

6.1.6.5 Guilt Trip

Similar to the argument from pity is the fallacy of *guilt trip*, which tries to get someone to accept some position or to do something by eliciting feelings of guilt. The difference between guilt trip and argument from pity is that argument from guilt trip attempts to make you feel bad for something *you* did. So, the general form of guilt trip is something like this:

1. Person A makes you feel at fault (guilty) for something.
2. Person A claims P.
3. Therefore, P (as if to 'make up' for the fault you committed)

Several things need to be noted about this fallacy. First, we should not accept a claim as being true because we feel as if we owe someone recompense for a (perceived) bad we've committed against them. While it may seem ridiculous to do so, and ridiculous to bring this up, it happens to all of us more than we'd like to admit. How many times have you accepted a "I should do X" as true simply because someone made you feel bad for not doing X? Feeling guilty for a perceived wrongdoing is not sufficient support for truth in most cases. Second, being guilty may at times indicate that some wrong has been done, or at least a *perceived* wrong. However, feeling guilty is not a *reason* for thinking that something is *in fact* wrong. If Sam feels guilty because she cheated on her math exam, then it may be that her guilt is indicative of her having done something wrong (in this case, she would have in fact done something wrong). But her guilt is *not the reason* why what she did is wrong: cheating on a math exam is wrong *not* because one feels guilty for doing it, but because it's unfair and deceitful. So, if we are to accept that cheating on the math exam is wrong, it should not be on the basis of our feeling of guilt that we should accept that it's wrong, but for reasons.

6.1.7 Groupthink Fallacy

This fallacy is very similar to the peer pressure fallacy. Groupthink fallacy occurs when pride of membership in a group is substituted for reason and deliberation in arriving at a position on an issue. What happens here is that a person becomes proud of belonging to a certain group, whether it is a nation, religion, a family, a school, or just a social group. The pride of membership in the group blinds you with a kind of fanatic zeal, resulting in clouded judgments, incorrect reasoning, and a failure to respond to reasons. So, a groupthink fallacy occurs when you accept or reject a certain claim because doing so is identified with one's group.

Nationalism is a particular kind of groupthink fallacy that occurs very widely and runs rampantly throughout countries. Being proud to be American in an extreme way may lead you to think that *anyone* who criticizes America is

obviously wrong, ought to get out of America, or must be crushed. This kind of fanaticism runs the risk of making you not care about *true* right and wrong. For example, suppose Sal is so proud of being American that he writes off anyone who criticizes America as utterly mistaken and even as an enemy. Sarah tells Sal that America has a growing problem with poverty and that it's getting worse, with more and more people falling below the poverty line. Sal might say "Screw you – you're wrong! If you don't like it here then leave." Notice that Sal is not even responsive to what Sarah is saying. In fact, what Sarah says may be *true* and still Sal just won't have anything to do with it. Sal just completely rejects Sarah's claim as false on the basis of his own nationalism, not on the basis of any reasons. Incidentally, this kind of nationalism actually ends up hurting and damaging the country.

The general form of this fallacy is as follows:

1. I am a proud member of group G.
2. Group G endorses or rejects P.
3. Therefore, I should endorse or reject P.

6.1.8 Red Herring

Red herrings try to prevent you from getting at the heart of an issue. In a sense, all of these fallacies are red herrings since the premises given are often irrelevant to the issue at hand. In any case, for our purpose, red herrings are going to be something like a distraction. Red herrings try to lead you away from the main issue. They might be arguments, or they might be ways of responding to questions. When a person brings up a topic during a conversation that distracts from the original issue at hand, especially when the new topic is brought up specifically for the purpose of distracting, then the person has introduced a *red herring*.

Suppose someone was trying to show that Sam stole a car. The main issue on the table is whether or not Sam stole a car. In the discussion, the person says that Sam has been a delinquent in school, and that he assaulted some teachers. These are distracters because they take us away from the issue at hand. The issue is not whether or not Sam is a violent person or whether or not Sam is a bad student, but whether or not Sam stole a car. Red herring fallacy has the following general form:

1. Topic A is being discussed.
2. Topic B is introduced (as relevant and important in some way).
3. Topic A ceases to be discussed and is abandoned altogether.
4. Topic B becomes the primary issue.

6.1.9 Appeal to Popularity

The phrase "Everyone knows...", "It's common sense that...", and "it's obvious that...", as well as others like them, can be used in three different ways. First, they can be used as proof surrogates, which we talked about in the last chapter. Second, they can be used as a form of peer pressure argument and groupthink fallacy. Third, they can be used as an *argument from/appeal to popularity*, or simply *fallacy of popularity*. This fallacy occurs when someone urges you to accept a claim simply on the grounds that some

substantial number of people (other than experts and authorities) believes the claim. This is slightly different from a proof surrogate in that a proof surrogate will try to slant a claim in such a way that it sounds like it is true by saying that some authority or expert believes it, or that it's obvious, etc. The argument from popularity, however, actually tries to argue for a claim on the premise that everyone believes the claim to be true. For example, you should accept that plastic water bottles can be harmful to your health. Why? *Because everyone believes it.* Is the conclusion shown to be true because everyone believes it to be true? If everyone believed that the Earth is flat, or the moon is made of cheese, or that people of a particular race are inferior, does that mean that you should also believe these things to be true? No. Appeal to popularity has the following general form:

- a. It is popular to believe that P.
- b. Therefore, P.

There are two variations of the appeal to popularity. First, there is the appeal to common practice. This fallacy occurs in the following way. "Everyone does X. Therefore, it is okay (for me) to do X." Just because everyone does something does not make it right or okay for you to do it also. If everyone was murdering children, that doesn't mean that it's right or okay to murder children. You might say "Wait, but that's an extreme." Well, it is, but if you believe that if everyone does X then it's okay or right to do X, then you have to accept this extreme case too! When you commit the argument from common practice, you end up adopting the bad principle that it's okay to do X, whatever X, if everyone else is doing it, whether you want to adopt that principle or not. The premise that everyone does something does not support or prove the conclusion that it's morally permissible to do something. In fact, the premise is totally irrelevant.

Second, there is the appeal to tradition. This fallacy occurs in the following way. "It is a tradition to do X (believe X). Therefore, it is okay to do X (X is true)." (It could also be "It is a common cultural practice to do X..."). This is clearly bad reasoning. For example, it is a tradition for children to believe that Santa Claus exists, but that doesn't make Santa Claus exist. Suppose it was tradition to have a human slave – clearly that by itself is *not* a reason for thinking that it's okay to have a human slave. In fact, that something is a tradition is irrelevant to the issue of whether or not that thing is morally permissible. Suppose you said "Look, it's wrong to burn cats alive – we shouldn't do that" and another person in the group said "It's tradition man! It's what we do every New Years Eve!" You would clearly see this as a bad reason. It being a tradition has no bearing on whether or not it's acceptable to burn cats.

6.1.10 Two Wrongs Make a Right

This has to do mainly with wrongful behavior. A person commits the two-wrongs fallacy when that person thinks that it is okay to do X to another person if that other person already did X to him/her. In other words, the fallacy is this: if someone does something wrong to you, then it is okay or right for you to do the same thing to them. The fallacy looks like this:

1. Person 1 did X to me.

2. Doing X is wrong.
3. Hence, doing X to Person 1 is right.

This argument should strike you as very bad reasoning. If X is wrong, then it's wrong, *period*. If someone hit someone in my family with his car, I'd become very angry because running people over with your car is *wrong*. If I think that, then it doesn't make any sense for me to run someone in *his* family over with my car. (We're not talking about justification – that's a different issue). To 'get even' begins with seeing what was done to them as *being wrong*. Then how can doing it be *right*? Two wrongs don't make a right.

6.1.11 The Subjectivist Fallacy

The *subjectivist fallacy* occurs when someone argues that something is true because they simply *think* or *believe* or *feel* that it is true. The general form of this fallacy is this:

1. I think/believe that X is true.
2. Therefore, X is true.

This is clearly bad reasoning. If someone thinks that $2+2=10$, does that make it true that $2+2=10$? No. Suppose you got an A grade on all of the exams and assignments for a class, but the professor gave you an F. When you point this out to the professor, the professor says "Well, I think that you still deserve an F. So, it's true that you deserve an F." This is clearly unacceptable. Thinking or believing so doesn't make it so.

6.1.12 The Relativist Fallacy

The *relativist fallacy* occurs when someone claims that something is true or right for a certain society or culture simply because that is what that society or culture *thinks* is true or right. This is like the subjectivist fallacy, except this time it's not what the individual *subject* thinks is true or right, but what a *culture* or *society* or some group thinks is true or right. If thinking something is true isn't enough to make something actually true, then certainly it doesn't matter how many people think or believe it. In fact, many believed at one point that the sun was a God, and that the earth was flat. That by itself didn't make it true.

6.1.13 Slippery Slope

The slippery slope fallacy tries to argue that we shouldn't do something or believe something because doing so will progress inevitably to some undesirable conclusion. The slippery slope fallacy has the following general form:

1. Inevitably, X will lead to Y; and Y will lead to Z; and ... will lead to B.
2. <B is a bad result, and no evidence is given to show 1 is true>
3. Therefore, not X.

Typically, either a single conditional statement or causal statement is offered, or a string of conditional or causal statements is offered, and no support or evidence is given to show why we should even believe such statements are even true. For example, "You shouldn't drink. Why? Because if you drink,

the next thing you know you'll be smoking pot, and then snorting speed, then cocaine, then shooting up heroin, and then by that time you'll have lost all your teeth, you'll have a bad complexion, and your brain will be deteriorated and you'll be homeless and have no friends." This argument is a slippery slope argument, and it's a fallacy. Are all of the conditional statements really true? Are we really given a conclusive argument for thinking that we shouldn't drink? Not at all. It's a slippery slope because it goes from A and slides down to B, and no reason or argument is given to show why A inevitably would lead to B.

Another version of slippery slope is when someone thinks that we should continue to do A because we already started A. So, for example, suppose we're spraying pesticides on the garden. John asks "Should we keep spraying pesticides on this garden? These pesticides are poisonous." Bill may say "Well, we've already sprayed gallons of it. We should just keep spraying." This is a slippery slope – just because we've already sprayed pesticides doesn't mean that we should continue to do so.

Sometimes a slippery slope argument is valid. The argument "If A, then B; if B, then C; Not C; Therefore not A" is a valid argument. What would make this a slippery slope is the conditional statements are unwarranted and without evidence. Often, slippery slopes can be identified by their hyperbolic sense.

6.1.14 Misplacing the Burden of Proof

Misplacing the burden of proof occurs when the burden of proof rests on a person who makes some claim, but then it is wrongly shifted onto someone else who challenges or disagrees with that claim. Misplacing the burden of proof can also occur more generally, when person A is the one who should be giving reasons for his or her position, but then person B is wrongly taken to be the one who should be giving reasons for his or her position. In short, the burden of proof is misplaced when someone requires the wrong side of an issue to make its case first. Consider, for example, the following case. Mary says "There is a pink elephant the size of a football in the next room." Jill replies "No there isn't." Mary responds "Tell me why I'm wrong." Now, here the burden of proof should be on Mary: *she* is the one who needs to give reasons for her belief. But the burden of proof has been mistakenly shifted over to Jill. The burden of proof is not on Jill, it's on Mary, and it has been *misplaced* on Jill.

So how do you determine who exactly has the burden of proof? Who is the one that really should be the one that provides reasons or an argument for her position? There are several rules of thumb that may help to determine where the burden of proof lays:

1. *Initial Plausibility*: The less initial plausibility that a claim has, the greater the burden of proof we place on someone who asserts that claim. If someone claims that she swam from Los Angeles to Hawaii, then she would be making a claim that has very low initial plausibility. Hence, the burden of proof would be on her. If, however, I claim that the chair I'm sitting on right now is a real

thing and not an imaginary thing, and Freddy disagrees, then the burden of proof rests on Freddy, not me, since the claim that the chair I'm sitting on right now is not a real thing but an imaginary thing has much less initial plausibility than the claim that I made. In other words, as a general rule, the more outlandish or unlikely or uncommon a claim is, the more likely the burden of proof is on the person who makes the claim.

2. *Affirmative/Negative:* The person who makes a positive or affirmative claim typically has the burden of proof, all things being equal. Suppose I said that methane gas from cows was a huge contributor to greenhouse gases, then the burden of proof is on me, not the person who denies what I say. The reason is that I am the one making a positive claim, whereas the person who disagrees with me is making a negative claim (i.e. a claim that just denies what I'm saying). Of course, if the positive claim has much more initial plausibility than the negative claim (like in the chair example in 1 above), then the burden of proof may actually be on the person making the negative claim. Note: a positive claim is one where something is being affirmed, e.g. "John is in the house", "Democrats care about the nation", and "God exists". Negative claims are statements where something is being negated or denied, e.g. "John is *not* in the house", "Democrats *don't* care about the nation", and "God does *not* exist." Sometimes the affirmative will have much more initial plausibility than the negative, in which case the person who makes the negative claim will have the burden of proof. (see 1 above).

Typically, those who claim that something exists have the burden of proof over those who claim that something doesn't exist. However, this is tricky. Someone who claims that I don't exist, or that the world doesn't exist, has the burden of proof, since what they are saying lacks initial plausibility. But when someone says that unicorns exist, or that small gremlins exist on the other side of the moon, or that ghosts exist, then the burden of proof would be on them.

3. *Special Circumstances:* Sometimes certain circumstances will specify where exactly the burden of proof is to be, regardless of the first or second rule above (initial plausibility and affirmative/negative). Consider, for example, in our criminal courts. The burden of proof lies with the prosecution: it is their job to show that the defendant is guilty. The burden of proof is not on the defendant to show that s/he is innocent. Special circumstances typically dictate where the burden of proof lies when the stakes are really high. In the court case, the burden of proof is on the prosecution because the stakes are high: a mistake could lead to the punishment or imprisonment of an innocent person, which is very bad. If someone proposes some policy that has serious risks, then the burden of proof is on that person to show us why the policy should be implemented since the stakes are very high. If my friend says that we should

invest in something that's very expensive, then the burden of proof is on her/him because the stakes are high: much money is at stake.

What happens quite frequently is that people make the following mistake in reasoning: a person says that nobody has proved the existence of X and, hence, that X exists, or that we should believe that X exists, or that it is rational to believe that X exists. When you think that something exists or that it's rational to think that something exists simply because no one has proved that X does not exist, then you are making an *appeal to ignorance*, which is a fallacy. The form of this argument is this:

1. Nobody has proved that P is false.
2. Therefore, P is true.

Ignorance about something doesn't prove anything except ignorance about that thing!

6.1.15 Begging the Question

Begging the question occurs when we ask our audience to accept premises that are as controversial as the conclusion we're arguing for, and that are controversial on the same grounds. In other words, person X begs the question when:

1. X claims C.
2. X argues for C by stating premise P
3. C is controversial for reason R, and P is controversial for reason R, i.e. C and P are controversial for the same reason, R.

In general, begging the question occurs when you try to prove or show that some claim P is true by an argument with a premise that is P or implies P. It's basically *circular reasoning*, when you try to prove something but you end up using what you're trying to prove as a premise. Consider the following argument:

1. The Bible says that God exists.
2. God wrote the Bible
3. God always tells the truth.
4. Therefore, God exists.

This argument begs the question: it tries to argue for the existence of God by assuming what it's trying to prove. It already assumes that God exists in premise 2, and then tries to show that God exists using premise 2! It argues in a circle.

Here's another example:

1. Murdering a human being is wrong.
2. A baby is a human being.
3. Abortion is murdering a baby.
4. Hence, abortion is wrong.

This argument begs the question because it's controversial to consider abortion as the murdering of a human being, and premise 3 seems to be controversial on that same ground. Hence, this argument begs the question.

One thing about begging the question that makes it a very peculiar argument is that they are *valid* arguments. Yes, that's right, circular

reasoning is a *valid form* of reasoning. The reason is that the conclusion is *assumed* among the premises, so it is *impossible* for the conclusion to be false while *all* of the premises to be true, since the conclusion is among the premises!

6.1.16 Grouping Ambiguity Fallacies

The *fallacy of division* occurs when you reason from a premise that makes a claim about a group *collectively* to a conclusion that makes a claim about the *members* of the group *individually*. If I said that UC Berkeley students *collectively* have the highest test scores in the nation compared to other college students, and then concluded that *any* UC Berkeley student you picked will have a score on the SAT than any student at some other university, then I'm committing the fallacy of division. We might say that citizens of country X have the highest income than any other country, but this claim might be about the income of the citizens *taken together*. To conclude from this that any one citizen of X makes more than any one citizen of a country other than X is to commit the fallacy of division.

The fallacy of division can occur when we ascribe a property to a group, and then conclude that the members of the group also have that property. You have a premise that says that a group *collectively* is P (where P is a property like 'cube-shaped' or 'best in the NBA' or 'smartest'). Then, the group is *divided* into its individual members, and then one mistakenly reasons that each *member* is also P. This fallacy has the following general form:

1. The group G has the property of P.
2. Therefore, each member of G has the property of P.

The *fallacy of composition* is like the previous fallacy but goes in the other direction. It's called the fallacy of composition because it starts with a statement about the *individual members* of a group, saying that each member has some property P, and then *composes* them together as a whole, and then concludes that the group *collectively* has the property P. Each person on a sports team may be very good at playing that sport, but that does not mean that taken together, the team itself is a good team. The general form of the fallacy of composition is as follows:

1. Members of group G each have the property P.
2. G has the property of P.

6.1.17 Appeal to Consequence of Belief

This fallacy attempts to persuade you into thinking that some statement P is true or false by appealing to the consequences of P, or by appealing to the consequences of *believing* that P. For existence, it's false that marijuana has benefits because believing that can lead to marijuana abuse. There are obvious problems with this. First, just because the consequences of believing something are unacceptable, wrong, or undesirable doesn't mean it's false. Second, just because there have been certain consequences associated with believing something fails to show that (1) the belief is a

significant *cause* of those consequences, and (2) the consequences are *necessary consequences* of holding to that belief.

This fallacy has the following general form:

1. Q is the consequence of (believing that) P.
2. Q is wrong, unacceptable, inefficient, etc.
3. Therefore, P is false.

6.1.18 Wishful Thinking

This fallacy occurs when we accept or reject, or urge acceptance or rejection, of a claim simply because it would be pleasant or unpleasant if it were true. Typically, wishful thinking appeals to some positive emotions – happiness, pleasure, etc. – but it need not. This fallacy is very similar to the appeal to the consequences of belief fallacy. The difference is that this one appeals to what it *wishes* to be the case, whereas the appeal to the consequences of belief is more focused on what would *happen* if something were the case (or believed to be the case). Wishful thinking has the following general form:

1. It would be wonderful if P were true.
2. Therefore, P is true.

For example, Charlie may believe that she has a soul that continues to live on after her body dies. Is this true? She may say “Well, I think it’s true. It would be so nice and reassuring to think that death in this world is the beginning of life in another world.” Notice here that she may *seem* to be giving reasons for the truth of her belief, but really she’s not. She just believes because it would be nice if her belief were true. Some people may believe that God exists because they want it to be true: it would so great and awesome if God does exist – hence, God does exist. This is pseudo-reasoning.

6.1.19 Appeal to Authority

This fallacy attempts to persuade you into thinking that some claim is true on the basis of some authority who has endorsed or made the claim. The basic form of this fallacy is this:

1. Person/source A claims that P
(where A is a perceived authority)
2. Therefore, P is true.

Usually the source is an authority figure, or at least perceived as an authority figure, by the people involved. So, parents might say something, and when children question them, the parents say “Because I am your parent!” Less obvious appeals to authority are cases when people appeal to some authority figure like a leader, someone with some expertise, a teacher, or someone older. The problem with an appeal to authority is that just because an authority figure makes or endorses a claim does *not* imply that the claim is true.

However, as we’ll see later, appealing to authority *can* be a legitimate way of supporting the truth of a claim. We all need to rely on experts: medical doctors, scientists, dentists, personal trainers, etc. Even now you are relying

on the authority of my expertise. For now, it is good to ask yourself the following questions when dealing with any sort of appeal to authority:

- a. *Does the person have sufficient expertise in the subject matter in question?* Some people have some expertise but may not have enough to make them a credible expert authority. Someone who majored in psychology, or received a master's degree in psychology from an unrecognized university may not have the sufficient knowledge, let alone the sufficient expertise, to make credible assessments of someone's psychological well-being.
- b. *Is the claim being made by the person within her area(s) of expertise?* Often times you'll find someone who is an expert in area A making claims that properly belong to area B. Someone who works in evolutionary biology and has a Ph.D. in that discipline, even many awards for her research in that area, might make claims about ethics. While that person is to *some* extent a credible authority, she is nevertheless making claims *outside* her subject area.
- c. *Is there an adequate degree of agreement among the other experts in the subject in question?* A scientist who has a doctoral degree from a well-known university may claim that the earth is *not* warming as a result of human activity and industry, but certainly they are of a minority position. Sometimes, there may not be any majority or minority view on an issue due to the dearth of research, or because of the inconclusive nature of current and past research. For instance, a well-known philosopher may argue that there is no such thing as free will. However, the question of free will is one that hotly debated by philosophers, and there is far from any consensus on the issue.
- d. *Is the person making the claim not significantly biased?* The authority may indeed be a credible expert, but have certain biases. Claims about the current state of global economy might be made by a renowned economist at Harvard University, but it may be that this person is making a claim based on certain biases or conflict of interest. If her research is being paid by the coal or oil industries, then perhaps we ought to be skeptical.
- e. *Is the area of expertise of the person making the claim a legitimate area or discipline?* Someone who is an 'expert' at astrology may claim that the position of the planets in the next few days will result in a catastrophic event somewhere in the USA. However, astrology itself is not a legitimate discipline.
- f. *Is the authority to whom is being appealed one that has been actually identified?* Oftentimes the authority figure is left unspecified, referred to as "some authority figure" or "a well-respected expert" in a certain field.

These are just tips. Remember, appeal to authority is *not* always a bad thing. We should give more weight to those who have legitimate expertise in a legitimate area so long as the claim they are making is within or close to the area of expertise. It would be foolish to follow the word of any perceived authority simply because we perceive them as an authority figure. It is extremely foolish to think that recognized and legitimate experts in a certain field only express 'opinions' that are no more or less credible than the

opinions of others. However, the point is that it is invalid to argue in this way: Authority A says P; therefore P.

6.1.20 Gambler's Fallacy

This fallacy occurs when someone assumes that departure from the average, or a departure of what happens in the long term, will be corrected or changed in the short term. This fallacy typically has the following form:

1. X has happened.
2. X isn't what happens on average, or it isn't what is expected to happen in the long term.
3. Therefore, X will come to an end soon.

You can see why this is called "Gambler's Fallacy". If Sal was gambling at the Roulette table, and placed bets on the ball landing on red fifty times in a row, and lost all fifty times, Sal might think that since the average is supposed to be 50/50 in terms of black and red, and 50 times in a row it landed on black, it *must* be the case that the next spin or the one after will be red. This isn't true. Sal is committing the gambler's fallacy. This is a case where the events involved (landing on black and landing on red) are *mutually independent* probabilistically speaking. The probability of a coin landing heads as opposed to tails is 50/50. This probability is *independent* of how it landed before. Just because it landed tails the previous twenty flips does not mean that the probability of its landing tails in the next flip is greater.

Sometimes this fallacy occurs with cases where the probabilities of occurrence are *not* mutually independent. This happens when someone takes the statistics and tries to distribute the number of occurrences according to the statistics, while failing to appreciate and consider ways that the occurrences are mutually related in such a way that could affect the way those occurrences come about. For instance, suppose in the past three years, a boxer has won 50% of all of his fights. This year, he has fought in 10 fights, and he lost 5 of them and won the other 5. He has 10 fights remaining. A few months passes, and he loses 5 of those remaining fights, and has 5 more fights left to finish the year. In order to keep the 50% record he has, he has to win all of the remaining 5 fights. However, if one were to predict that he will in fact win the remaining 5 fights because of the fact that he has won 50% of his fights in the past three years, and that to keep to the 50% average, he must win the remaining 5 fights, and then conclude that he will indeed win the next 5 fights, one is then committing the gambler's fallacy. It could be that the previous 5 losses have really taken a toll on his body, weakening it and thereby *decreasing* the probability of winning the remaining 5 fights.

6.1.21 Spotlight Fallacy

This fallacy occurs when someone assumes that everyone or every case of a certain class/type is significantly like those of the same class/type that gets attention or coverage from the media. This fallacy has the following form:

1. Members or cases of type X with the property P receive a great deal of attention and coverage from the media.
2. Therefore, all members or cases of type X have the property P.

For example, the media may show people of some religion R as committing violent crime. Someone might then conclude that *everyone* who is a member of R is a violent criminal. This is an assumption and gains *no* support from the media coverage. It may very well be the case that the media tends to cover *only* those members of R that commit a violent crime, or *only* those cases of violent crime committed by members of R. The media may be ignoring members of R who do *not* commit violent crime, or violent crimes that are *not* committed by members of R.

6.1.22 General Non Sequiturs

A *non sequitur* is Latin for *does not follow*. In the context of assessing arguments, what this means is that the conclusion *does not follow* the premises/reasons given. So, clearly weak and invalid arguments are both *non sequiturs*. However, there are various tactics that people commonly use to make it seem that conclusion follows. One common tactic is when someone presents his position, but does *not* give reasons for his position. Instead, he presents some argument *against* his position and then attacks that argument. After, he makes it appear that he has argued *for* his position. The general form of this strategy is this:

1. I claim P
2. <I present an argument *against* P, and then attack that argument>
3. Therefore P

Notice that P is not supported at all. Perhaps in step 2 the person has successfully attacked the argument against P, but no reason has been given to *support* P, i.e. show that it is in fact true. Suppose my position is that we are dreaming. Then I present an argument that attempts to show that we are not dreaming. I then attack that argument and show that that argument is bad. Does it now follow that I have given positive reasons for my view? Not at all.

6.2 Formal Fallacies

Formal fallacies are fallacies that occur as a result of faulty argument structure. The argument *looks* like one of the valid argument forms, but in fact it is not. If these are given as deductive arguments, they are always invalid.

Fallacy of Denying the Antecedent (FDA)

The form of this fallacy is as follows

:

- i. If P, then Q:
- ii. It is not the case that P.
- iii. Therefore, it is not the case that Q.

This looks like a valid argument, namely, modus tollens (see section 3). However, in a modus tollens argument, a negation of the antecedent is inferred (concluded) based on two premises: a conditional, and a negation of its *consequent*. With FDA, the negation of the *consequent* is inferred (concluded) based on two premises: a conditional, and a negation of its *antecedent*. This is an invalid argument.

Example 1: Jin's eyes are not red; if Jin's eyes are red, then she is on drugs; therefore, Jin is not on drugs.

Example 2: if Sara is in Paris, France, then Sara is in Europe; Sara is not in Paris, France; therefore, Sara is not in Europe.

FDA can also occur without explicit conditionals. In particular, it can occur when dealing arguments involving universal claims. Here are some examples:

Example 3: All X are Y. Z is not X (or all Z are not X). Therefore, Z is not Y (or all Z are not Y).

It would be a valid move to infer that Z is not X (or all Z are not X) if the premises were 'All X are Y' and 'All Z are not Y'. Otherwise, however, it is invalid.

Fallacy of Affirming the Consequent (FAC)

The form of this fallacy is as follows:

- i. If P, then Q.
- ii. Q.
- iii. Therefore, P.

This looks like a valid argument, namely, modus ponens (see section 3). However, in a modus ponens argument, the consequent is inferred (concluded) based on two premises: a condition, and an affirmation of its *antecedent*. With FAC, the *antecedent* is inferred (concluded) based on two premises: a conditional, and an affirmation of its *consequent*. This is an invalid argument.

Example 1: Jin is on drugs; if Jin's eyes are red, then she is on drugs; therefore, Jin's eyes are red.

Example 2: if Sara is in Paris, France, then Sara is in Europe; Sara is in Europe; therefore, Sara is in Paris, France.

FAC can also occur without explicit conditionals. In particular, it can occur when dealing arguments involving universal claims. Here are some examples:

Example 3: All X are Y. Z is a Y (or all Z are Y). Therefore, Z is an X (or all Z are X).

It would be a valid move to infer that Z is a Y (or all Z are Y) if the premises were 'All X are Y' and 'All Z are X'. Otherwise, however, it is invalid.

Fallacy of Undistributed Middle (FUM)

This fallacy occurs when one infers that two things are related to each other in some way because they are related in a certain way to a common thing. There are different ways that this fallacy occurs. Here are a few ways:

- X is F; Y is F; therefore, X is Y.
- If X, then Z; If Y, then Z; therefore, If X, then Y.
- X has properties P1, P2, P3, etc.; Y has properties P1, P2, P3, etc.; therefore, X is Y.
- All X are Y; All Z are Y; therefore, all X are Z.

All of these are invalid arguments.

6.3 Fallacies in Language

1. Fallacy of Composition
2. Fallacy of Division
3. Equivocation
4. Amphiboly

6.4 Inductive Fallacies

Inductive Fallacies are fallacies that occur in giving inductive arguments. These fallacies typically consist in making an inference from premises that provide weak or at best questionable support for the inference.

6.4.1 Generalization Fallacies

6.4.1.1 *Hasty Generalization*

- 6.4.1.2 Argument from Anecdote
- 6.4.1.3 Argument from Small Sample Size
- 6.4.1.1 Generalizing from Exceptional Cases
- 6.4.1.2 Fallacy of Biased Sample
- 6.4.1.3 Self-Selection Fallacy
- 6.4.1.4 Accident Fallacy
- 6.4.1.2 Fallacies in Causal Reasoning
 - Cum Hoc Ergo Propter Hoc
 - Post Hoc Ergo Propter Hoc
 - Causal Anecdote